UNIT I

INTRODUCTION

DEFINITION :

Research in common parlance refers to a search for knowledge. Once can also define research as a scientific and systematic search for pertinent information on a specific topic. In fact, research is an art of scientific investigation. The Advanced Learner's Dictionary of Current English lays down the meaning of research as "a careful investigation or inquiry specially through search for new facts in any branch of knowledge."1 Redman and Mory define research as a "systematized effort to gain new knowledge."2 Some people consider research as a movement, a movement from the known to the unknown.

OBJECTIVES:

The purpose of research is to discover answers to questions through the application of scientific procedures. The main aim of research is to find out the truth which is hidden and which has not been discovered as yet. Though each research study has its own specific purpose, we may think of research objectives as falling into a number of following broad groupings:

1. To gain familiarity with a phenomenon or to achieve new insights into it (studies with this object in view are termed as *exploratory* or *formulative* research studies);
2. To portray accurately the characteristics of a particular individual, situation or a group (studies with this object in view are known as *descriptive* research studies);
3. To determine the frequency with which something occurs or with which it is associated with something else (studies with this object in view are known as *diagnostic* research studies);
4. To test a hypothesis of a causal relationship between variables (such studies are known as *hypothesis-testing* research studies).

TYPES OF RESEARCH

The basic types of research are as follows:

(i) ***Descriptive vs. Analytical:*** *Descriptive research* includes surveys and fact-finding enquiries of different kinds. The main characteristic of this method is that the researcher has

no control over the variables; he can only report what has happened or what is happening. The methods of research utilized in descriptive research are survey methods of all kinds, including comparative and correlational methods. In *analytical research*, on the other hand, the researcher has to use facts or information already available, and analyze these to make a critical evaluation of the material.

(ii) ***Applied vs. Fundamental :***Research can either be applied (or action) research or fundamental (to basic or pure) research. *Applied research* aims at finding a solution for an immediate problem facing a society or an industrial/business organization, whereas *fundamental research* is mainly concerned with generalisations and with the formulation of a theory.

(iii) ***Quantitative vs. Qualitative :***Quantitative research is based on the measurement of quantity or amount. It is applicable to phenomena that can be expressed in terms of quantity. Qualitative research, on the other hand, is concerned with qualitative phenomenon, i.e., phenomena relating to or involving quality or kind.

(iv) ***Conceptual vs. Empirical :***Conceptual research is that related to some abstract idea(s) or theory. It is generally used by philosophers and thinkers to develop new concepts or to reinterpret existing ones. On the other hand, empirical research relies on experience or observation alone, often without due regard for system and theory. It is data-based research, coming up with conclusions which are capable of being verified by observation or experiment.

(v) ***Some Other Types of Research****:* All other types of research are variations of one or more of the above stated approaches, based on either the purpose of research, or the time required to accomplish research, on the environment in which research is done, or on the basis of some other similar factor.

VARIOUS STEPS IN RESEARCH PROCESS:

Before embarking on the details of research methodology and techniques.

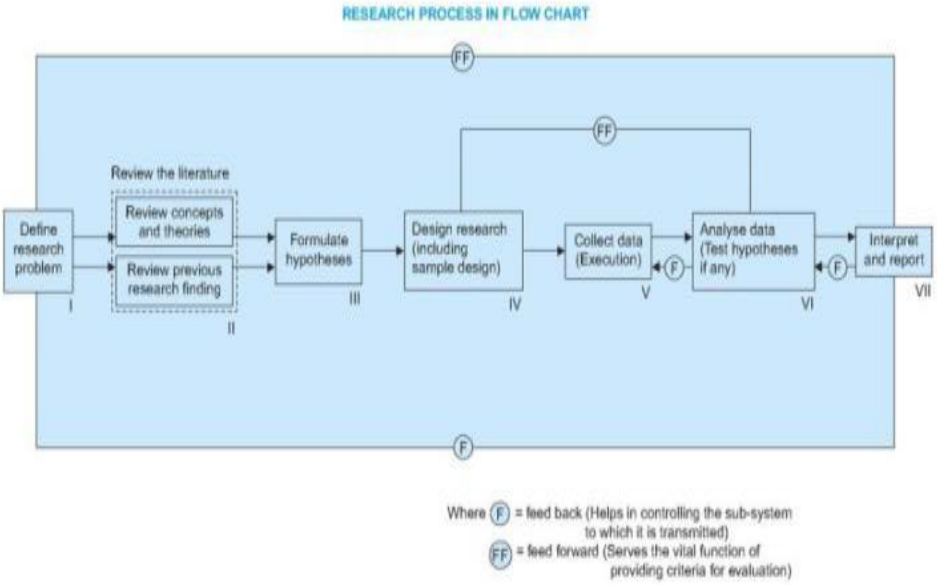


Fig. 1.1

The chart shown in Figure 1.1 well illustrates a research process. Ts a research process chart shown in Figure 1.1 well illustrates a research process.

However, the following order concerning various steps provides a useful procedural guideline regarding the research process:

(1) Formulating the research problem;

(2) Extensive literature survey;

(3) Developing the hypothesis;

(4) Preparing the research design;

(5) Determining sample design;

(6) Collecting the data;

(7) Execution of the project;

(8) Analysis of data;

(9) Hypothesis testing;

(10) Generalisations and interpretation, and

(11) Preparation of the report or presentation of the results,

1. Formulating the research problem:

There are two types of research problems, viz., those which relate to states of nature and those which relate to relationships between variables. At the very outset the researcher must single out the problem he wants to study, i.e., he must decide the general area of interest or aspect of a subject-matter that he would like to inquire into. Initially the problem may be stated in a broad general way and then the ambiguities, if any, relating to the problem be resolved. Then, the feasibility of a particular solution has to be considered before a working formulation of the problem can be set up. The formulation of a general topic into a specific research problem, thus, constitutes the first step in a scientific enquiry. Essentially two steps are involved in formulating the research problem, viz., understanding the problem thoroughly, and rephrasing the same into meaningful terms from an analytical point of view. The best way of understanding the problem is to discuss it with one's own colleagues or with those having some expertise in the matter. In an academic institution the researcher can seek the help from a guide who is usually an experienced man and has several research problems in mind. Often, the guide puts forth the problem in general terms and it is up to the researcher to narrow it down and phrase the problem in operational terms. In private business units or in governmental organisations, the problem is usually earmarked by the administrative agencies with whom the researcher can discuss as to how the problem originally came about and what considerations are involved in its possible solutions.

2. **Extensive literature survey:**

Once the problem is formulated, a brief summary of it should be written down. It is compulsory for a research worker writing a thesis for a Ph.D. degree to write a synopsis of the topic and submit it to the necessary Committee or the Research Board for approval. At this juncture the researcher should undertake extensive literature survey connected with the problem. For this purpose, the abstracting and indexing journals and published or unpublished bibliographies are the first place to go to. Academic journals, conference proceedings, government reports, books etc., must be tapped depending on the nature of the problem. In this process, it should be remembered that one source will lead to another. The earlier studies, if any, which are similar to the study in hand should be carefully studied. A good library will be a great help to the researcher at this stage.

3.Development of working hypotheses:

After extensive literature survey, researcher should state in clear terms the working hypothesis or hypotheses. Working hypothesis is tentative assumption made in order to draw out and test its logical or empirical consequences.

How does one go about developing working hypotheses? The answer is by using the following approach:

(a) Discussions with colleagues and experts about the problem, its origin and the objectives in seeking a solution;

(b) Examination of data and records, if available, concerning the problem for possible trends, peculiarities and other clues;

(c) Review of similar studies in the area or of the studies on similar problems; and

(d) Exploratory personal investigation which involves original field interviews on a limited scale with interested parties and individuals with a view to secure greater insight into the practical aspects of the problem

4.Preparing the research design:

The research problem having been formulated in clear cut terms, the researcher will be required to prepare a research design, i.e., he will have to state the conceptual structure within which research would be conducted. In other words, the function of research design is to provide for the collection of relevant evidence with minimal expenditure of effort, time and

money. But how all these can be achieved depends mainly on the research purpose. Research purposes may be grouped into four categories, viz.,

(i) Exploration,

(ii) Description,

(iii) Diagnosis, and

(iv) Experimentation.

The preparation of the research design, appropriate for a particular research problem, involvesusually the consideration of the following:

(i) the means of obtaining the information;

(ii) the availability and skills of the researcher and his staff (if any);

(iii)explanation of the way in which selected means of obtaining information will be organisedand the reasoning leading to the selection;

(iv) the time available for research; and

(v) the cost factor relating to research, i.e., the finance available for the purpose.

5.Determining sample design:

All the items under consideration in any field of inquiry constitute a 'universe' or 'population'. A complete enumeration of all the items in the 'population' is known as a census inquiry. A brief mention of the important sample designs is as follows:

(i) ***Deliberate sampling:***Deliberate sampling is also known as purposive or non-probability sampling. This sampling method involves purposive or deliberate selection of particular units of the universe for constituting a sample which represents the universe. When population elements are selected for inclusion in the sample based on the ease of access, it can be called *convenience sampling*.

(ii) ***Simple random sampling:***This type of sampling is also known as chance sampling or probability sampling where each and every item in the population has an equal chance of inclusion in the sample and each one of the possible samples, in case of finite universe, has the same probability of being selected.

(iii) ***Systematic sampling****:* In some instances the most practical way of sampling is to select every 15th name on a list, every 10th house on one side of a street and so on. Sampling of this type is known as systematic sampling.

(iv) ***Stratified sampling****:* If the population from which a sample is to be drawn does not constitute

a homogeneous group, then stratified sampling technique is applied so as to obtain a representative sample. If the items selected from each stratum is based on simple random sampling the entire procedure, first stratification and then simple random sampling, is known as *stratified random sampling*.

(v) *Quota sampling:* In stratified sampling the cost of taking random samples from individual strata is often so expensive that interviewers are simply given quota to be filled from different strata, the actual selection of items for sample being left to the interviewer's judgement . This is called quota sampling.

(vi) *Cluster sampling and area sampling:* Cluster sampling involves grouping the population and then selecting the groups or the clusters rather than individual elements for inclusion in the sample.

(vii) *Multi-stage sampling*: This is a further development of the idea of cluster sampling. This technique is meant for big inquiries extending to a considerably large geographical area like an entire country.

(viii) *Sequential sampling:* This is somewhat a complex sample design where the ultimate size of the sample is not fixed in advance but is determined according to mathematical decisions on the basis of information yielded as survey progresses. This design is usually adopted under acceptance sampling plan in the context of statistical quality control.

6.Collecting the data:

In dealing with any real life problem it is often found that data at hand are inadequate, and hence, it becomes necessary to collect data that are appropriate. There are several ways of collecting the appropriate data which differ considerably in context of money costs, time and other resources at the disposal of the researcher. But in the case of a survey, data can be collected by any one or more of the following ways:

(i) *By observation:*This method implies the collection of information by way of investigator's own observation, without interviewing the respondents. The information obtained relates to what is currently happening and is not complicated by either the past behaviour or future intentions or attitudes of respondents. This method is no doubt an expensive method and the information provided by this method is also very limited. As such this method is not suitable in inquiries where large samples are concerned.

(ii) *Through personal interview:*The investigator follows a rigid procedure and seeks answers to a set of pre-conceived questions through personal interviews. This method of collecting

data is usually carried out in a structured way where output depends upon the ability of the interviewer to a large extent.

(iii) ***Through telephone interviews:*** This method of collecting information involves contacting the respondents on telephone itself. This is not a very widely used method but it plays an important role in industrial surveys in developed regions, particularly, when the survey has to be accomplished in a very limited time.

(iv) ***By mailing of questionnaires:*** The researcher and the respondents do come in contact with each other if this method of survey is adopted. Questionnaires are mailed to the respondents with a request to return after completing the same. It is the most extensively used method in various economic and business surveys. Before applying this method, usually a Pilot Study for testing the questionnaire is conduced which reveals the weaknesses, if any, of the questionnaire. Questionnaire to be used must be prepared very carefully so that it may prove to be effective in collecting the relevant information.

(v) ***Through schedules:*** Under this method the enumerators are appointed and given training. They are provided with schedules containing relevant questions. These enumerators go to respondents with these schedules. Data are collected by filling up the schedules by enumerators on the basis of replies given by respondents. Much depends upon the capability of enumerators so far as this method is concerned. Some occasional field checks on the work of the enumerators may ensure sincere work.

7.Execution of the project:

Execution of the project is a very important step in the research process. If the execution of the project proceeds on correct lines, the data to be collected would be adequate and dependable. The researcher should see that the project is executed in a systematic manner and in time. If the survey is to be conducted by means of structured questionnaires, data can be readily machine-processed. In such a situation, questions as well as the possible answers may be coded. If the data are to be collected through interviewers, arrangements should be made for proper selection and training of the interviewers. The training may be given with the help of instruction manuals which explain clearly the job of the interviewers at each step. Occasional field checks should be made to ensure that the interviewers are doing their assigned job sincerely and efficiently.

A careful watch should be kept for unanticipated factors in order to keep the survey as much realistic as possible. This, in other words, means that steps should be taken to ensure that the

survey is under statistical control so that the collected information is in accordance with the pre-defined standard of accuracy. If some of the respondents do not cooperate, some suitable methods should be designed to tackle this problem. One method of dealing with the non-response problem is to make a list of the non-respondents and take a small sub-sample of them, and then with the help of experts vigorous efforts can be made for securing response.

8. Analysis of data:

After the data have been collected, the researcher turns to the task of analyzing them. The analysis of data requires a number of closely related operations such as establishment of categories, the application of these categories to raw data through coding, tabulation and then drawing statistical inferences. The unwieldy data should necessarily be condensed into a few manageable groups and tables for further analysis. Thus, researcher should classify the raw data into some purposeful and usable categories. *Coding* operation is usually done at this stage through which the categories of data are transformed into symbols that may be tabulated and counted. *Editing* is the procedure that improves the quality of the data for coding. With coding the stage is ready for tabulation. *Tabulation* is a part of the technical procedure wherein the classified data are put in the form of tables. The mechanical devices can be made use of at this juncture. A great deal of data, specially in large inquiries, is tabulated by computers. Computers not only save time but also make it possible to study large number of variables affecting a problem simultaneously.

9.Hypothesis-testing:

After analysing the data as stated above, the researcher is in a position to test the hypotheses, if any, he had formulated earlier. Do the facts support the hypotheses or they happen to be contrary? This is the usual question which should be answered while testing hypotheses. Various tests, such as Chi square test, *t*-test, *F*-test, have been developed by statisticians for the purpose. The hypotheses may be tested through the use of one or more of such tests, depending upon the nature and object of research inquiry. Hypothesis-testing will result in either accepting the hypothesis or in rejecting it. If the researcher had no hypotheses to start with, generalisations established on the basis of data may be stated as hypotheses to be tested by subsequent researches in times to come.

10.Generalisations and interpretation:

If a hypothesis is tested and upheld several times, it may be possible for the researcher to arrive at generalisation, i.e., to build a theory. As a matter of fact, the real value of research lies in its ability to arrive at certain generalisations. If the researcher had no hypothesis to start with, he might seek to explain his findings on the basis of some theory. It is known as interpretation. The process of interpretation may quite often trigger off new questions which in turn may lead to further researches.

11.Preparation of the report or the thesis:

Finally, the researcher has to prepare the report of what has been done by him. Writing of report must be done with great care keeping in view the following:

1. The layout of the report should be as follows:

(*i*) the preliminary pages;

(*ii*) the main text, and

(*iii*) the end matter.

In its preliminary pages the report should carry title and date followed by acknowledgements and foreword. Then there should be a table of contents followed by a list of tables and list of graphs and charts, if any, given in the report. *The main text of the report* should have the following parts:

(a) ***Introduction:***It should contain a clear statement of the objective of the research and an explanation of the methodology adopted in accomplishing the research. The scope of the study along with various limitations should as well be stated in this part.

(b) ***Summary of findings:***After introduction there would appear a statement of findings and recommendations in non-technical language. If the findings are extensive, they should be summarised.

(c) ***Main report:***The main body of the report should be presented in logical sequence and broken-down into readily identifiable sections.

(d) ***Conclusion:***Towards the end of the main text, researcher should again put down the results of his research clearly and precisely. In fact, it is the final summing up.

2. Report should be written in a concise and objective style in simple language avoiding vague expressions such as ‘it seems,’ ‘there may be’, and the like.

3. Charts and illustrations in the main report should be used only if they present the information more clearly and forcibly.

4. Calculated ‘confidence limits’ must be mentioned and the various constraints experienced in conducting research operations may as well be stated.

MATHEMATICAL TOOLS FOR ANALYSIS

we can write that scales of measurement can be considered in terms of their mathematical properties. The most widely used classification of measurement scales are:

(a) Nominal scale;

(b) Ordinal scale;

(c) Interval scale;

(d) Ratio scale.

(a) Nominal scale: Nominal scale is simply a system of assigning number symbols to events in order to label them. The usual example of this is the assignment of numbers of basketball players in order to identify them. Such numbers cannot be considered to be associated with an ordered scale for their order is of no consequence; the numbers are just convenient labels for the particular class of events and as such have no quantitative value. Nominal scales provide convenient ways of keeping track of people, objects and events. One cannot do much with the numbers involved. For example, one cannot usefully average the numbers on the back of a group of football players and come up with a meaningful value. Neither can one usefully compare the numbers assigned to one group with the numbers assigned to another. The counting of members in each group is the only possible arithmetic operation when a nominal scale is employed. Accordingly, we are restricted to use mode as the measure of central tendency. There is no generally used measure of dispersion for nominal scales. Chi-square test is the most common test of statistical significance that can be utilized, and for the measures of correlation, the contingency coefficient can be worked out.

Nominal scale is the least powerful level of measurement. It indicates no order or distance relationship and has no arithmetic origin. A nominal scale simply describes differences between things by assigning them to categories. Nominal data are, thus, counted data. The scale wastes any information that we may have about varying degrees of attitude, skills, understandings, etc. In spite of all this, nominal scales are still very useful and are widely used in surveys and other ex-post-facto research when data are being classified by major sub-groups of the population.

(b) Ordinal scale: The lowest level of the ordered scale that is commonly used is the ordinal scale. The ordinal scale places events in order, but there is no attempt to make the intervals of the scale equal in terms of some rule. Rank orders represent ordinal scales and are frequently used in research relating to qualitative phenomena. A student's rank in his graduation class involves the use of an ordinal scale. One has to be very careful in making statement about scores based on ordinal scales. For instance, if Ram's position in his class is 10 and Mohan's position is 40, it cannot be said that Ram's position is four times as good as that of Mohan. The statement would make no sense at all. Ordinal scales only permit the ranking of items from highest to lowest. Ordinal measures have no absolute values, and the real differences between adjacent ranks may not be equal. All that can be said is that one person is higher or lower on the scale than another, but more precise comparisons cannot be made.

Thus, the use of an ordinal scale implies a statement of 'greater than' or 'less than' (an equality statement is also acceptable) without our being able to state how much greater or less. The real difference between ranks 1 and 2 may be more or less than the difference between ranks 5 and 6. Since the numbers of this scale have only a rank meaning, the appropriate measure of central tendency is the median. A percentile or quartile measure is used for measuring dispersion. Correlations are restricted to various rank order methods. Measures of statistical significance are restricted to the non-parametric methods.

(c) Interval scale: In the case of interval scale, the intervals are adjusted in terms of some rule that has been established as a basis for making the units equal. The units are equal only in so far as one accepts the assumptions on which the rule is based. Interval scales can have an arbitrary zero, but it is not possible to determine for them what may be called an absolute zero or the unique origin. The primary limitation of the interval scale is the lack of a true zero; it does not have the capacity to measure the complete absence of a trait or characteristic. The Fahrenheit scale is an example of an interval scale and shows similarities in what one can and cannot do with it. One can say that an increase in temperature from 30° to 40° involves the same increase in temperature as an increase from 60° to 70°, but one cannot say that the temperature of 60° is twice as warm as the temperature of 30° because both numbers are dependent on the fact that the zero on the scale is set arbitrarily at the temperature of the freezing point of water. The ratio of the two temperatures, 30° and 60°, means nothing because zero is an arbitrary point.

Interval scales provide more powerful measurement than ordinal scales for interval scale also incorporates the concept of equality of interval. As such more powerful statistical measures

can be used with interval scales. Mean is the appropriate measure of central tendency, while standard deviation is the most widely used measure of dispersion. Product moment correlation techniques are appropriate and the generally used tests for statistical significance are the 't' test and 'F' test.

(d) Ratio scale: Ratio scales have an absolute or true zero of measurement. The term 'absolute zero' is not as precise as it was once believed to be. We can conceive of an absolute zero of length and similarly we can conceive of an absolute zero of time. For example, the zero point on a centimeter scale indicates the complete absence of length or height. But an absolute zero of temperature is theoretically unobtainable and it remains a concept existing only in the scientist's mind. The number of minor traffic-rule violations and the number of incorrect letters in a page of type script represent scores on ratio scales. Both these scales have absolute zeros and as such all minor traffic violations and all typing errors can be assumed to be equal in significance. With ratio scales involved one can make statements like "Jyoti's" typing performance was twice as good as that of "Reetu." The ratio involved does have significance and facilitates a kind of comparison which is not possible in case of an interval scale.

Ratio scale represents the actual amounts of variables. Measures of physical dimensions such as weight, height, distance, etc. are examples. Generally, all statistical techniques are usable with ratio scales and all manipulations that one can carry out with real numbers can also be carried out with ratio scale values. Multiplication and division can be used with this scale but not with other scales mentioned above. Geometric and harmonic means can be used as measures of central tendency and coefficients of variation may also be calculated.

Thus, proceeding from the nominal scale (the least precise type of scale) to ratio scale (the most precise), relevant information is obtained increasingly. If the nature of the variables permits, the researcher should use the scale that provides the most precise description. Researchers in physical sciences have the advantage to describe variables in ratio scale form but the behavioural sciences are generally limited to describe variables in interval scale form, a less precise type of measurement.

Sources of Error in Measurement

Measurement should be precise and unambiguous in an ideal research study. This objective, however, is often not met with in entirety. As such the researcher must be aware about the

sources of error in measurement. The following are the possible sources of error in measurement.

(a) Respondent: At times the respondent may be reluctant to express strong negative feelings or it is just possible that he may have very little knowledge but may not admit his ignorance. All this reluctance is likely to result in an interview of 'guesses.' Transient factors like fatigue, boredom, anxiety, etc. may limit the ability of the respondent to respond accurately and fully.

(b) Situation: Situational factors may also come in the way of correct measurement. Any condition which places a strain on interview can have serious effects on the interviewer-respondent rapport. For instance, if someone else is present, he can distort responses by joining in or merely by being present. If the respondent feels that anonymity is not assured, he may be reluctant to express certain feelings.

(c) Measurer: The interviewer can distort responses by rewording or reordering questions. His behaviour, style and looks may encourage or discourage certain replies from respondents. Careless mechanical processing may distort the findings. Errors may also creep in because of incorrect coding, faulty tabulation and/or statistical calculations, particularly in the data-analysis stage.

(d) Instrument: Error may arise because of the defective measuring instrument. The use of complex words, beyond the comprehension of the respondent, ambiguous meanings, poor printing, inadequate space for replies, response choice omissions, etc. are a few things that make the measuring instrument defective and may result in measurement errors. Another type of instrument deficiency is the poor sampling of the universe of items of concern.

Researcher must know that correct measurement depends on successfully meeting all of the problems listed above. He must, to the extent possible, try to eliminate, neutralize or otherwise deal with all the possible sources of error so that the final results may not be contaminated.

DEVELOPING A RESEARCH QUESTION

What is a Research Question?

A research question guides and centers your research. It should be clear and focused, as well as synthesize multiple sources to present your unique argument. Even if your instructor has given you a specific assignment, the research question should ideally be something that you are interested in or care about. Be careful to avoid the "all-about" paper and questions that can be answered in a few factual statements.

How Do I Develop A Research Question

Think about a topic that you would like to study. A more narrow focus on the topic will help drive your research.

A quick search of some of the literature on the topic will help guide you towards questions about your topic. If you were interested in gender roles in the Native American Southwest, what aspect of it are you interested in? What time period? What age group?

These questions will flush out your interest and will help you create a research question.

Choice of Questions

Ask the following 8 questions to evaluate the quality of your research question and the ease with which you should be able to answer it:

1. Does the question deal with a topic or issue that interests me enough to spark my own thoughts and opinions?

2. Is the question easily and fully researchable?

3. What type of information do I need to answer the research question? Qualitative? Quantitative? Original research?

4. Is the scope of this information reasonable (e.g., can I really research 30 online writing programs developed over a span of 10 years?)

5. Given the type and scope of the information that I need, is my question too broad, too narrow, or okay?

6. What sources will have the type of information that I need to answer the research question (journals, books, Internet resources, government documents, people)?

7. Can I access these sources?

8. Given my answers to the above questions, do I have a good quality research question that I actually will be able to answer by doing research?

LITERATURE REVIEW

Based on personal observations, comments by students, and the criticisms of thesis assessors, the literature review is often the hardest part of a thesis to write well. That this is so is quite understandable given the fact that writing a literature review is a bit like putting the pieces of a large and complex jigsaw puzzle together without the benefit of a “picture on the box” to indicate where the pieces might or should go!

Another cause of difficulty are the questions of what exactly needs reviewing and what should go where because there are a considerable range of choices.

To help you write effective literature reviews, this section looks at:

- **Goals of literature reviews**
- **Common problems**
- **An idealised process for conducting a literature review**
- **Common difficulties and how to deal with them**

Literature reviews - common problems

Common problems with literature reviews include:

List-like writing that lacks synthesis

(e.g. Smith (2003) investigated X and found Y. Jones (2000) looked at B and found C. Adams (1995) verified that M causes N. ...).

This is a bit like giving the reader one piece of a jigsaw puzzle at a time. Your job as a writer is to put all the "pieces of the jigsaw puzzle" together and then describe the resulting picture to the reader, pointing out what parts of the picture are clear, what parts are fuzzy, and what parts are missing altogether. You would then identify the goal of your research as being to bring one of the "fuzzy areas" into "sharper focus", to "fill in one of the holes", or to "develop the picture into new directions".

Not being sufficiently critical

Here it is important to note that the purpose of a literature review is **not** just to summarise what is currently known about a topic, but is also to provide a detailed **justification** for your research. That is, ultimately it is in the form of an **argument**, and this argument is developed by pointing out "holes" in the jigsaw puzzle that need filling or "fuzzy" parts that need clarifying see example 2. It might also put some disparate pieces together which suggest a promising line of research see example 3.

As research is supposed to make an "original" contribution to human knowledge, one of the things you need to do as a writer is to **demonstrate** that you are not simply repeating what has already been done before (or if you are, that there is a reason to have some doubt about the previous findings, perhaps as a result of some methodological weakness, thus necessitating the need for a confirmatory study with an improved methodology, or that the previous results are likely to need updating). Thus, as you review the existing literature, you need to identify any limitations, deficiencies, or gaps in existing knowledge or practice that need to be addressed.

Not discriminating between relevant and irrelevant materials

A literature review is **not** supposed to simply demonstrate how much you've read, but provide a description of how **certain parts** of what you have read provide the foundation for, motivate, and frame **your** research. As such, the relevance to your thesis of what you review needs to be clear to the reader. With "background material", which is important for orientating the reader, it is particularly important to ask: "What of this background does the reader **really** need to appreciate what comes next, and what is ‘trivia' or unimportant detail which can be left out?" see example 5.

Lack of a clear organizational structure

Using a mind map or plan can help to address this problem.

Exclusion of landmark studies

This suggests to the reader that the student "isn't on top of the field". **But note, your literature review needs to be focussed on your thesis; examiners don't want to read material that is not relevant**.

Relying on material that is likely to be out-of-date

For PhD and MPhil students, note that the literature review that you did for confirmation will need to be updated for your final thesis.

Adopting a parochial perspective

This might happen if you were to mostly cite papers produced by the research group you are a member of or if you were to look mainly at the literature on a topic produced in one country or continent.

SURVEYING

The essence of survey method can be explained as "questioning individuals on a topic or topics and then describing their responses"[1]. In business studies survey method of primary data collection is used in order to test concepts, reflect attitude of people, establish the level of customer satisfaction, conduct segmentation research and a set of other purposes. Survey method can be used in both, quantitative, as well as, qualitative studies.
Survey method pursues two main purposes:

1. Describing certain aspects or characteristics of population and/or
2. Testing hypotheses about nature of relationships within a population.

Survey method can be broadly divided into three categories: mail survey, telephone survey and personal interview. The descriptions of each of these methods are briefly explained on the following table [2]:

Survey method	Description
Mail survey	A written survey that is self-administered
Telephone survey	A survey conducted by telephone in which the questions are read to the respondents
Personal interview	A face-to-face interview of the respondent

Major survey methods and their descriptions

Alternatively, from the viewpoint of practicality, the most popular variations of surveys include questionnaires, interviews and documentation review. The main advantages and disadvantages associated with these primary data collection methods are explained by Denscombe (2010)[3] in the following manner:

Method	Purpose	Advantages	Disadvantages
Questionnaires	Conducted in order to gather large size of information in a short period of time	Members of the sample group can remain anonymous Considerably cheaper than most of the other primary data collection methods Possibility of generating large amount of data	Difficulties of ensuring greater depth for the research The problem of the 'first choice selection'
Interviews	Conducted in order to reflect emotions and experiences, and explore issues with a greater focus	The possibility to direct the process of data collection The possibility to collect the specific type of information required	Great amount of time required in order to arrange and conduct interviews and primary data collection. Additional costs might be incurred associated with arranging and conducting interviews, travelling etc. Potential for interviewee bias

Documentation review	Conducted in order to study issues that have developed over a specific period of time	Possibility to retrieve comprehensive information	Challenges associated with access to documentation Inflexibility of the research process

Advantages and disadvantages of survey data collection methods

Advantages of Survey Method

1. Surveys can be conducted faster and cheaper compared to other methods of primary data collection such as observation and experiments
2. Primary data gathered through surveys are relatively easy to analyse

Disadvantages of Survey Method

1. In some cases, unwillingness or inability of respondents to provide information
2. Human bias of respondents, i.e. respondents providing inaccurate information
3. Differences in understanding: it is difficult to formulate questions in such a way that it will mean exactly same thing to each respondent

SYNTHESIZING

Mixed research synthesis is the latest addition to the repertoires of mixed methods research and systematic review. Mixed research synthesis requires that the problems generated by the methodological diversity within and between qualitative and quantitative studies be resolved. Three basic research designs accommodate this diversity, including the segregated, integrated, and contingent designs. Much work remains to be done before mixed research synthesis can secure its place in the repertoires of mixed methods research and systematic review, but the effort is well worth it as it has the potential to enhance both the significance and utility for practice of the many qualitative and quantitative studies constituting shared domains of research.

Mixed research synthesis is the latest addition to the repertoires of mixed methods research and systematic review. *Mixed research synthesis* is our name for the type of systematic review

aimed at the integration of results from both qualitative and quantitative studies in a shared domain of empirical research. In contrast to mixed methods research in which the data set subject to analysis and interpretation is composed of the qualitative and quantitative *data* (e.g., from interviews, observations, questionnaires, physiologic measures, and the like) obtained directly from research participants within a single study or program of research, the data in mixed research synthesis studies are the *findings* of primary qualitative and quantitative studies in a designated body of empirical research. The focus of mixed research synthesis studies is on researchers integrations of their data, or the results they report; the products of mixed research synthesis studies are other researchers' (i.e., reviewers of research) integrations of those results to "sum up" what is known about a target phenomenon and, thereby, to direct both practice and future research.

The Challenges of Mixed Research Synthesis

Advances in qualitative and quantitative research synthesis and the increasing prominence of mixed methods research as the "third research paradigm" (Johnson & Onwuegbuzie, 2004, p. 14) for the practice disciplines have contributed to the current interest in mixed research synthesis. But for mixed research synthesis to advance, researchers must solve the problems generated by the methodological diversity within and between qualitative and quantitative studies. Difference has recurrently been identified as the most important factor complicating both the qualitative and quantitative research synthesis enterprises (Cooper, 1998; Mulrow, Langhorne, Grimshaw, 1997; Sandelowski, Docherty, & Emden, 1997; Sandelowski, Voils, & Barroso, 2006). Even studies of ostensibly the same variables or target events, experiences, or phenomena in ostensibly similar groups of people employing ostensibly the same methodological approaches will have differences sufficient to require finding the means to enable their findings to be compared and combined.

Qualitative Research Synthesis

In the qualitative research synthesis literature, the difference problem is most often addressed in relation to the philosophical differences among research traditions and the singularity of every research participant and research encounter (Sandelowski et al., 1997). The qualitative research methods literature is characterized, in part, by efforts to differentiate among ontological positions (e.g., realist, idealist, and relativist); epistemological positions (e.g., objectivist, constructionist, subjectivist); paradigms of inquiry (e.g., neo-positivism, interpretivism, critical theory, postmodernism); foundational theories and philosophies (e.g., symbolic interactionism,

Heideggerian phenomenology, Foucaultian geneology); and methodologies (e.g., grounded theory, phenomenology, ethnography, narrative/discourse study; Crotty; 1998; Guba & Lincoln, 2005).[3] These differences can make the synthesis of qualitative findings alone a daunting enterprise. In addition, the qualitative research emphasis on delineating the complexities and contradictions of "N=1 experiences" (Eisner, 1991, p. 197) seems at odds with the conventional research synthesis emphasis on simplification and summary. Moreover, the diversity in the implementation and reporting of qualitative research complicates both the identification of the methods actually used in a study and the findings produced from those methods

Quantitative Research Synthesis

In the quantitative research literature, the problem of difference is referred to as heterogeneity and is addressed as the methodological diversity within and between observational and experimental studies, and in terms of the contrast between the "real" differences in the target phenomena and "artifactual" differences owing to the way these phenomena were studied (Glasziou & Sanders, 2002). The constellation of statistical techniques known as meta-analysis, which is most identified with quantitative research synthesis and intended to accommodate these differences, continues to engender debate. One continuing criticism is that, because no two studies can ever be perfectly identical, meta-analysis are comparing apples and oranges (Glass, 2000; Hunt, 1997), thereby, calling into question the validity and generalizability of meta-analyses (Matt, 2003). Another criticism is that meta-analysis involves testing hypotheses about parameters in populations of studies, yet the criteria for inferential statistics (e.g., random sampling) are rarely met. Yet other concerns include the management of variable primary study quality (Conn & Rantz, 2003) and of results from diverse research designs (e.g., combining findings from single group pre-post designs and randomized controlled trials, from cross-sectional and longitudinal studies, or from experimental and observational studies). The Cochrane Collaboration and Library, an icon of the evidence-based practice movement in health care, includes only randomized controlled trials in their research syntheses, but most studies conducted that could serve as the basis for clinical practice are not randomized controlled trials.

Mixed Research Synthesis

Mixed research synthesis complicates the difference problem as qualitative and quantitative research are themselves viewed as exemplifying difference. Mixed research synthesis entails the "mixing" of the differences characterizing efforts to integrate qualitative research findings with the

differences characterizing efforts to integrate quantitative research findings (Sandelowski, Voils, & Barroso, 2006). Scholars have debated whether these differences preclude mixed research synthesis.

The Future of Mixed Research Synthesis

The viability of the mixed research synthesis enterprise rests on finding ways to make the seemingly incomparable comparable in order to make the seemingly uncombinable combinable (i.e., assimilable, arrangeable, or some other process of putting or using qualitative and quantitative findings together). Given the complexity of these goals, it is not surprising that mixed research synthesis methods have yet to be developed that satisfactorily accommodate the singularity, descriptive precision, and intricacy of qualitative research findings and the generality, numerical precision, and single-dimensionality of quantitative research findings (Buchanan, 1992; Sivesind, 1999). Although much work remains before mixed research synthesis can secure its place in the repertoires of mixed methods research and systematic review, the effort is well worth it as it has the potential to enhance both the significance and utility for practice of the many qualitative and quantitative research studies constituting shared domains of research.

CRITICAL ANALYSIS

Research Methodology - Introduction

In this Research Methodology Critical Analysis assignment the researcher is attempting to critically asses the **research methodology** of a empirical study and attempting to identify and analyze the weakness of the study. The objective of this exercise is to identify the areas where improvement can be made in the study and also understand the limitations of an empirical research study. The Research Methodology Critical Analysis report is also aiming to provide an explanation for the weaknesses which has been identified in the empirical study. In this report the emphasis of the researcher is not on judging the quality of the study but to highlight only those points which has a scope for improvement and changes. In this report we will be first briefly discuss what is a research methodology and its various aspects and then we will be highlighting different sections of the research which is reviewed with its weakness and its brief analysis

Critical analysis and assessment of research methodology for empirical study

Research purpose' objective and research questions

Weakness and explanation:-

It has been observed from the research objective and aims of study section of the empirical study that there is a certain level of ambiguity associated with research questions paper and research objective paper. They are not fully linked with the aims of the study and one of the research objective and research questions is not addressing the research aim properly. This is misdirecting the entire study from its real objectives. As it is discussed in the introduction part of the assignment the research is aimed to study the relationship existing between the employee and the brand equity of the organization and its products which the organization is manufacturing. As one of the research objective stated by the researcher is **"To assess the perception of the EQUATE employee commitment to the organization"** it can be observed and argued that even if a employee is committed to the organization he has been working for it does not necessarily ensures that his attitude' actions etc are representing the brad equity of the organization. A commitment level is slightly different from the knowledge and perception of brand equity. Even in research question part the researcher is attempting to address the questions like commitment level of employee and their knowledge of brand equity but as earlier described the researcher's study is aimed to investigate the relationship between the employee and brand equity. The research ideally should be more qualitative in nature and the elements described in the research questions should be more towards analyzing the relationship development and its aspects rather than finding the causative factors of creations (Amaratunga, D., Baldry, D., Sarshar , M. and Newton, R. 2002).

Suggestion for improvement:-

Ghauri and Gronhaug (2005) are of the opinion that a collection of aims, objectives and goals in order to study a problem may be roughly termed as research, which includes determining the right methods of data collection, a close analysis of all available data and an elaborate discussion on it. Hence research is a better understanding of the problem by a close analysis of all available data (Brown, J. D. and Rodgers, T. 2002). What can be suggested here is to modify the research objective slightly so that they can be in resonance with the research purpose fully. For a study to be moved in right direction it is always prudent to go through the research objective' research purpose and research questions and ensure that they are

complementing each other. The same research objective can be rewritten as **"to assess and investigate the perception and knowledge of brand equity in employees"**. The purpose of a research objective is to answer the problem statement posed by the research study title and hypothesis and it should be developed as crisply and accurately as possible.

READING MATERIALS

The nature of papers

The starting research student assumes that all research papers are well-written and describe good research that is both correct and important. This is not true.

Research papers can report **poor research**. The review process is supposed to prevent this, but it is not a perfect process and there is a fair amount of published work out there that is not good. The publication venue is a guide to likely quality. A top-rank journal or conference is unlikely to publish poor research, because the review process is fierce. A lower-rank journal or conference is more likely to let poor research through.

Published research is supposed to be correct. However, there are cases where the authors and reviewers have honestly failed to notice that their data, assumptions, proofs, or conclusions are **wrong**.

A big mistake made by many new researchers is to think that every single paper is important. Of course, it was important to the authors, but it may well be reporting on a dead end, a trivial result, a system that does not work well, or an idea that has been superceded. So you will find that many papers are **unimportant**, at least with respect to your particular research area.

Even if a paper is good research, correct, and important, it may not be well written. Sadly, there are some papers that are **incomprehensible**. It is wrong to assume that the authors are trying to hide something in their poor writing, though this may be the case. Often it is just poor writing skills. Part of this course is to help you to learn how to ensure that *your* papers are well written.

A way to read

CEM Joad wrote *Teach Yourself Philosophy* in 1944. His advice can be applied to reading scientific papers. He says:

- **Make judicious selection.** In philosophy Joad says "Never try to read a whole book at once." In our case, do not try to read all the papers on a subject at once. Joad says "pick out certain chapters, two or three perhaps, which deal with matters that seem to you to be of particular importance or which relate to topics on which you have already read something." We need to pick out those papers that are of most importance to the problem in hand. Of philosophy books, Joad says "Use the introduction and preface to help you identify these." In our case, we need to use something else to determine which papers we should attend to. See Fred Brooks' note (above) for more on that.
- **Indulge in intelligent skipping.** You do not have to read everything. You do not need to read every word of every paper. Do not be embarrassed to skip. You cannot possibly read every paper in the world; you need to ensure that the ones you spend your time on are the important ones. For some papers you need only look at the introduction and conclusion, with a glance elsewhere. For some, you will need to read the whole thing, but maybe not today. Joad says "omit the unintelligible; skip the repetitions; jump over the boring bits". You are allowed to do this!
- **Ensure you understand.** Joad says stop reading when you stop understanding, but he also says to interpret "understand" generously. If you do not understand, there is sometimes point in pressing on a little further to see if it becomes clear, but if you really do not get it, there is no point in reading on for the sake of reading on. You need to stop and say "why do I not understand?"; "is this important enough that I need to learn something else so that I can understand?"
- **Make notes.** The process of writing notes helps enormously with remembering. The process of having to summarise someone else's work helps enormously with understanding. Exercise 1 asks you to precis the content of an eight page paper in just one page. This is a great skill to have. Also you need to ensure that you have made good enough notes that you will be able to find any paper again in two years time when it turns out to be important. Too many times have I heard the cry "I know that I have read a paper on exactly this problem, but I cannot remember where!"

REVIEWING

How to Review a Technical Paper Peer review of journal articles and other technical reports is a key element in the maintenance of academic integrity. This article assists the reader in the efficient preparation of constructive reviews. The parts of a typical review are listed, as well as formats for the most common situations. Common defects of technical papers are discussed.

Introduction

At one time or another, every academic is asked to review papers submitted for publication in journals. These reviews play a key role in maintaining the integrity of a journal. In addition, the exercise exposes the referee and the author to new ideas and perspectives. Unfortunately, nascent academics are never formally taught the art and skills needed to referee a technical paper. As a result, most reviews take more time than necessary, while contributing little constructive knowledge to the author. The following text offers some tips to the referee to assist in the preparation of a written review. Learning the mechanics of review writing can never substitute for full comprehension of the material, but it can transform the review into a constructive document. At the same time, there are simple rules for identifying flaws in the paper that greatly simplify review preparation and allow the referee to concentrate on the paper's content. This guide focuses on technical papers, but some of the advice also applies to papers in the social sciences and liberal arts.

Why is a review necessary?

The peer review serves several roles, although the precise combination varies with the type of review. The most important reasons for review include finding deficiencies in: ™ technical approach and analysis; ™ computation; ™ ignorance of related research.

Reviews are useful to detect a second kind of problem.

Two examples are:

1. style and grammar that confuse the reader
2. patent or legal issues.

Types of reviews

There are three types of reviews: “anonymous”, “friendly”, and “internal”. In an anonymous review, the editor solicits a referee to review the article. The referee returns the review to the editor who, Page 2 of 5 after removing any identification, gives it to the author. Academic journals typically use the anonymous review, but it is also used for books, articles in proceedings, and some reports.

Most reviews have four parts

Before reviewing a paper, it is useful to consider the desired output. In this way, you can categorize your comments for later inclusion in the best part. The four parts of a review are:

1. referee’s review form;
2. additional comments;
3. original paper;
4. cover letter to editor.

What to write if there is no form

There will be circumstances where no review form is provided. Here is a format to use in such cases.

(1) Title and author of paper
(2) Summary of paper
(3) Good things about the paper (one paragraph)
(4) Major comments
(5) Minor comments
(6) Recommendations

What makes a good paper?

Good papers contain something of merit. You, an expert in the subject, should be able to find it (if it exists). However, the item of merit may be poorly presented, which can undermine the paper’s value. A logical structure is the first element of a good presentation.

A standard structure for technical papers has evolved as follows:

(1) Abstract

(2) Introduction

(3) Body of the Paper (technique, results, discussion)

(4) Conclusions

(5) References

(6) Tables

(7) Figures (and captions)

When to decline

Most editors ask the referee to finish a review within a specified time. Unfortunately, a good review takes many hours to prepare and it must compete with other obligations. Therefore, you can (and should) decline to review a paper if you cannot devote the necessary time before the deadline. But tell the editor immediately so that he can find a substitute referee.

RETHINKING

Over the past few decades, the manner in which social scientists have addressed the question of *how* to conduct research has shifted tremendously. Early ethnographic studies set out to examine and elaborate "objects" of inquiry, sometimes perceived as a historic and rigidly bounded fieldwork sites that could be comprehensively "known." As the social sciences began to transition away from positivist frameworks, research methodologies were compelled to adapt to the shifting boundaries between "the field," fieldwork, and the fieldworker. Anthropology, in particular, encountered several theoretical turns, challenging previous ways of knowing and bringing to the fore questions of methodological import. With every so-called turn that a discipline endures, new boundaries are constructed and limitations are tested and established. The aim of this special edition is to explore and expand these boundaries by focusing on the theoretical and practical complexities of research methods.

All of the papers in this issue make important contributions to our understanding of research methods and broader methodological concerns. The papers examine specific methods employed during fieldwork as well as comment on their connection to methodological paradigms that are often largely shaped by disciplinary histories, practices, and deeply entrenched epistemologies. The impetus for bringing these diverse papers together emerged after we organised an international conference on the same topic.

The papers selected for this special edition reflect not only on the diversity of methods, but also on the ways in which methods and their ethical implications evolve during the course of conducting research. Therefore a key issue in this collection is that research methods are as much about the tools researchers employ as they are about the implications, disturbances and even failures that result in their use.

CRITICAL EVALUATION :

What is a critical review?

A critical review (sometimes called a critique, critical commentary, critical appraisal and critical analysis) is a detailed commentary on and critical evaluation of a text. Critical review may carry out a as a stand-alone exercise, or as part of your research and preparation for writing a literature review.

Critical Review of a Research Article

Research critique is a planned, careful critical evaluation of a piece of research work against the prespecified criteria to judge the strengths and weaknesses of the research study. Critique should be balanced, where alternative suggestions must be provided to further enforce the strengths and eliminate the weaknesses of the study to improve overall quality of the research project.

Purposes

- To provide inputs regarding the strength and weakness of a study to the researchers.
- To provide suggestions to the students regarding the methodological flaws in their research project and also to evaluate the understanding of research by the students.
- To judge the scientific merits of the study.
- To take a decision whether to publish the study in journal or not.

Guidelines for Writing a Research Critique

- Read and understand the research report carefully.
- Carryout the critical appraisal of all the aspects of the research report before writing the critique of a research report.
- Avoid general vague statements; be objective and sensitive while framing the negative comments, and be practical by considering all the limitations of the research.
- Keep a balanced approach in the research critique by presenting both strong and weak points of report, because certainly no report without any negative aspects.
- Positive and negative points must be supported with examples to make a clear stand about the strengths and weaknesses of the research report.

Guidelines for critiquing the research report

Some of the positive points of the study can be gathered from the questions formulated; more the number of questions answered with 'yes', the better the study is considered. A rating scale may be used for critiquing a research report.

CRITIQUE OF THE STUDY

Research problem

- Is the problem statement clearly and concisely articulated?
- Is the problem significant for nursing profession and has the researcher provided good argument for the significance?
- Is the research problem within the professional domain?
- Is the problem not complex to study and appropriately delimited in its scope?
- Does the problem statement clearly depict the variables, population, and place of the study?
- Is the research problem feasible to study in reference to time, resources, researcher's ability and availability of respondents?
- Is the research problem directed to develop or test the nursing theories?

Objectives

- Are the objectives in accordance with research problem?
- Are the objectives stated in clear and concise form?
- Are the objectives stated using accepted action verbs in logical manner?

Operational definitions

- Are all the study variables operationally defined?
- Are operational definitions stated in logical and measurable form?

INTERPRETATION

After collecting and analyzing the data, the researcher has to accomplish the task of drawing inferences followed by report writing. This has to be done very carefully, otherwise misleading conclusions may be drawn and the whole purpose of doing research may get vitiated. It is only through interpretation that the researcher can expose relations and processes that underlie his findings. In case of hypotheses testing studies, if hypotheses are tested and upheld several times, the researcher may arrive at generalizations. But in case the researcher had no hypothesis to start with, he would try to explain his findings on the basis of some theory. This may at times result in new questions, leading to further researches. All this analytical information and consequential inference(s) may well be communicated, preferably

through research report, to the consumers of research results who may be either an individual or a group of individuals or some public/private organisation.

Meaning of Interpretation

Interpretation refers to the task of drawing inferences from the collected facts after an analytical and/or experimental study. In fact, it is a search for broader meaning of research findings. The task of interpretation has two major aspects viz., (i) the effort to establish continuity in research through linking the results of a given study with those of another, and (ii) the establishment of some explanatory concepts. "In one sense, interpretation is concerned with relationships within the collected data, partially overlapping analysis. Interpretation also extends beyond the data of the study to include the results of other research, theory and hypotheses." [1] Thus, interpretation is the device through which the factors that seem to explain what has been observed by researcher in the course of the study can be better understood and it also provides a theoretical conception which can serve as a guide for further researches.

WHY INTERPRETATION?

Interpretation is essential for the simple reason that the usefulness and utility of research findings lie in proper interpretation. It is being considered a basic component of research process because of the following reasons:

Interpretation and Report Writing

1. It is through interpretation that the researcher can well understand the abstract principle that works beneath his findings. Through this he can link up his findings with those of other studies, having the same abstract principle, and thereby can predict about the concrete world of events.

Fresh inquiries can test these predictions later on. This way the continuity in research can be maintained.

2. Interpretation leads to the establishment of explanatory concepts that can serve as a guide for future research studies; it opens new avenues of intellectual adventure and stimulates the quest for more knowledge.
3. Researcher can better appreciate only through interpretation why his findings are what they are and can make others to understand the real significance of his research findings.
4. The interpretation of the findings of exploratory research study often results into hypotheses for experimental research and as such interpretation is involved in the transition from exploratory to experimental research. Since an exploratory study does not have a hypothesis to start with, the findings of such a study have

to be interpreted on a *post-factum* basis in which case the interpretation is technically described as ‘ *post factum*’ interpretation.

TECHNIQUE OF INTERPRETATION

The task of interpretation is not an easy job, rather it requires a great skill and dexterity on the part of researcher. Interpretation is an art that one learns through practice and experience. The researcher may, at times, seek the guidance from experts for accomplishing the task of interpretation.

The technique of interpretation often involves the following steps:

1. Researcher must give reasonable explanations of the relations which he has found and he must interpret the lines of relationship in terms of the underlying processes and must try to find out the thread of uniformity that lies under the surface layer of his diversified research findings. In fact, this is the technique of how generalization should be done and concepts be formulated.
2. Extraneous information, if collected during the study, must be considered while interpreting the final results of research study, for it may prove to be a key factor in understanding the problem under consideration.
3. It is advisable, before embarking upon final interpretation, to consult someone having insight into the study and who is frank and honest and will not hesitate to point out omissions and errors in logical argumentation. Such a consultation will result in correct interpretation and, thus, will enhance the utility of research results.
4. Researcher must accomplish the task of interpretation only after considering all relevant factors affecting the problem to avoid false generalization. He must be in no hurry while interpreting results, for quite often the conclusions, which appear to be all right at the beginning, may not at all be accurate.

RESEARCH PURPOSES:

The process used to collect information and data for the purpose of making business decisions. The methodology may include publication research, interviews, surveys and other research techniques, and could include both present and historical information.

Purposes of Research: Exploratory, Descriptive & Explanatory. There is a parallel between how people come to understand something and the process of researching an idea. This lesson explores the purposes of research as well as three approaches to research in psychology: exploratory, descriptive, and explanatory. The process used to collect information and data for the purpose of making business decisions. The methodology may include publication research, interviews, surveys and other research techniques, and could include both present and historical information. Research is what gives the researchers, storyteller, student or scientist the means to make a conclusion. When you've done proper research, you're then able to spend time analyzing all of the moving parts, the variables and thus make an informed observation or conclusion on the subject. When you're exploring the topic, you want to gather as much evidence, experience and observation for you to begin building your theories. By familiarising yourself as much as possible with the subject you're researching you begin to have objective and subjective observations. Exploration is the first phase of the research. The exploration aspect stokes the curiosity of the researcher, which then brings up questions that spur into further discovery, digging and study.

ETHICS IN RESEARCH :

APA's Ethics Code mandates that psychologists who conduct research should inform participants about: The purpose of the research, expected duration and procedures. Participants' rights to decline to participate and to withdraw from the research once it has started, as well as the anticipated consequences of doing so. Ethical standards also require that researchers not put participants in a situation where they might be at risk of harm as a result of their participation

Research ethics is specifically interested in the analysis of ethical issues that are raised when people are involved as participants in research.

1. **Discuss intellectual property frankly**

Academe's competitive "publish-or-perish" mindset can be a recipe for trouble when it comes to who gets credit for authorship. The best way to avoid disagreements about who should get credit and in what order is to talk about these issues at the beginning of a working relationship, even though many people often feel uncomfortable about such topics.

2. **Be conscious of multiple roles**

APA's Ethics Code says psychologists should avoid relationships that could reasonably impair their professional performance or could exploit or harm others. But it also notes that many

kinds of multiple relationships aren't unethical--as long as they're not reasonably expected to have adverse effects.

3. **Follow informed-consent rules:**

APA's Ethics Code mandates that psychologists who conduct research should inform participants about:

1The purpose of the research, expected duration and procedures.

2Participants' rights to decline to participate and to withdraw from the research once it has started, as well as the anticipated consequences of doing so.

4. **Respect confidentiality and privacy :**

Upholding individuals' rights to confidentiality and privacy is a central tenet of every psychologist's work. However, many privacy issues are idiosyncratic to the research population, writes Susan Folkman, PhD, in "Ethics in Research with Human Participants" (APA, 2000)

Other steps researchers should take include:

- **Discuss the limits of confidentiality.** Give participants information about how their data will be used, what will be done with case materials, photos and audio and video recordings, and secure their consent.
- **Know federal and state law.** Know the ins and outs of state and federal law that might apply to your research. For instance, the Goals 2000: Education Act of 1994 prohibits asking children about religion, sex or family life without parental permission.

5. **Tap into ethics resources:**
   - One of the best ways researchers can avoid and resolve ethical dilemmas is to know both what their ethical obligations are and what resources are available to them.
   - "Researchers can help themselves make ethical issues salient by reminding themselves of the basic underpinnings of research and professional ethics," says Bullock.

The following is a rough and general summary of some ethical principals that various codes address

Honesty

Strive for honesty in all scientific communications. Honestly report data, results, methods and procedures, and publication status. Do not fabricate, falsify, or misrepresent data. Do not deceive colleagues, research sponsors, or the public.

Objectivity

Strive to avoid bias in experimental design, data analysis, data interpretation, peer review, personnel decisions, grant writing, expert testimony, and other aspects of research where objectivity is expected or required. Avoid or minimize bias or self-deception. Disclose personal or financial interests that may affect research.

Integrity

Keep your promises and agreements; act with sincerity; strive for consistency of thought and action.

Carefulness

Avoid careless errors and negligence; carefully and critically examine your own work and the work of your peers. Keep good records of research activities, such as data collection, research design, and correspondence with agencies or journals.

Openness

Share data, results, ideas, tools, resources. Be open to criticism and new ideas..

Confidentiality

Protect confidential communications, such as papers or grants submitted for publication, personnel records, trade or military secrets, and patient records.

UNIT 2

QUANTATIVE METHODS FOR PROBLEM SOLVING

QUANTITATIVE RESEARCH

Quantitative methods emphasize objective measurements and the statistical, mathematical, or numerical analysis of data collected through polls, questionnaires, and surveys, or by manipulating pre-existing statistical data using computational techniques. Quantitative research focuses on gathering numerical data and generalizing it across groups of people or to explain a particular phenomenon..

Characteristics of Quantitative Research

Your goal in conducting quantitative research study is to determine the relationship between one thing [an independent variable] and another [a dependent or outcome variable] within a population. Quantitative research designs are either descriptive [subjects usually measured once] or experimental [subjects measured before and after a treatment]. A descriptive study establishes only associations between variables; an experimental study establishes causality.

Quantitative research deals in numbers, logic, and an objective stance. Quantitative research focuses on numeric and unchanging data and detailed, convergent reasoning rather than divergent reasoning [i.e., the generation of a variety of ideas about a research problem in a spontaneous, free-flowing manner].

Its main characteristics are:

- The data is usually gathered using structured research instruments.
- The results are based on larger sample sizes that are representative of the population.
- The research study can usually be replicated or repeated, given its high reliability.
- Researcher has a clearly defined research question to which objective answers are sought.
- All aspects of the study are carefully designed before data is collected.
- Data are in the form of numbers and statistics, often arranged in tables, charts, figures, or other non-textual forms.
- Project can be used to generalize concepts more widely, predict future results, or investigate causal relationships.
- Researcher uses tools, such as questionnaires or computer software, to collect numerical data.

Things to keep in mind when reporting the results of a study using quantitative methods:

1. Explain the data collected and their statistical treatment as well as all relevant results in relation to the research problem you are investigating. Interpretation of results is not appropriate in this section.
2. Report unanticipated events that occurred during your data collection. Explain how the actual analysis differs from the planned analysis. Explain your handling of missing data and why any missing data does not undermine the validity of your analysis.
3. Explain the techniques you used to "clean" your data set.
4. Choose a minimally sufficient statistical procedure; provide a rationale for its use and a reference for it. Specify any computer programs used.
5. Describe the assumptions for each procedure and the steps you took to ensure that they were not violated.
6. When using inferential statistics, provide the descriptive statistics, confidence intervals, and sample sizes for each variable as well as the value of the test statistic, its direction, the degrees of freedom, and the significance level [report the actual *p* value].
7. Avoid inferring causality, particularly in nonrandomized designs or without further experimentation.
8. Use tables to provide exact values; use figures to convey global effects. Keep figures small in size; include graphic representations of confidence intervals whenever possible.
9. Always tell the reader what to look for in tables and figures.

STATISTICAL MODELING AND ANALYSIS

INTRODUCTION

In the Information Age, data is no longer scarce – it's overpowering. The key is to sift through the overwhelming volume of data available to organizations and businesses and correctly interpret its implications. But to sort through all this information, you need the right statistical data analysis tools.

With the current obsession over “big data,” analysts have produced a lot of fancy tools and techniques available to large organizations. However, there are a handful of basic data analysis tools that most organizations aren’t using…to their detriment.

We suggest starting your data analysis efforts with the following five fundamentals – and learn to avoid their pitfalls – before advancing to more sophisticated techniques.

1. Mean

The arithmetic mean, more commonly known as “the average,” is the sum of a list of numbers divided by the number of items on the list. The mean is useful in determining the overall trend of a data set or providing a rapid snapshot of your data. Another advantage of the mean is that it’s very easy and quick to calculate.

Pitfall:

Taken alone, the mean is a dangerous tool. In some data sets, the mean is also closely related to the mode and the median (two other measurements near the average). However, in a data set with a high number of outliers or a skewed distribution, the mean simply doesn’t provide the accuracy you need for a nuanced decision.

2. Standard Deviation

The standard deviation, often represented with the Greek letter sigma, is the measure of a spread of data around the mean. A high standard deviation signifies that data is spread more widely from the mean, where a low standard deviation signals that more data align with the mean. In a portfolio of data analysis methods, the standard deviation is useful for quickly determining dispersion of data points.

Pitfall:

Just like the mean, the standard deviation is deceptive if taken alone. For example, if the data have a very strange pattern such as a non-normal curve or a large amount of outliers, then the standard deviation won’t give you all the information you need.

3. Regression

Regression models the relationships between dependent and explanatory variables, which are usually charted on a scatterplot. The regression line also designates whether those relationships are strong or weak. Regression is commonly taught in high school or college statistics courses with applications for science or business in determining trends over time.

Pitfall:
Regression is not very nuanced. Sometimes, the outliers on a scatterplot (and the reasons for them) matter significantly. For example, an outlying data point may represent the input from your most critical supplier or your highest selling product. The nature of a regression line, however, tempts you to ignore these outliers. As an illustration, examine a picture of ANSCOMBE'S QUARTET, in which the data sets have the exact same regression line but include widely different data points.

4. Sample Size Determination
When measuring a large data set or population, like a workforce, you don't always need to collect information from every member of that population – a sample does the job just as well. The trick is to determine the right size for a sample to be accurate. Using proportion and standard deviation methods, you are able to accurately determine the right sample size you need to make your data collection statistically significant.

Pitfall:
When studying a new, untested variable in a population, your proportion equations might need to rely on certain assumptions. However, these assumptions might be completely inaccurate. This error is then passed along to your sample size determination and then onto the rest of your statistical data analysis

5. Hypothesis Testing
Also commonly called t testing, hypothesis testing assesses if a certain premise is actually true for your data set or population. In data analysis and statistics, you consider the result of a hypothesis test statistically significant if the results couldn't have happened by random chance. Hypothesis tests are used in everything from science and research to business and economic

Pitfall:
To be rigorous, hypothesis tests need to watch out for common errors. For example, the placebo effect occurs when participants falsely expect a certain result and then perceive (or actually attain) that result. Another common error is the Hawthorne effect (or observer effect), which happens when participants skew results because they know they are being studied.
Overall, these methods of DATA ANALYSIS add a lot of insight to your DECISION-MAKING PORTFOLIO, particularly if you've never analyzed a process or data set with statistics before. However, avoiding the common pitfalls associated with each method is just as important. Once you master these fundamental techniques for statistical data analysis, then you're ready to advance to more powerful data analysis tools.

The 13 Steps for Statistical Modeling in any Regression or ANOVA

No matter what statistical model you're running, you need to go through the same 13 steps. The order and the specifics of how you do each step will differ depending on the data and the type of model you use.

These 13 steps are in 3 major parts. Most people think of only Part 3 as modeling. However, if you do all 3 parts, and think of them all as part of the analysis, the modeling process will be faster, easier, and make more sense.

Part 1: Define and Design

In the first 4 steps, the object is clarity. You want to make everything as clear as possible to yourself. The more clear things are at this point, the smoother everything will be.

1. Write out research questions in theoretical and operational terms

A lot of times, when researchers are confused about the right statistical method to use, the real problem is they haven't defined their research questions. They have a general idea of the relationship they want to test, but it's a bit vague. You need to be very specific.

For each research question, write it down in both theoretical and operational terms.

2. Design the study or define the design

Depending on whether you are collecting your own data or doing secondary data analysis, you need a clear idea of the design. Design issues are about randomization and sampling:

- Nested and Crossed Factors
- Potential confounders and control variables
- Longitudinal or repeated measurements on a study unit
- Sampling: simple random sample or stratification or clustering

3. Choose the variables for answering research questions and determine their level of measurement

Every model has to take into account both the design and the level of measurement of the variables.

Level of measurement, remember, is whether a variable is nominal, ordinal, or interval. Within interval, you also need to know if variables are discrete counts or continuous.

It's absolutely vital that you know the level of measurement of each response and predictor variable, because they determine both the type of information you can get from your model and the family of models that is appropriate.

4. Write an analysis plan

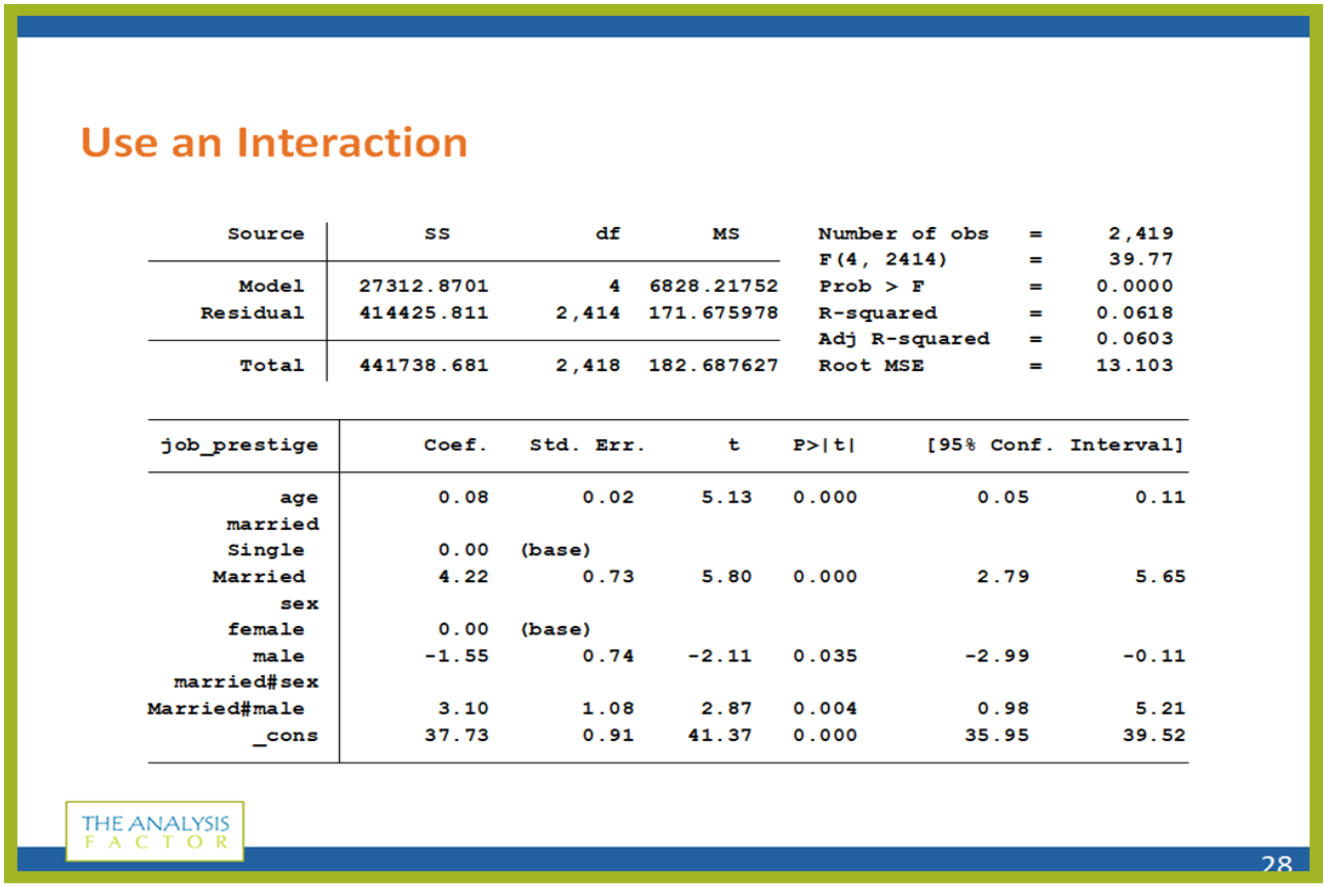
Write your best guess for the statistical method that will answer the research question, taking into account the design and the type of data.

It does not have to be final at this point—it just needs to be a reasonable approximation.

5. Calculate sample size estimations

This is the point at which you should calculate your sample sizes–before you collect data and after you have an analysis plan. You need to know which statistical tests you will use as a basis for the estimates.

And there really is no point in running post-hoc power analyses–it doesn't tell you anything.



Four Critical Steps in Building Linear Regression Models
Take the frustration out of statistical model building.

Part 2: Prepare and explore

6. Collect, code, enter, and clean data

The parts that are most directly applicable to modeling are entering data and creating new variables.

For data entry, the analysis plan you wrote will determine how to enter variables. For example, if you will be doing a linear mixed model, you will want the data in long format.

7. Create new variables

This step may take longer than you think–it can be quite time consuming. It's pretty rare for every variable you'll need for analysis to be collected in exactly the right form. Create indices, categorize, reverse code, whatever you need to do to get variables in their final form, including running principal components or factor analysis.

8. Run Univariate and Bivariate Statistics

You need to know what you're working with. Check the distributions of the variables you intend to use, as well as bivariate relationships among all variables that might go into the model.

You may find something here that leads you back to step 7 or even step 4. You might have to do some data manipulation or deal with missing data.

More commonly, it will alert you to issues that will become clear in later steps. The earlier you are aware of issues, the better you can deal with them. But even if you don't discover the issue until later, it won't throw you for a loop if you have a good understanding of your variables.

9. Run an initial model

Once you know what you're working with, run the model listed in your analysis plan. In all likelihood, this will not be the final model.

But it should be in the right family of models for the types of variables, the design, and to answer the research questions. You need to have this model to have something to explore and refine.

Part 3: Refine the model

10. Refine predictors and check model fit

If you are doing a truly exploratory analysis, or if the point of the model is pure prediction, you can use some sort of stepwise approach to determine the best predictors.

If the analysis is to test hypotheses or answer theoretical research questions, this part will be more about refinement. You can

- Test, and possibly drop, interactions and quadratic or explore other types of non-linearity
- Drop nonsignificant control variables
- Do hierarchical modeling to see the effects of predictors added alone or in blocks.
- Check for overdispersion
- Test the best specification of random effects

11. Test assumptions

Because you already investigated the right family of models in Part 1, thoroughly investigated your variables in Step 8, and correctly specified your model in Step 10, you should not have big surprises here. Rather, this step will be about confirming, checking, and refining. But what you learn here can send you back to any of those steps for further refinement.

12. Check for and resolve data issues

Steps 11 and 12 are often done together, or perhaps back and forth. This is where you check for data issues that can affect the model, but are not exactly assumptions. These include:

Data issues are about the data, not the model, but occur within the context of the model
• Multicollinearity
• Outliers and influential points
• Missing data
• Truncation and censoring

Once again, data issues don't appear until you have chosen variables and put them in the model.

13. Interpret Results

Now, finally, interpret the results.

You may not notice data issues or misspecified predictors until you interpret the coefficients. Then you find something like a super high standard error or a coefficient with a sign opposite what you expected, sending you back to previous steps.

TIME SERIES ANALYSIS

INTRODUCTION

Time series data often arise when monitoring industrial processes or tracking corporate business metrics. The essential difference between modeling data via time series methods or using the process monitoring methods discussed earlier in this chapter is the following:

Time series analysis accounts for the fact that data points taken over time may have an internal structure (such as autocorrelation, trend or seasonal variation) that should be accounted for.

This section will give a brief overview of some of the more widely used techniques in the rich and rapidly growing field of time series modeling and analysis.

Time series analysis is a statistical technique that deals with time series data, or trend analysis. Time series data means that data is in a series of particular time periods or intervals. The data is considered in three types:

Time series data: A set of observations on the values that a variable takes at different times.

Cross-sectional data: Data of one or more variables, collected at the same point in time.

Pooled data: A combination of time series data and cross-sectional data.

Terms and concepts:

Dependence: Dependence refers to the association of two observations with the same variable, at prior time points.

Stationarity: Shows the mean value of the series that remains constant over a time period; if past effects accumulate and the values increase toward infinity, then stationarity is not met.

Differencing: Used to make the series stationary, to De-trend, and to control the auto-correlations; however, some time series analyses do not require differencing and over-differenced series can produce inaccurate estimates.

Specification: May involve the testing of the linear or non-linear relationships of dependent variables by using models such as ARIMA, ARCH, GARCH, VAR, Co-integration, etc.

Exponential smoothing in time series analysis: This method predicts the one next period value based on the past and current value. It involves averaging of data such that the nonsystematic components of each individual case or observation cancel out each other. The exponential smoothing method is used to predict the short term predication. Alpha, Gamma, Phi, and Delta are the parameters that estimate the effect of the time series data. Alpha is used when seasonality is not present in data. Gamma is used when a series has a trend in data. Delta is used when seasonality cycles are present in data. A model is applied according to the pattern of the data. Curve fitting in time series analysis: Curve fitting regression is used when data is in a non-linear relationship. The following equation shows the non-linear behavior:

Dependent variable, where case is the sequential case number.

Curve fitting can be performed by selecting “regression” from the analysis menu and then selecting “curve estimation” from the regression option. Then select “wanted curve linear,” “power,” “quadratic,” “cubic,” “inverse,” “logistic,” “exponential,” or “other.”

ARIMA:

ARIMA stands for autoregressive integrated moving average. This method is also known as the Box-Jenkins method.

Identification of ARIMA parameters:

Autoregressive component: AR stands for autoregressive. Autoregressive paratmeter is denoted by p. When p =0, it means that there is no auto-correlation in the series. When p=1, it means that the series auto-correlation is till one lag.

Integrated: In ARIMA time series analysis, integrated is denoted by d. Integration is the inverse of differencing. When d=0, it means the series is stationary and we do not need to take the difference of it. When d=1, it means that the series is not stationary and to make it stationary, we need to take the first difference. When d=2, it means that the series has been differenced twice. Usually, more than two time difference is not reliable.

Moving average component: MA stands for moving the average, which is denoted by q. In ARIMA, moving average q=1 means that it is an error term and there is auto-correlation with one lag.

In order to test whether or not the series and their error term is auto correlated, we usually use W-D test, ACF, and PACF.

Decomposition: Refers to separating a time series into trend, seasonal effects, and remaining variability Assumptions:

Stationarity: The first assumption is that the series are stationary. Essentially, this means that the series are normally distributed and the mean and variance are constant over a long time period.

Uncorrelated random error: We assume that the error term is randomly distributed and the mean and variance are constant over a time period. The Durbin-Watson test is the standard test for correlated errors.

No outliers: We assume that there is no outlier in the series. Outliers may affect conclusions strongly and can be misleading.

Random shocks (a random error component): If shocks are present, they are assumed to be randomly distributed with a mean of 0 and a constant variance.

Statistics Solutions can assist with your quantitative analysis by assisting you to develop your methodology and results chapters. The services that we offer include:

Data Analysis Plan

Edit your research questions and null/alternative hypotheses.Write your data analysis plan; specify specific statistics to address the research questions, the assumptions of the statistics, and justify why they are the appropriate statistics; provide references

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Two hours of additional support with your statistician

PROBABILITY DISTRIBUTIONS

INTRODUCTION

In probability theory and statistics, a probability distribution is a mathematical function that, stated in simple terms, can be thought of as providing the probabilities of occurrence of different possible outcomes in an experiment. For instance, if the random variable X is used to denote the outcome of a coin toss ("the experiment"), then the probability distribution of X would take the value 0.5 for X = heads, and 0.5 for X = tails (assuming the coin is fair).

In more technical terms, the probability distribution is a description of a random phenomenon in terms of the probabilities of events. Examples of random phenomena can include the results of an experiment or survey. A probability distribution is defined in terms of an underlying sample space, which is the set of all possible outcomes of the random phenomenon being observed. The sample space may be the set of real numbers or a higher-dimensional vector space, or it may be a list of non-numerical values; for example, the sample space of a coin flip would be {heads, tails} .

Probability distributions are generally divided into two classes. A discrete probability distribution (applicable to the scenarios where the set of possible outcomes is discrete, such as a coin toss or a roll of dice) can be encoded by a discrete list of the probabilities of the outcomes, known as a probability mass function. On the other hand, a continuous probability distribution (applicable to the scenarios where the set of possible outcomes can take on values in a continuous range (e.g. real numbers), such as the temperature on a given day) is typically described by probability density functions (with the probability of any individual outcome actually being 0). The normal distribution is a commonly encountered continuous probability distribution. More complex experiments, such as those involving stochastic processes defined in continuous time, may demand the use of more general probability measures.

A probability distribution whose sample space is the set of real numbers is called univariate, while a distribution whose sample space is a vector space is called multivariate. A univariate distribution gives the probabilities of a single random variable taking on various alternative values; a multivariate distribution (a joint probability distribution) gives the probabilities of a random vector – a list of two or more random variables – taking on various combinations of values. Important and commonly encountered univariate probability distributions include the binomial distribution, the hypergeometric distribution, and the normal distribution. The multivariate normal distribution is a commonly encountered multivariate distribution.

Discrete probability distribution

A discrete probability distribution is a probability distribution characterized by a probability mass function. Thus, the distribution of a random variable X is discrete, and X is called a discrete random variable, $\sum P(X=u)=1$ as u runs through the set of all possible values of X. A discrete random variable can assume only a finite or countably infinite number of values. For the number of potential values to be countably infinite, even though their probabilities sum to 1, the probabilities have to decline to zero fast enough. For example, If $P(X=u) = 1/2n$ for n=1,2,3…we have the sum of probabilities $1/2 + 1/4 + 1/8 + ... = 1$.

Well-known discrete probability distributions used in statistical modeling include the Poisson distribution, the Bernoulli distribution, the binomial distribution, the geometric distribution, and the negative binomial distribution.

When a sample (a set of observations) is drawn from a larger population, the sample points have an empirical distribution that is discrete and that provides information about the population distribution.

Continuous probability distribution

A continuous probability distribution is a probability distribution that has a cumulative distribution function that is continuous. Most often they are generated by having a probability density function. Mathematicians call distributions with probability density functions absolutely continuous, since their cumulative distribution function is absolutely continuous with respect to the Lebesgue measure λ. If the distribution of X is continuous, then X is called a continuous random variable. There are many examples of continuous probability distributions: normal, uniform, chi-squared, and others.

Intuitively, a continuous random variable is the one which can take a continuous range of values—as opposed to a discrete distribution, where the set of possible values for the random variable is at most countable. While for a discrete distribution an event with probability zero is impossible (e.g., rolling 31/2 on a standard dice is impossible, and has probability zero), this is not so in the case of a continuous random variable. For example, if one measures the width of an oak leaf, the result of 3½ cm is possible; however, it has probability zero because uncountably many other potential values exist even between 3 cm and 4 cm. Each of these individual outcomes has probability zero, yet the probability that the outcome will fall into the interval (3 cm, 4 cm) is nonzero.

Formally, if X is a continuous random variable, then it has a probability density function $f(x)$, and therefore its probability of falling into a given interval, say $[a, b]$ is given by the integral.

P[a<X<b]=$\int_a^b f(x)dx$ In particular, the probability for X to take any single value a (that is $a \leq X \leq a$) is zero, because an integral with coinciding upper and lower limits is always equal to zero.

The definition states that a continuous probability distribution must possess a density, or equivalently, its cumulative distribution function be absolutely continuous. This requirement is stronger than simple continuity of the cumulative distribution function, and there is a special

class of distributions, singular distributions, which are neither continuous nor discrete nor a mixture of those. An example is given by the Cantor distribution. Such singular distributions however are never encountered in practice.

Binomial distribution

In probability theory and statistics, the binomial distribution with parameters *n* and *p* is the discrete probability distribution of the number of successes in a sequence of *n* independent experiments, each asking a yes–no question, and each with its own boolean. A single success/failure experiment is also called a Bernoulli trial or Bernoulli experiment and a sequence of outcomes is called a Bernoulli process; for a single trial, i.e., $n = 1$, the binomial distribution is a Bernoulli distribution. The binomial distribution is the basis for the popular binomial test of statistical significance.

The binomial distribution is frequently used to model the number of successes in a sample of size *n* drawn with replacement from a population of size *N*. If the sampling is carried out without replacement, the draws are not independent and so the resulting distribution is a hypergeometric distribution, not a binomial one. However, for *N* much larger than *n*, the binomial distribution remains a good approximation, and is widely used.

FUNDAMENTALS OF STATISTICAL ANALYSIS OF INFERENCE

Statistical inference and random sampling

Random sampling is the second great concept in inference, and it distinguishes probabilistic statistical inference from non-statistical inference as well as from non-probabilistic inference based on statistical data.

Let's begin the discussion with a simple though unrealistic situation. Your friend Arista a) looks into a cardboard carton, b) reaches in, c) pulls out her hand, and d) shows you a green ball. What might you reasonably infer?

You might at least be fairly sure that the green ball came from the carton, though you recognize that Arista might have had it concealed in her hand when she reached into the carton. But there is not much more you might reasonably conclude at this point except that there was at least one green ball in the car- ton to start with. There could be no more balls; there could be many green balls and no others; there could be a thousand red balls and just one green ball; and there could be one green ball, a hundred balls of different colors, and two pounds of

mud—given that she looked in first, it is not improbable that she picked out the only green ball among other material of different sorts.

There is not much you could say with confidence about the probability of yourself reaching into the same carton with your eyes closed and pulling out a single green ball. To use other language (which some philosophers might say is not appropriate here as the situation is too specific), there is little basis for induction about the contents of the box. Nor is the situa- tion very different if your friend reaches in three times in a row and hands you a green ball each time.

So far we have put our question rather vaguely. Let us frame a more precise inquiry: What do we predict about the next item(s) we might draw from the carton? If we assume—based on who-knows-what information or notions—that another ball will emerge, we could simply use the principle of sameness and (until we see a ball of another color) predict that the next ball will be green, whether one or three or 100 balls is (are) drawn.

But now what about if Arista pulls out nine green balls and one red ball? The principle of sameness cannot be applied as simply as before. Based on the last previous ball, the next one will be red. But taking into account all the balls we have seen, the next will "probably" be green. We have no solid basis on which to go further. There cannot be any "solution" to the "problem" of reaching a general conclusion on the basis of these specific pieces of evidence.

Now consider what you might conclude if you were told that a single green ball had been drawn with a random sampling procedure from a box containing nothing but balls. Knowledge that the sample was drawn randomly from a given universe is grounds for belief that one knows much more than if a sample were not drawn randomly. First, you would be sure—if you had reasonable basis to believe that the sampling really was random, which is not easy to guarantee—that the ball came from the box. Second, you would guess that the proportion of green balls is not very small, because if there are only a few green balls and many other-colored balls, it would be un- usual—that is, the event would have a low probability—to draw a green ball. Not impossible, but unlikely. And we can compute the probability of drawing a green ball—or any other com- bination of colors—for different assumed compositions within the box. So the knowledge that the sampling process is random greatly increases our ability—or our confidence in our ability—to infer the contents of the box.

Let us note well the strategy of the previous paragraph: Ask about the probability that one or more various possible contents of the box (the "universe") will produce the observed sample, on the assumption that the sample was drawn randomly. This is the central strategy of all statistical inference, though I do not find it so stated elsewhere. We shall come back to this idea shortly.

There are several kinds of questions one might ask about the contents of the box. One general category includes questions about our best guesses of the box's contents—that is, questions of estimation. Another category includes questions about our surety of that description, and

our surety that the contents are similar or different from the contents of other boxes; the consideration of surety follows after estimates are made. The estimation questions can be subtle and unexpected , but do not cause major controversy about the foundations of statistics. So we can quickly move on to questions about the extent of surety in our estimations.

Consider your reaction if the sampling produces 10 green balls in a row, or 9 out of 10. If you had no other information (a very important assumption that we will leave aside for now), your best guess would be that the box contains all green balls, or a proportion of 9 of 10, in the two cases respectively. This estimation process seems natural enough.

You would be surprised if someone told you that instead of the box containing the proportion in the sample, it contained just half green balls. How surprised? Intuitively, the extent of your surprise would depend on the probability that a half-green "universe" would produce 10 or 9 green balls out of 10. This surprise is a key element in the logic of the hypothesis-testing branch of statistical inference.

We learn more about the likely contents of the box by asking about the probability that various specific populations of balls within the box would produce the particular sample that we received. That is, we can ask how likely a collection of 25 per- cent green balls is to produce (say) 9 of 10 green ones, and how likely collections of 50 percent, 75 percent, 90 percent (and any other collections of interest) are to produce the observed sample. That is, we ask about the consistency between any particular hypothesized collection within the box and the sample we observe. And it is reasonable to believe that those universes which have greater consistency with the observed sample— that is, those universes that are more likely to produce the ob- served sample—are more likely to be in the box than other universes. This (to repeat, as I shall repeat many times) is the basic strategy of statistical investigation. If we observe 9 of 10 green balls, we then determine that universes with (say) 9/10 and 10/10 green balls are more consistent with the observed evidence than are universes of 0/10 and 1/10 green balls. So by this process of considering specific universes that the box might contain, we make possible more specific inferences about the box's probable contents based on the sample evidence than we could without this process.

Please notice the role of the assessment of probabilities here: By one technical means or another (either simulation or formulas), we assess the probabilities that a particular universe will produce the observed sample, and other samples as well.

It is of the highest importance to recognize that without additional knowledge (or assumption) one cannot make any statements about the probability of the sample having come from any particular universe, on the basis of the sample evidence. (Better read that last sentence again.) We can only speak about the probability that a particular universe will produce the observed sample, a very different matter. This issue will arise again very sharply in the context of confidence intervals.

Let us generalize the steps in statistical inference:

1.Frame the original question as: What is the chance of get- ting the observed sample x from population X? That is, what is probability of (If x then X)?

2.Proceed to this question: What kinds of samples does X pro- duce, with which probability? That is, what is the probability of this particular x coming from X? That is, what is p(x|X)?

3.Actually investigate the behavior of X with respect to x and other samples. One can do this in two ways:

a. Use the formulaic calculus of probability, perhaps resorting to Monte Carlo methods if an appropriate formula does not exist. Or,

b. Use resampling (in the larger sense), the domain

of which equals (all Monte Carlo experimentation) minus (the use of Monte Carlo methods for approximations, investigation of complex functions in statistics and other theoretical mathematics, and uses elsewhere in science). Resampling in its more restricted sense includes the bootstrap, permutation tests, and other non-parametric methods.

4. Interpretation of the probabilities that result from step 3 in terms of i) acceptance or rejection of hypotheses, ii) surety of conclusions, or iii) inputs to decision theory.

Here is a short definition of statistical inference: The selection of a probabilistic model that might resemble the process you wish to investigate, the investigation of that model's behavior, and the interpretation of the results.

Summary and conclusions

A statistical question asks about the probabilities of a sample having arisen from various source universes in light of the evidence of a sample. In every case, the statistical answer comes from considering the behavior of particular specified universes in relation to the sample evidence and to the behavior of other possible universes. That is, a statistical problem is an exercise in postulating universes of interest and interpreting the probabilistic distributions of results of those universes. The preceding sentence is the key operational idea in statistical inference.

Some key points:

1) In statistical inference as in all sound thinking, one's purpose is central. All judgments should be made relative to that purpose, and in light of costs and benefits. (This is the spirit of the Neyman-Pearson approach).

2) One cannot avoid making judgments; the process of statistical inference cannot ever be perfectly routinized or objectified. Even in science, fitting a model to experience requires judgment.

3) The best ways to infer are different in different situations—economics, psychology, history, business, medicine, engineering, physics, and so on.

4) Different tools must be used when the situations call for them—sequential vs. fixed sampling, Neyman-Pearson vs. Fisher, and so on.

5) In statistical inference it is wise not to argue about the proper conclusion when the data and procedures are ambiguous. Instead, whenever possible, one should go back and get more data, hence lessening the importance of the efficiency of statistical tests. In some cases one cannot easily get more data, or even conduct an experiment, as in biostatistics with cancer patients. And with respect to the past one cannot produce more historical data. But one can gather more and different kinds of data, e.g. the history of research on smoking and lung cancer

MULTIVARIATE METHODS

Multivariate Methods

Multivariate statistical methods are used to analyze the joint behavior of more than one random variable. There are a wide range of mulitvariate techniques available, as may be seen from the examples below. These techniques can be done using the Statgraphics multivariate statistical analysis software.

All statistical techniques which simultaneously analyse more than two variables on a sample of observations can be categorized as multivariate techniques. We may as well use the term 'multivariate analysis' which is a collection of methods for analyzing data in which a number of observations are available for each object. In the analysis of many problems, it is helpful to have a number of scores for each object. For instance, in the field of intelligence testing if we start with the theory that general intelligence is reflected in a variety of specific performance measures, then to study intelligence in the context of this theory one must administer many tests of mental skills, such as vocabulary, speed of recall, mental arithmetic, verbal analogies and so on. The score on each test is one variable, Xi, and there are several, k, of such scores for each object, represented as X1, X2...Xk. Most of the research studies involve more than two variables in which situation analysis is desired of the association between one (at times many) criterion variable and several independent variables, or we may be required to study the association between variables having no dependency relationships. All such analyses are termed as multivariate analyses or multivariate techniques. In brief, techniques that take account of the various relationships among variables are termed multivariate analyses or multivariate techniques.

Characteristics And Applications

Multivariate techniques are largely empirical and deal with the reality; they possess the ability to analyse complex data. Accordingly in most of the applied and behavioural researches, we

generally resort to multivariate analysis techniques for realistic results. Besides being a tool for analyzing the data, multivariate techniques also help in various types of decision-making. For example, take the case of college entrance examination wherein a number of tests are administered to candidates, and the candidates scoring high total marks based on many subjects are admitted. This system, though apparently fair, may at times be biased in favour of some subjects with the larger standard deviations.

Multivariate techniques may be appropriately used in such situations for developing norms as to who should be admitted in college. We may also cite an example from medical field. Many medical examinations such as blood pressure and cholesterol tests are administered to patients. Each of the results of such examinations has significance of its own, but it is also important to consider relationships between different test results or results of the same tests at different occasions in order to draw proper diagnostic conclusions and to determine an appropriate therapy. Multivariate techniques can assist us in such a situation. In view of all this, we can state that "if the researcher is interested in making probability statements on the basis of sampled multiple measurements, then the best strategy of data analysis is to use some suitable multivariate statistical technique."1The basic objective underlying multivariate techniques is to represent a collection of massive data in a simplified way. In other words, multivariate techniques transform a mass of observations into a smaller number of composite scores in such a way that they may reflect as much information as possible contained in the raw data obtained concerning a research study. Thus, the main contribution of these techniques is in arranging a large amount of complex information involved in the real data into a simplified visible form. Mathematically, multivariate techniques consist in "forming a linear composite vector in a vector subspace, which can be represented in terms of projection of a vector onto certain specified subspaces."

For better appreciation and understanding of multivariate techniques, one must be familiar with fundamental concepts of linear algebra, vector spaces, orthogonal and oblique projections and univariate analysis. Even then before applying multivariate techniques for meaningful results, one must consider the nature and structure of the data and the real aim of the analysis. We should also not forget that multivariate techniques do involve several complex mathematical computations and as such can be utilized largely with the availability of computer facility.

Multivariate Techniques

A brief description of the various multivariate techniques named above (with special emphasis on factor analysis) is as under:

(i) **Multiple regression**: In multiple regression we form a linear composite of explanatory variables in such way that it has maximum correlation with a criterion variable. This technique is appropriate when the researcher has a single, metric criterion variable. Which is supposed to be a function of other explanatory variables. The main objective in using this technique is to predict the variability the dependent variable based on its covariance with all the independent variables. One can predict the level of the dependent phenomenon through multiple regression analysis model, given the levels of independent variables. Given a dependent variable, the linear-multiple regression problem is to estimate constants B1, B2, ... Bk and A

(ii) **Multiple discriminant analysis**: Through discriminant analysis technique, researcher may classify individuals or objects into one of two or more mutually exclusive and exhaustive groups on the basis of a set of independent variables. Discriminant analysis requires interval independent variables and a nominal dependent variable. For example, suppose that brand preference (say brand x or y) is the dependent variable of interest and its relationship to an individual's income, age, education, etc. is being investigated, then we should use the technique of discriminant analysis. Regression analysis in such a situation is not suitable because the dependent variable is, not intervally scaled. Thus discriminant analysis is considered an appropriate technique when the single dependent variable happens to be non-metric and is to be classified into two or more groups, depending upon its relationship with several independent variables which all happen to be metric. The objective in discriminant analysis happens to be to predict an object's likelihood of belonging to a particular group based on several independent variables. In case we classify the dependent variable in more than two groups, then we use the name multiple discriminant analysis; but in case only two groups are to be formed, we simply use the term discriminant analysis.

(iii) **Multivariate analysis of variance:** Multivariate analysis of variance is an extension of bivariate analysis of variance in which the ratio of among-groups variance to within-groups variance is calculated on a set of variables instead of a single variable. This technique is considered appropriate when several metric dependent variables are involved in a research study along with many non-metric explanatory variables. (But if the study has only one metric dependent variable and several non-metric explanatory variables, then we use the ANOVA

technique as explained earlier in the book.)In other words, multivariate analysis of variance is specially applied whenever the researcher wants to test hypotheses concerning multivariate differences in group responses to experimental manipulations. For instance, the market researcher may be interested in using one test market and one control market to examine the effect of an advertising campaign on sales as well as awareness, knowledge and attitudes. In that case he should use the technique of multivariate analysis of variance for meeting his objective.

(iv) **Canonical correlation analysis:** This technique was first developed by Hotelling wherein an effort is made to simultaneously predict a set of criterion variables from their joint co-variance with a set of explanatory variables. Both metric and non-metric data can be used in the context of this multivariate technique. The procedure followed is to obtain a set of weights for the dependent and independent variables in such a way that linear composite of the criterion variables has a maximum correlation with the linear composite of the explanatory variables. For example, if we want to relate grade school adjustment to health and physical maturity of the child, we can then use canonical correlation analysis, provided we have for each child a number of adjustment scores (such as tests, teacher's ratings, parent's ratings and so on) and also we have for each child a number of health and physical maturity scores (such as heart rate, height, weight, index of intensity of illness and so on).The main objective of canonical correlation analysis is to discover factors separately in the two sets of variables such that the multiple correlation between sets of factors will be the maximum possible. Mathematically, in canonical correlation analysis, the weights of the two sets viz., a1, a2, ... ak

CONCEPTS OF CORRELATION AND REGRESSION

Correlation

Methods of correlation and regression can be used in order to analyze the extent and the nature of relationships between different variables. Correlation analysis is used to understand the nature of relationships between two individual variables. For example, if we aim to study the impact of foreign direct investment (FDI) on the level of economic growth in Vietnam, then two variables can be specified as the amounts of FDI and GDP for the same period.

Correlation coefficient 'r' is calculated through the following formula:

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left(n\sum x^2 - (\sum x)^2\right)\left(n\sum y^2 - (\sum y)^2\right)}}$$

Where, *x* and *y* are values of variables, and *n* is size of the sample.

The value of correlation coefficient can be interpreted in the following manner:

If 'r' is equal to 1, then there is perfect positive correlation between two values;

If 'r' is equal to -1, then there is perfect negative correlation between two values;

If 'r' is equal to zero, then there is no correlation between the two values.

In practical terms, the closer the value of 'r' to 1, the higher positive impact of FDI on GDP growth in Vietnam. Similarly, if the value of 'r' is less than 0, the closer it is to – 1, the greater the negative impact of FDI on GDP growth in Vietnam. If 'r' is equal to zero, then FDI is perceived to have no impact on GDP change in Vietnama within the given sample.

The Pearson product-moment correlation is calculated by taking the ratio of the sample of the two variables to the product of the two standard deviations and illustrates the strength of linear relationships. In Pearson product-moment correlation the correlation coefficient is not robust due to the fact that strong linear relationships between the variables are not recognized. The correlation coefficient is sensitive to outlying points therefore the correlation coefficient is not resistant.

Spearman Rank correlation requires the data to be sorted and the value to be assigned a specific rank with 1 to be assigned as the lowest value. Moreover, in case of data value appearing more than once, equal values will be specified their average rank.

Autocorrelation (serial correlation) implies the correlation among the values of the same variables but at various times. Autocorrelation coefficient is calculated by changing lagged data with the formula for the Pearson product-moment correlation coefficient. Also, because a series of unshifted data will express perfect correlation, the function begins with the coefficient of 1.

Correlation coefficient 'r' illustrated above is just a mathematical formula and you don't have to calculate correlation coefficient manually. For a bachelor's degree dissertation most supervisors accept correlation tests that have been run on a simple Excel spreadsheet. For

master's or PhD level studies, on the other hand, you will have to use more advanced statistical software such as SPSS or NCSS for your correlation analysis.

Correlation analysis as a research method offers a range of advantages. This method allows data analysis from many subjects simultaneously. Moreover, correlation analysis can study a wide range of variables and their interrelations. On the negative side, findings of correlation does not indicate causations i.e. cause and effect relationship

Regression:

Regression analysis is a which is used when the study involves modelling and analysing several variables, where the relationship includes a dependent variable and one or more independent variables. In simple terms, regression analysis is a quantitative method used to test the nature of relationships between a dependent variable and one or more independent variables.

The basic form of regression models includes unknown parameters (β), independent variables (X), and the dependent variable (Y).Regression model, basically, specifies the relation of dependent variable (Y) to a function combination of independent variables (X) and unknown parameters (β)

$$Y \approx f(X, \beta)$$

Regression equation can be used to predict the values of 'y', if the value of 'x' is given, and both 'y' and 'x' are the two sets of measures of a sample size of 'n'. The formulae for regression equation would be

$$b = \frac{n\sum xy - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

$$a = \frac{\sum y - b\sum x}{n}$$

Do not be intimidated by visual complexity of correlation and regression formulae above. You don't have to apply the formula manually, and correlation and regression analyses can be run

with the application of popular analytical software such as Microsoft Excel, Microsoft Access, SPSS and others.

Linear regression analysis is based on the following set of assumptions:

1. Assumption of linearity: There is a linear relationship between dependent and independent variables.

2. Assumption of homoscedasticity: Data values for dependent and independent variables have equal variances.

3. Assumption of absence of collinearity or multicollinearity:There is no correlation between two or more independent variables.

4. Assumption of normal distribution: The data for the independent variables and dependent variable are normally distributed.

FUNDAMENTALS OF TIME SERIES ANALYSIS

Time is one of most important factors on which our businesses and real life depends. But, technology has helped us manage the time with continuous innovations taking place in all aspects of our lives. Don't worry, we are not talking about anything which doesn't exist. Let's be realistic here!

Here, we are talking about the techniques of predicting & forecasting future strategies. The method we generally use, which deals with time-based data that is nothing but "Time Series Data" & the models we build ip for that is "Time Series Modeling**"**. As the name indicates, it's basically working on time (years, days, hours, and minutes) based data, to explore hidden insights of the data and trying to understand the unpredictable nature of the market which we have been attempting to quantify. There are many statistical techniques available for time series forecast however we have found few effectives ones which are listed below:

Techniques of Forecasting

- Simple Moving Average (SMA)

- Exponential Smoothing (SES)
- Autoregressive Integration Moving Average (ARIMA)
- Neural Network (NN)
- Croston

METHOD-I: SIMPLE MOVING AVERAGE (SMA)

Introduction:

A simple moving average (SMA) is the simplest type of technique of forecasting. Basically, a simple moving average is calculated by adding up the last 'n' period's values and then dividing that number by 'n'. So the moving average value is considering as the forecast for next period.

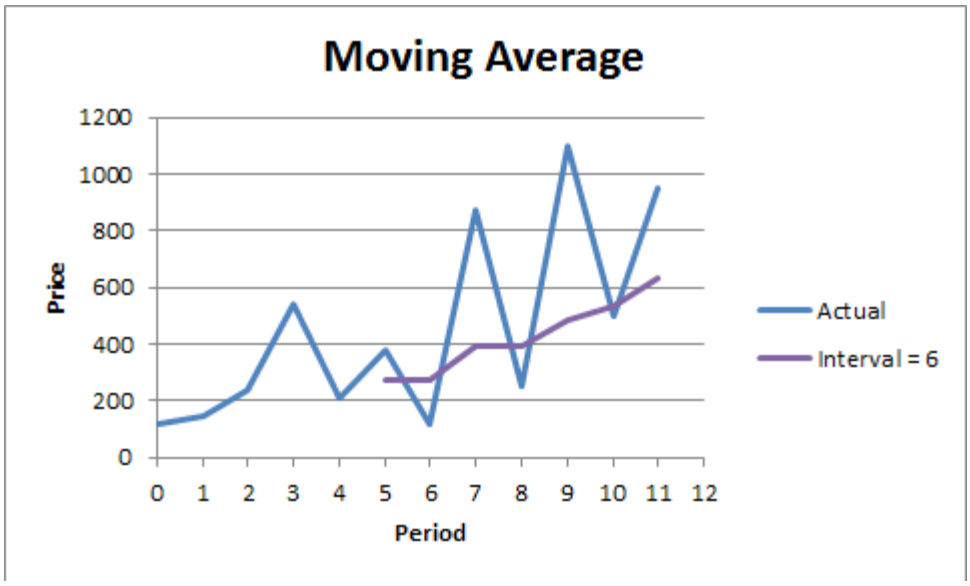
Why Do We Use SMA?

Moving averages can be used to quickly identify whether selling is moving in an uptrend or a downtrend depending on the pattern captured by the moving average.

i.e. A moving average is used to smooth out irregularities (peaks and valleys) to easily recognize trends.

SMA Working Example:

Let us suppose, we have a time series data, to have a better understanding on SMA, Where, we have the graphical view of our data, in that we have twelve observations of Price with equal interval of time. After plotting our data, it seems that it has upward trend with lot of peaks and valleys.



Conclusion: The larger the interval, the more the peaks, and valleys are smoothed out. The smaller the interval, the closer the moving averages are to the actual data points. The SMA is basically deal with historical data having more and more peak and valleys. Probably it would be stock data, retail data etc.

METHOD-II: EXPONENTIAL SMOOTHING

Introduction:

This is the second well known method to produce a smoothed Time Series. Exponential Smoothing assigns exponentially decreasing weights as the observation get older.

Why Do We Use Exponential Smoothing?

Exponential smoothing is usually a way of "smoothing" out the data by removing much of the "noise" (random effect) from the data by giving a better forecast.

Types of Exponential Smoothing Methods

- **Simple Exponential Smoothing: –**

If you have a time series that can be described using an additive model with constant level and no seasonality, you can use simple exponential smoothing to make short-term forecast.

- **Holt's Exponential Smoothing: –**

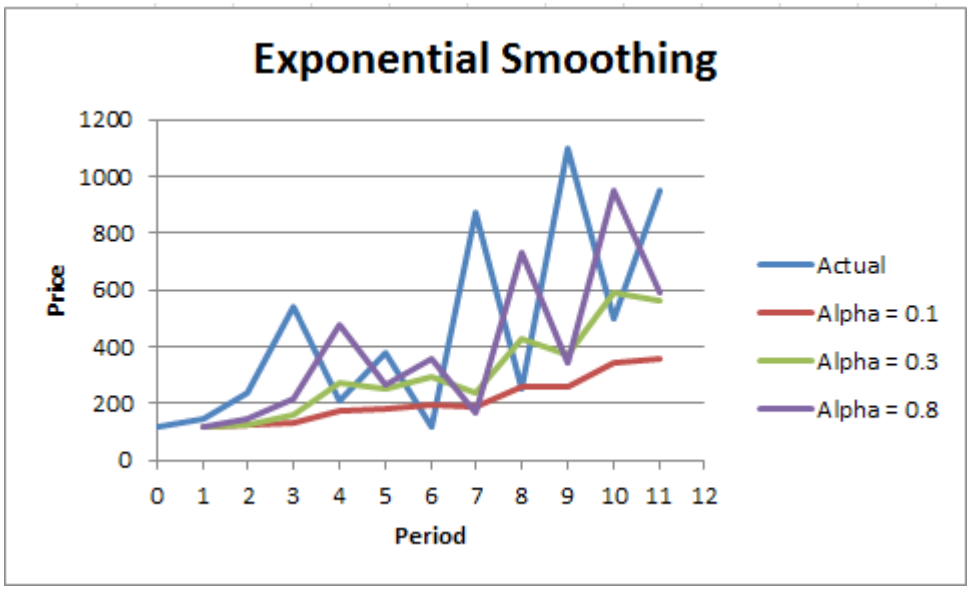
If you have a time series that can be described using an additive model with increasing or decreasing trend and no seasonality, you can use Holt's exponential smoothing to make short-term forecasts.

- **Winters' Three Parameter Linear and Seasonal Exponential Smoothing: –**

If you have a time series that can be described using an additive model with increasing or decreasing trend and seasonality, you can use Holt-Winters exponential smoothing to make short-term forecasts.

Graphical Views:

Exponential Smoothing:



Here, we have alpha value that is smoothing constant and this method is called simple exponential smoothing method which consider other two factors as constant (i.e. Seasonality & Trend). Double's (Holt's) Exp. Smoothing & Winter's Exp. Smoothing Methods dealing two factors i.e. Trend and Seasonality (i.e. Beta & Gamma).

Conclusion: Larger the alpha, closer to the actual data points and vice versa. This method is suitable for forecasting data with no trend or seasonal pattern (alpha = Smoothing Constant).

METHOD-III AUTOREGRESSIVE INTEGRATED MOVING AVERAGE (ARIMA)

Autoregressive Integrated Moving Average (ARIMA): –

A statistical technique that uses time series data to predict future. The parameters used in the ARIMA is (P, d, q) which refers to the autoregressive, integrated and moving average parts of the data set, respectively. ARIMA modeling will take care of trends, seasonality, cycles, errors and non-stationary aspects of a data set when making forecasts.

Understanding ARIMA Model in General Term: –

How to Understand ARIMA model?

To understand this, we can refer real time scenario that is sugar cane juicer, from juicer it is difficult to extract all the juice in one go, so the shopkeeper repeats the process for several times till there is no more juice left in the residual. That's how ARIMA works, the idea with ARIMA models is that the final residual should look like white noise otherwise there is juice or information available in the data to extract.

How Do We Use ARIMA Model?

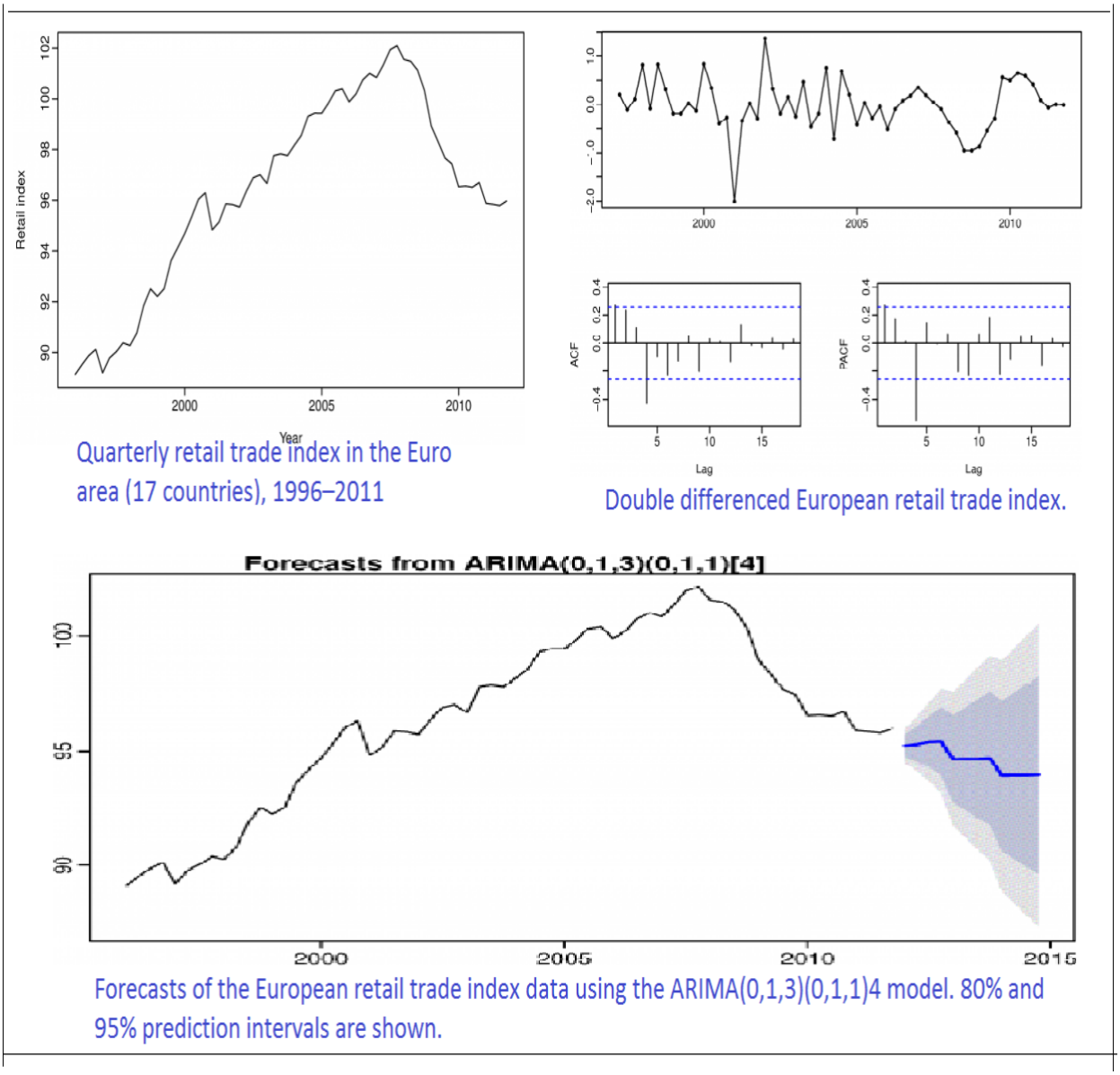
ARIMA checks stationarity availability in the data, the data should also show a constant variance in its fluctuations over time. To get the proper information about the parameter used in ARIMA is based on "identification process" which was purposed by Box-Jenkins.

When Do We Use ARIMA Model?

As we all know ARIMA is mainly used to project future values using historical time series data. Its main application is in short forecasting with minimum 38-40 historical data points with minimum number of outliers. If you do not have at least 38 data points, then it advisable to look for some other methods.

Working Example of ARIMA

Here, we are trying to understand ARIMA using quarterly European retail trade data from 1996 to 2011. The data are clearly non-stationary, with some seasonality, so we will first take a seasonal difference. The seasonally differenced data are shown in Fig. These also appear to be non-stationary, and so we take an additional first difference and may be next if require.



Quarterly retail trade index in the Euro area (17 countries), 1996–2011

Double differenced European retail trade index.

Forecasts of the European retail trade index data using the ARIMA(0,1,3)(0,1,1)4 model. 80% and 95% prediction intervals are shown.

As we considered seasonal ARIMA model which first checks their basic requirements and is ready for forecasting. Forecasts from the model for the next three years are shown in Figure. Notice how the forecasts follow the recent trend in the data (this occurs because of the double differencing).

Conclusion: – It works best when your data exhibits a stable or consistent pattern over time with a minimum amount of outliers.

METHOD-IV NEURAL NETWORK

Introduction:

ANN: – Artificial neural network (ANN) is basically machine learning approach that models human brain and consists of a number of artificial neurons. Their ability to learn by example makes them very flexible and powerful.

Why Do We Use Neural Networks?

Neural networks, has its own strength to derive meaning from complicated or imprecise data, and most of the time can be used to detect the pattern and trend in the data, which cannot be detectable easily from human eye or any computer techniques. We also have some of the advantage of NN like Adaptive learning, self-organization, real-time operation, fault tolerance.

Applications of neural networks

Now a day, in every field NN is equally important, for example, some of the fields I have listed below: –

- Sales Forecasting
- Industrial Process Control
- Customer Research
- Data Validation
- Risk Management
- Target Marketing

Conclusion:

We can use NN in any type of industry and get benefited, as it is very flexible and also doesn't requires any algorithms. They are regularly used to model parts of living organisms and to investigate the internal mechanisms of the brain.

METHOD-V CROSTON

Introduction:

It's modification of exponential smoothing for sporadic demand product time series suggested by Croston in 1972. The core value of this method is not only the estimation of average demand volume, but also estimation of time interval length between two non-zero demands, term called as intermittent demand.

The Croston method works in two steps, First, separate exponential smoothing estimates are made of the average size of a demand. Second, the intermittent demands are calculated. This is then used in a form of the constant model to predict the future demand.

How Croston's Work?

Croston's has a complex formula, however, what it output is actually very simple. The screenshot below explains what Croston's does in a very simple way for the sake of understanding.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
Demand History	7	0	0	0	4	0	0	5	0	2	0	3	1.75	Average
	7	4	5	2	3								4.2	Croston Average
				Percentage of time there is demand									42%	
				Percentage of demand periods x average									1.75	

Above is the 12-month average vs. Croston's vs, while below is the 5-month average vs. Croston's.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
Demand History	7	0	0	0	4	0	0	5	0	2	0	3	2	5 Month Average
	7	4	5	2	3								4.2	Croston Average
				Percentage of time there is demand									42%	
				Percentage of demand periods x average									1.75	

As you can see, Croston's removes the periods that have no demand only averaging the periods that have demand. Next Croston's calculates the frequency of the demand. The math behind this is complex, but the output is extremely similar to performing an exponential smoothing.

Why Do We Use CROSTON's?

In the given fig. we have two Croston's forecast based on demand histories, with more non-zero data-points. Here Croston's will come into the picture and show its benefits.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Demand History	7	0	0	0	4	0	0	5	0	2	0	3	
	0	0	1.8	0	0	1.8	0	0	1.8	0	0	1.8	Croton's Forecast
												5.5	Average without 0s
												1.8	Average with 0s

There are two examples of Croston's compared to a long duration moving average. However, look how things change based upon changes in timing.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Demand History	7	0	0	0	4	0	0	0	0	5	0	0	*Changed Months*
	0	0	5.5	0	0	0	0	5.5	0	0	0	0	Croton's Forecast
												5.5	Average without 0s
												3.5	Average with 0s

*Notice the significant difference in the forecast provided by Coston's in the second forecast.

- At the very beginning, Croston's starts detecting cyclic and periodicity into the data points of demand patterns. In this case, it is suggested that a demand could occur possibly after 3.5 (4 after roundup) zero period.
- The second most important thing which Croston's does is, it adjusts the next occurrence from the last non-zero period, if the recent periods are zero periods.

Conclusion:

The Croston method is a forecast strategy for products with intermittent demand. In the univariate forecast profile, choose forecast strategy.

Croston's can be easily emulated with exponential smoothing and any timing benefit is usually adjusted by order lot sizing, and or safety stock in supply planning. Therefore, demand history must not only be lumpy, but must also be very low in order for Croston's to be of value.

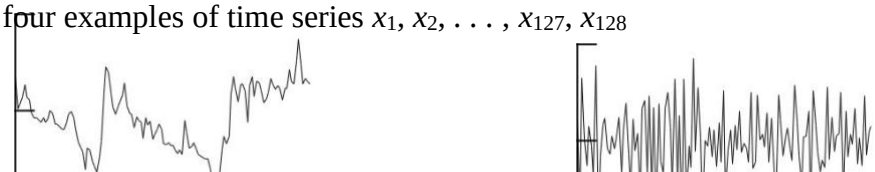
FUNDEMENTALS OF SPECTRAL ANALYSIS

- Q: what is spectral analysis?
  one of the most widely used methods for data analysis in geophysics, oceanography, atmospheric science, astronomy, engineering (all types), . . .

- method is used with time series
- let x_t denote value of time series at time t; for example,

  $x_{10} = 43^\circ$ = temperature at GHS at 8AM on 10th day of 2003

- four examples of time series $x_1, x_2, \ldots, x_{127}, x_{128}$



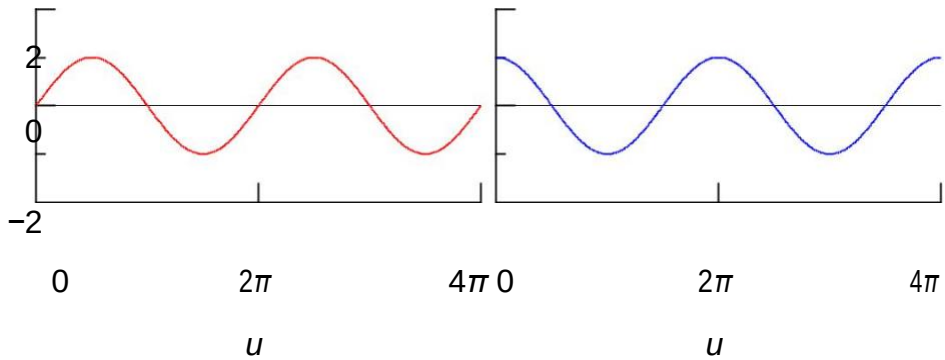
- Q: how would you describe these 4 series?

spectral analysis describes x_t's by comparing them to sines and cosines

Sines and Cosines: I

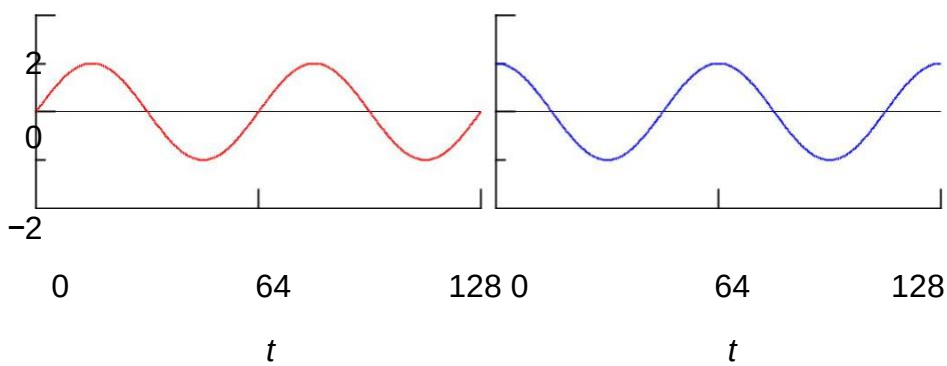
- Q: what do sines and cosines have to do with time series?

- plots of sin(u) and cos(u) versus u as u goes from 0 to 4π



- let $u = 2\pi\ {}_{128}{}^{2}\ t$ for $t = 1, 2, \ldots, 128$

- plots of $\sin(2\pi\ {}_{128}{}^{2}\ t)$ and $\cos(2\pi\ {}_{128}{}^{2}\ t)$ versus t



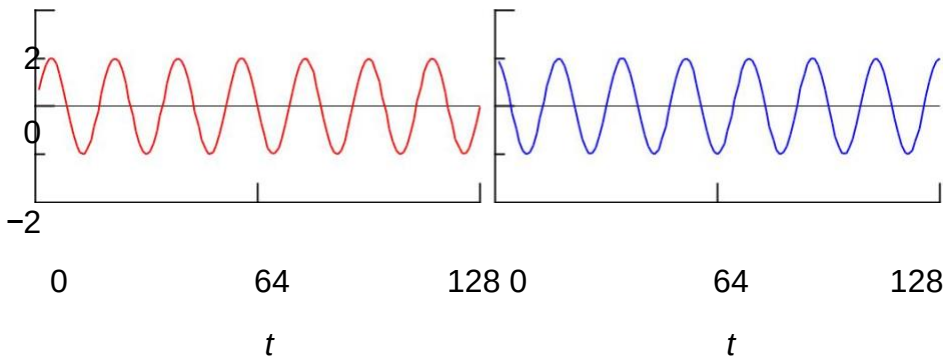
- can regard above as artificial time series: $s_t = \sin(2\pi\ {}_{128}{}^{2}\ t)$ etc

- can interpret ${}_{128}{}^{2}$ as '2 cycles over time span of 128'

Sines and Cosines: II

• now let $u = 2\pi\, {}_{128}{}^{7}\, t$ for $t = 1, 2, \ldots, 128$

• plots of $\sin(2\pi\, {}_{128}{}^{7}\, t)$ and $\cos(2\pi\, {}_{128}{}^{7}\, t)$ versus t



• now get series with 7 cycles over time span of 128

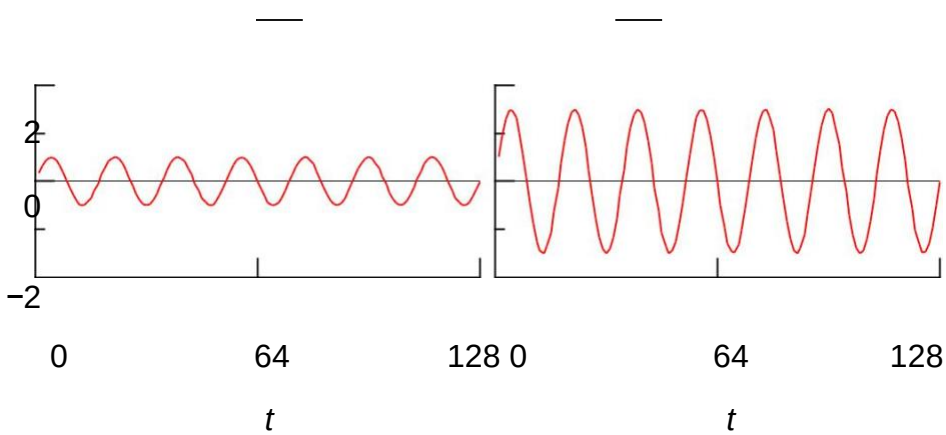
• in general plots of $\sin(2\pi\, {}_{128}{}^{k}\, t)$ etc versus t have k cycles

• quantity ${}_{128}{}^{k}$ is called frequency of sine or cosine (usually use variable f to denote frequency)

if k is small, sine time series is said to have low frequency if k is large, sine time series is said to have high frequency

• amplitude (maximum range of variation) of $s_t = \sin(2\pi\, {}_{128}{}^{k}\, t)$ is 1

• plots of $0.5 \sin(2\pi\, {}_{128}{}^{7}\, t)$ and $1.5 \sin(2\pi\, {}_{128}{}^{7}\, t)$



Goal of Spectral Analysis

0 given a time series x_t, figure out how to construct it using sines and cosines; i.e., to write

$$x_t = a_k \sin(2\pi \, {}_{128}{}^{k} \, t) + b_k \cos(2\pi \, {}_{128}{}^{k} \, t)\text{k}$$

- above called 'Fourier representation' for a time series (named after 19th Century French mathematician)
- allows us to reexpress time series in a standard way
- different time series will need different a_k's and b_k's
- can compare different time series by comparing the a_k's and b_k's

ERROR ANALYSIS

An error is systematic .It is likely to occur repeatedly and is not recognized by the learner as an error. The learner has incorporated a particular erroneous from the perspective of the target language into his/her own system. The learner has created a systematic entity called an inter language.

Table 3.1. The Procedures of Error Analysis

No	Steps	Explanations
1	Collection of samples of learner language	Deciding what samples of language learner to use for the analysis and how to collect these samples
2	Identification of errors	Identifying the errors by underlying the errors the learner made
3	Classification of errors	Grouping the errors that have been found and stating the classes of the errors
4	Explanation of errors	Explaining the errors by establishing the source of the errors and calculating how often the errors appear
5	Evaluation of errors	Evaluating the errors step involves tabulating the errors and drawing conclusion

Referring to the steps of error analysis method above, the data will be analyzed as follows:

1. **Identification of errors**

In this step, the researcher studied the gained data and tried to find out the grammatical errors by underlying the errors. She tried to analyze the data as objective as possible.

2. **Classification of errors**

Once the errors have been identified, she classified them in the following categories of grammatical rules, for example:

a. errors in using verb forms
b. errors in agreement between subject and verb
c. errors in the use of article
d. errors in the use of preposition
e. errors in pluralization
f. errors in the use of pronoun
g. errors in the use of conjunction
h. errors in the use of adverb

3. **Calculation of errors**

In this step, she calculated the errors in order to know how frequent those errors have been made by the college students of first year at IAIN Tulungagung. In calculating the frequency of each error, she employed the following formula:

$P = \frac{n1}{n} \times 100\%$

P: percentage of each error

N1: total of the given error

ΣN: total of the whole errors

By calculating the frequency of each error, she could identify the most frequent error and the least frequent error made by the college students.

4. **Tabulating result**
   Once the errors were calculated and arranged, she tabulated the result of the analysis. This table was meant to ease the identification of the percentage of each error. Therefore, the result of the analysis of the grammatical errors in narrative text made by the college students of IAIN Tulungagung was presented in the form of a table.

5. **Drawing a conclusion**
   The last step would be drawing a conclusion based on the analysis. In this step, she had to make a valid conclusion in the form of a brief error description.

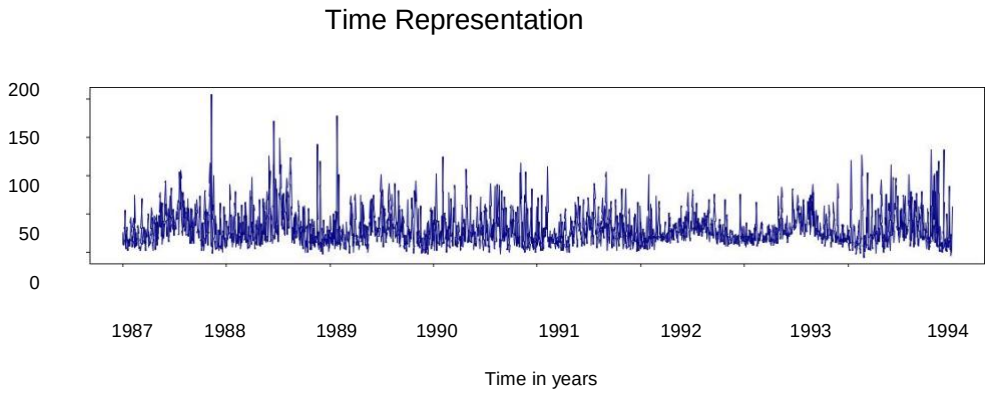
APPLICATIONS OF SPECTRAL ANALYSIS

The basic concepts and application of spectral analysis are explained. Stationary time series and autocorrelation are first defined. Autocorrelation is related to the familiar concepts of variance and covariance. The use of autocorrelation analysis is explained in estimating the interdependent relationship of a time series over discrete time lags. In order to measure the behavior of the time series using autocorrelation, it would be necessary to examine a very large number of autocorrelation lags. Alternatively, the technique of Fourier analysis can be used to transform the autocorrelation function of the time series into a continuous function, termed a spectrum. The spectrum has a one to one correspondence to the autocorrelation for the time series and has the advantage of representing all possible autocorrelations over the discrete time lags. The spectrum can then be examined as a measure of the behavior of the time series. Spectral analysis indicates the reliability of the analysis of autocorrelated variables when familiar statistical techniques such as sample means and variances are used.

The application of spectral analysis to management science problems in three general areas is illustrated:
(1) Inventory demand,
(2) Transportation simulation,
(3) Stock market price behavior.

Spectral analysis was used to detect cycles and trends in the data. Analyses were focused on the spectrum which provides a measure of the relative contribution of cycles in a band of frequencies to the total variance of the data.

Sometimes it is useful to describe the properties of the time series in a frequency domain. The spectrum is defined as there is a one-to-one correspondence between the spectrum and the auto-covariance function.



An Application for spectral Analysis:

ECoG signal for two channels for a subject that has received a sensory stimulus at some point during the recording. A straightforward way of estimating the spectrum of a stationary process is the periodogram Thus it seems more appropriate to model the signal as a non-stationary processes and study the time-varying spectral density.

A straightforward estimate of a time-varying spectral density would be the dynamic periodogram.

UNIT 3

TABULAR AND GRAPHICAL DESCRIPTION OF DATA

TABLES AND GRAPHS OF FREQUENCY DATA OF ONE VARIABLE

The most common displays for the values collected on a single variable are considered. The final section discusses the relative merits of these displays.

ONE VARIABLE - Frequency distributions and bar charts

The frequency distribution shows the frequency with which the values of a variable are distributed amongst the individuals.

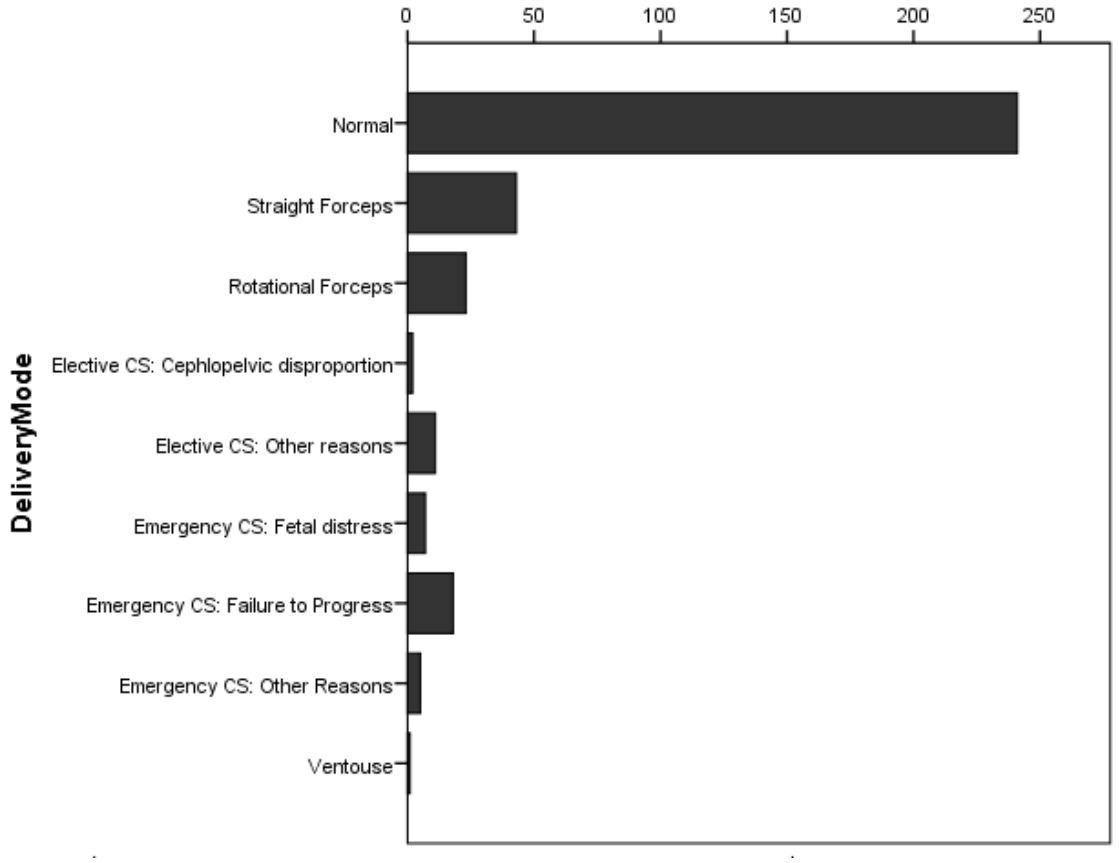
Here is an example of a categorical variable assessed in a group of individuals and the results displayed in this way:

Ref: Frame S, Moore J, Peters A & Hall D, Maternal height and shoe size as predictors of pelvic disproportion: an assessment, British Journal of Obstetrics and Gynaecology, Dec 1985, Vol. 92, 1239-1245.

The variable being displayed here is the mode of delivery at birth ('type of delivery'):

Type of Delivery	N	%
Normal	241	68.7
Straight forceps	43	12.3
Rotational forceps	23	6.6
Elective CS for:		
Cephlopelvic disproportion	2	0.6
Other reasons	11	3.1
Emergency CS for:		
Fetal distress	7	2.0
Failure to progress	18	5.1
Other reasons	5	1.4
All CS	43	12.3
Venthouse	1	0.3
Total	351	100.1

Bar charts are commonly used to display frequency distributions. The heights of the bars show the frequency with which observations fall within certain intervals. Here is the same data shown as a bar-chart:



None of the relevant information is lost with either of these displays. If the information is given in either of these formats, as opposed to giving the list of categorisations obtained in their raw form, then the only thing that cannot be determined is the order in which the 351 patients presented. In this instance, the order in which the women presented is probably unimportant so that this loss is not of concern.

The bar chart possibly makes the information given more easily digestible in that it illustrates the relative frequency of the birth types graphically. Therefore, the bar chart may be preferred to the frequency distribution as a means of displaying this data.

The bar chart can sometimes be made more effective as a means of display by ordering the categories according to frequency. For example:

Ref: O'Connor M, Europe and nutrition: prospects for public health, BMJ 1992; 304, 178-80.

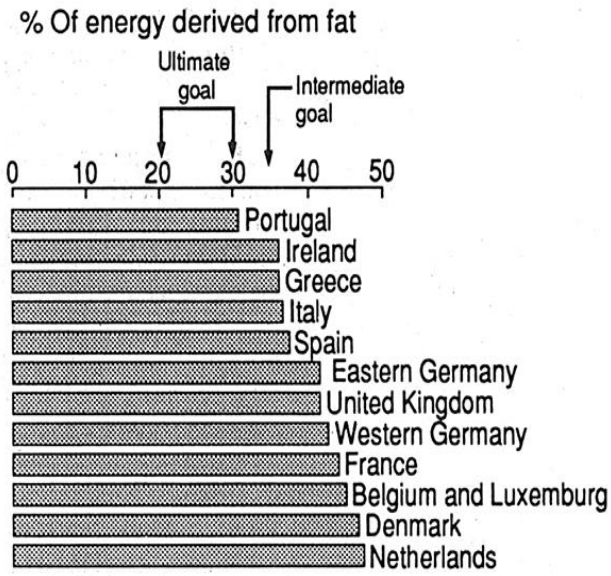


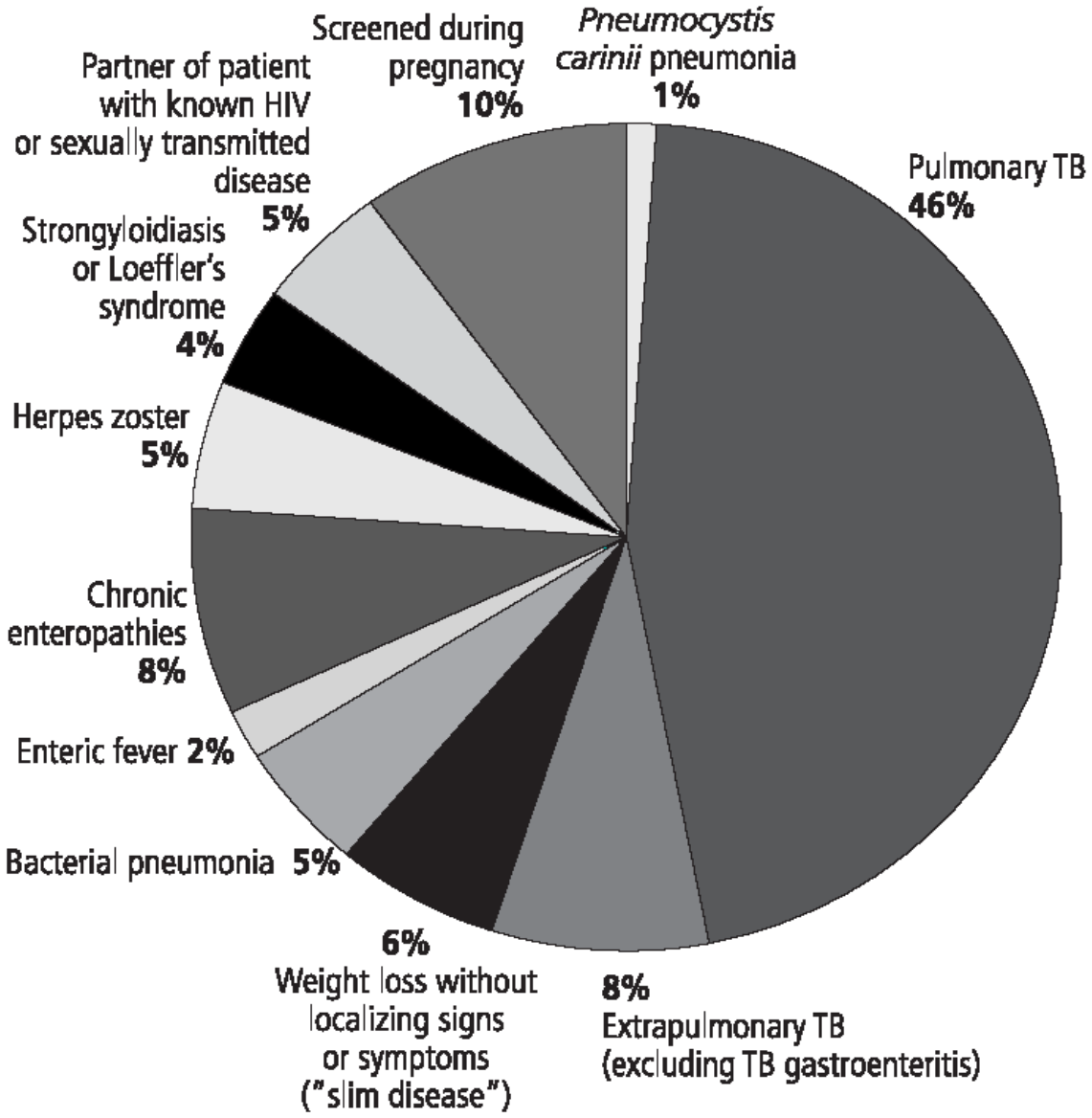
FIG 2—*Estimated percentage of total energy derived from fat in some European countries*
Source: World Health Organisation.

ONE VARIABLE - Pie charts

A pie diagram shows the proportion of a sample falling into each of the categories of a categorical variable. The proportions are shown relative to one another by dividing a circle accordingly and naming the segments. For example, in the first pie diagram below, 46% of the 200 patients diagnosed with HIV presented initially with pulmonary TB, therefore they are depicted by a segment encompassing almost half the circle.

(1) Ref: Farmer P eg al. Community-based treatment of advanced HIV disease: introducing DOT-HAART (directly observed therapy with highly active antiretroviral therapy). Bulletin of the World Health Organisation, 2001, 79(12) 1145-1151.

Fig. 2. **Presenting diagnoses in 200 patients with HIV disease, Clinique Bon Sauveur, Haiti, 1993–95**



Presenting diagnoses in 200 patients with HIV disease Clinique Bon Sauveut, Haiti, 1993 -1995

(2) Ref: Platt and Pharoah, Child health statistical review; 1996; Arch Dis Childhood, 1996,75, 527-533

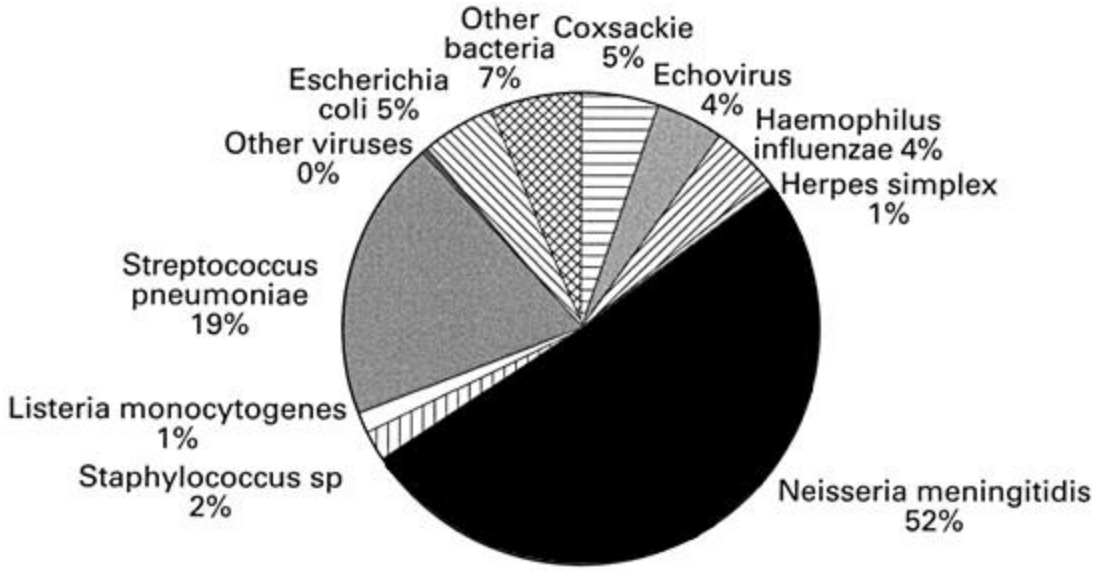


Figure 5 Organisms associated with meningitis among children age 0–5 years, England and Wales, 1994. Source: Office of National Statistics.[8]

NOTE: Pie diagrams are useful only for displaying data collected on NOMINALvariables. If there were an order to the categories then it would effectively be lost with this type of display.

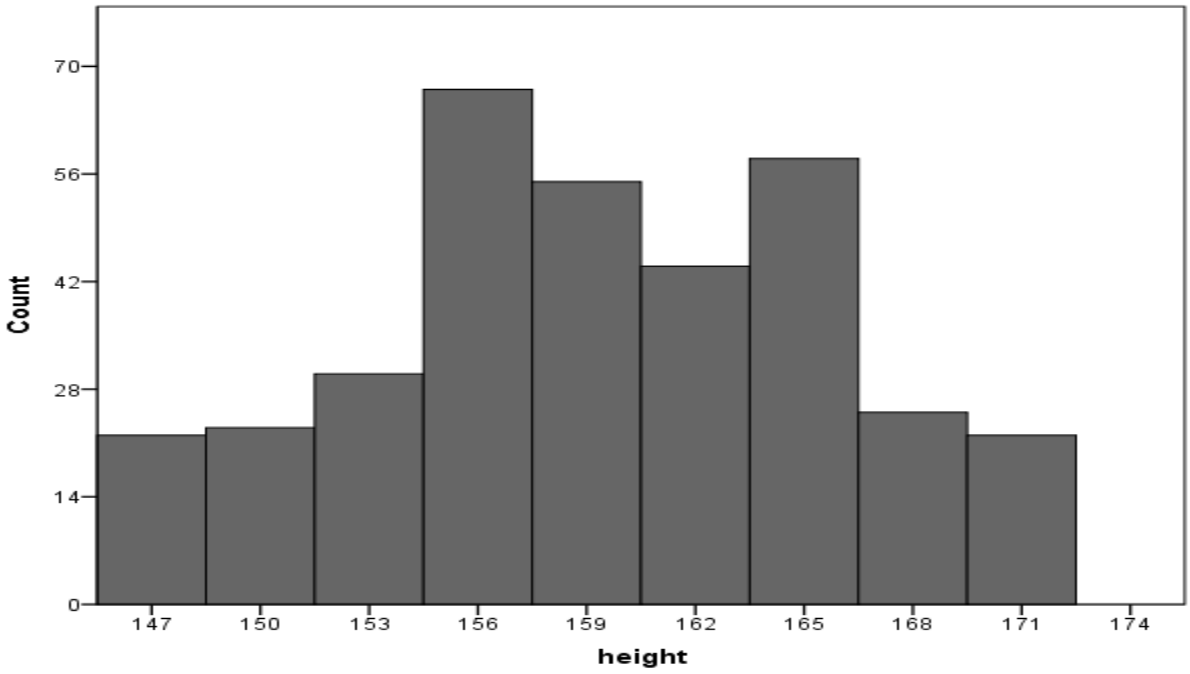
ONE VARIABLE - histogram

For numeric variables the frequency distribution of the individual values is not very helpful. It is more useful to categorise the values and to report the frequencies for each category. For example, the heights of 346 of the 351 women in the previous example were also recorded. If the heights are separated into ordinal categories, then these can be displayed in the following frequency distribution:

Height (cm)	Number of women
147	22
150	23
153	30
156	67
159	55
162	44
165	58
168	25
171	22
174	0
Total	346 *

*In 5 cases height was not recorded

Using these frequencies, we can create a graphical summary similar to a bar chart. However, this graph is known as a histogram:



ONE variable - Bar charts vs. Histograms

Bar charts are usually used to show the frequencies of a categorical variable. Therefore, the categories in a bar chart can often be reordered, separated or grouped together. The height of each bar shows the frequency within each category.

Histograms are used to show the distribution of a numeric variable. Histograms usually separate the variable into ordinal categories. These categories are often called bins.

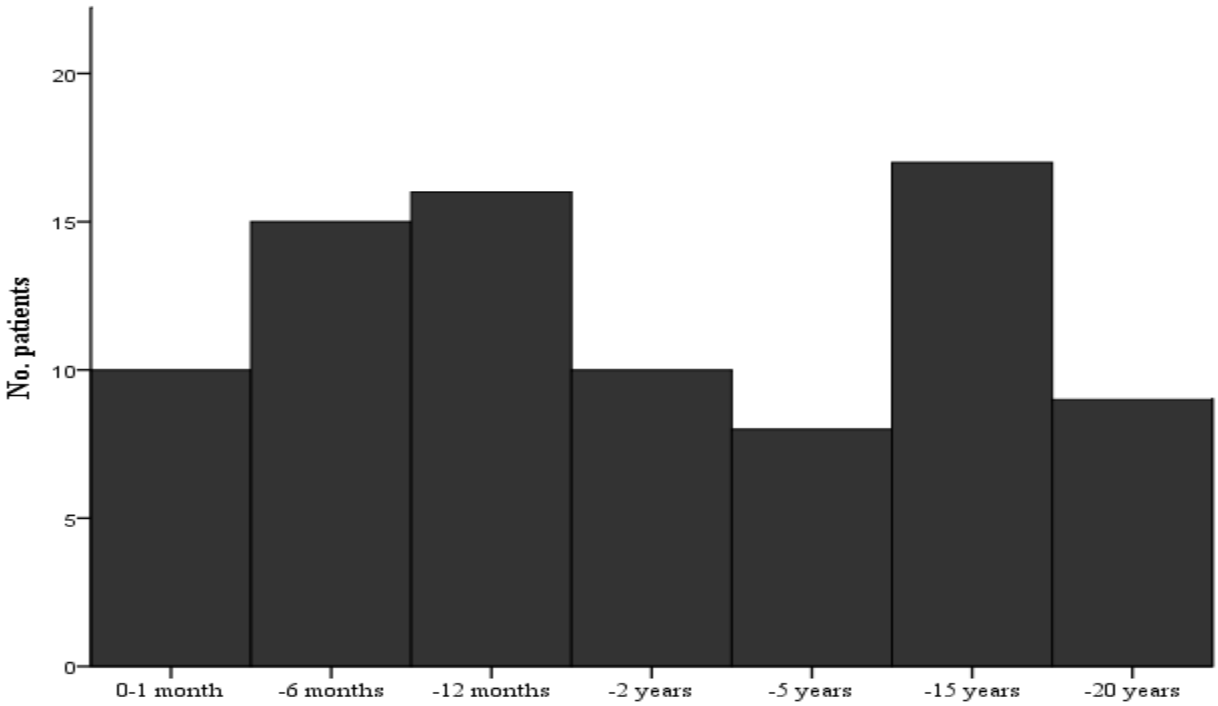
Histograms usually separate the numeric data into categories with an equal range (just like in the histogram above where each height category has a range of 3cm). However, this is not always the case, as we will see in the following example.

The ages of 85 pediatric patients presenting for cardiac operations are given in the table below as both the number per category and also the number per month within category:-

Age	Number	Number per month
0 - 1 month	10	10.00
- 6 months	15	3.00
- 12 months	16	2.67
- 2 years	10	0.83
- 5 years	8	0.22
- 15 years	17	0.14
- 20 years	9	0.15
Total	85	0.35

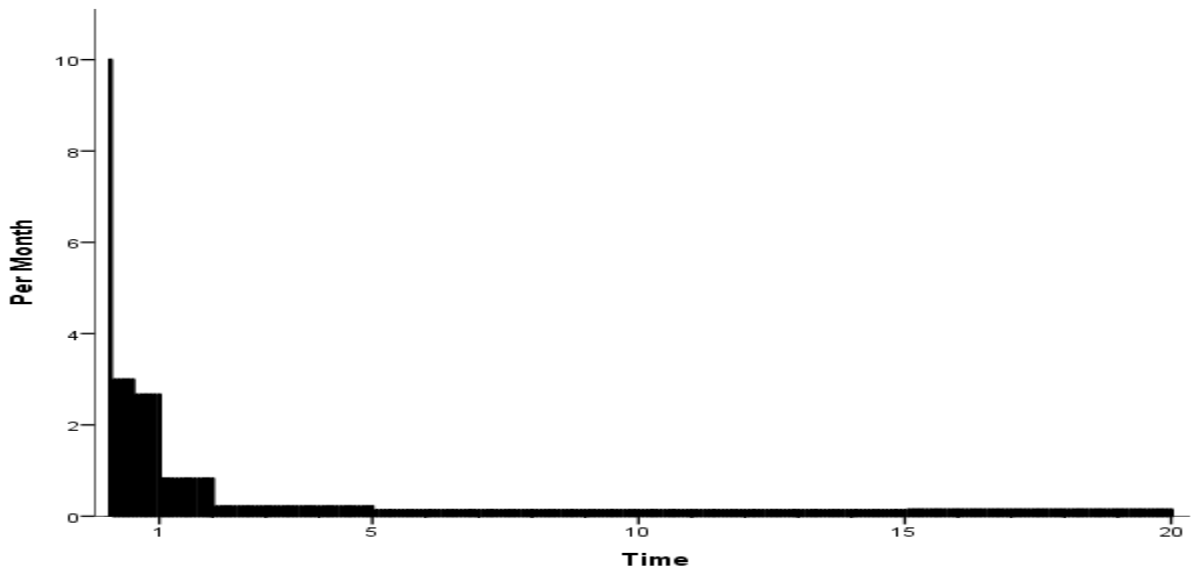
Age in the first column is grouped 0-1 month, 2-6 months, 6-12 months, 1-2 years, 3-5 years, 5-15 years, and 16+ years.

The number per month within category shows a pattern of declining frequency with age which is harder to see within the number per category display. A bar chart would consider the number per category with each category being given equal width in the display:



In forming a histogram the width of the bar is changed in proportion to the range of the continuous variable being covered by that particular bar. It is now the area of the bar which shows the frequency with which observations fall within certain intervals.

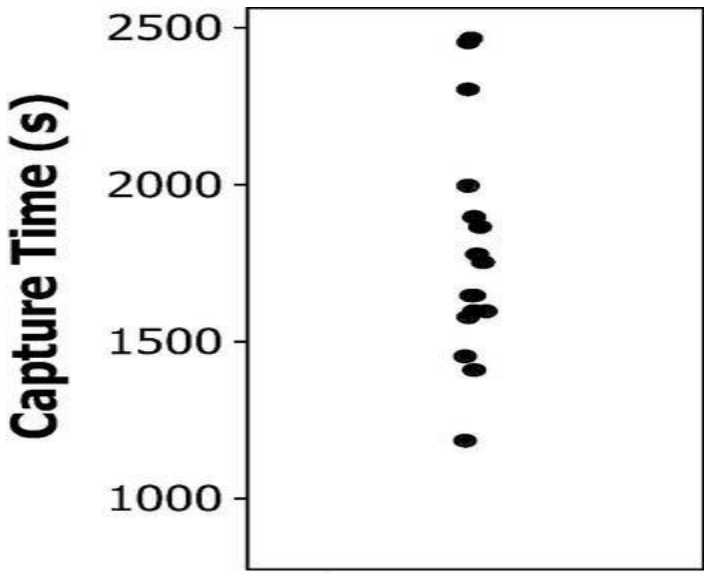
A histogram of the data considers the frequency per month. In effect this is giving the bars their representative distance on the horizontal axis (ie. the bar for the 10 babies in the 0-1 month category, which covers only one month, is stretched upwards to be tall and thin; the 9 patients within the 15-20 years category, which spans 5 years, is stretched out across a width 60 times wider (12 x 1 month x 5 years) to make it short and wide:



The bar chart masks the true nature of the distribution. The histogram clearly shows that the majority of operations occurred in children of less than one year of age, there were few in those over two years. This pattern is less readily identified from either the frequency table or the bar chart.

ONE VARIABLE - Dot diagrams

A useful and under-used display of continuous numeric data is the dot-diagram. One point represents the value for each individual or item, no information is lost (apart from the order of collecting measurements and outliers are clearly highlighted. For example, below are the times taken for a scorpion to capture its prey (a cricket) in a laboratory experiment.



Ref: Edmunds MC, Sibly RM. Optimal sting use in the feeding behavior of the scorpion Hadrurus spadix. Journal of Arachnology. 2010 Apr;38(1):123-5.

Capture time. Jitter has been applied to x coordinates to make overlapping points visible.

Summary of displays for one variable

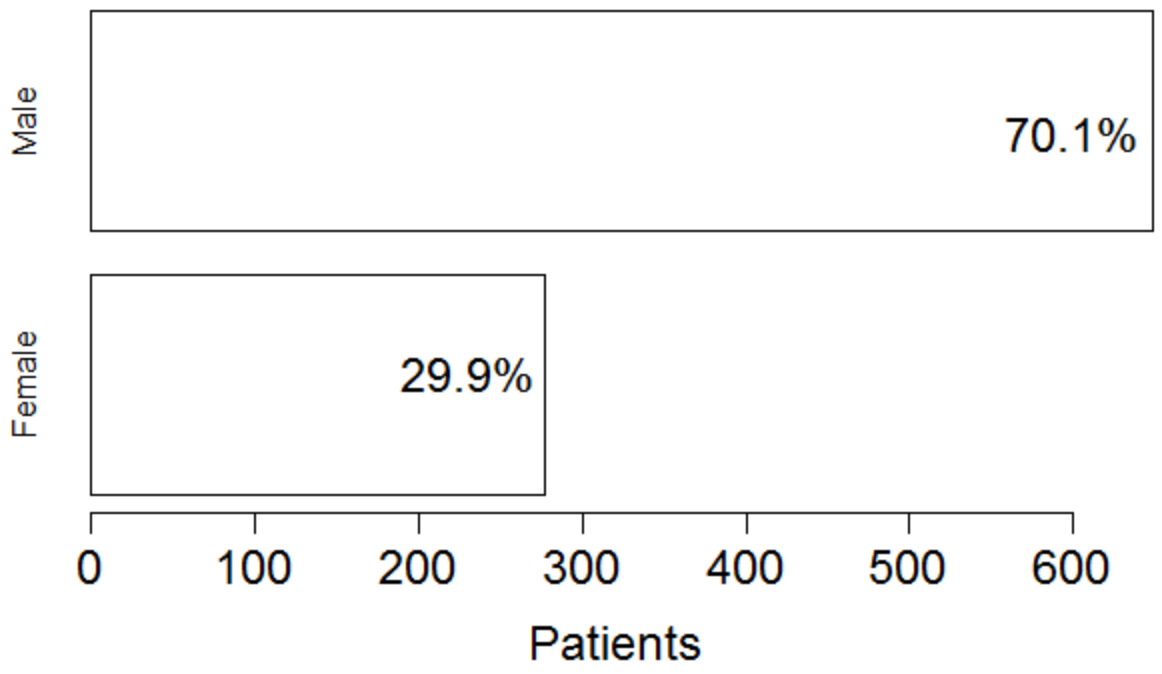
The previous 5 types of display (frequency distribution, bar-chart, histogram, dot-diagram and pie-chart) are useful when only one variable is being considered.

The type of variable may influence the appropriate means of display:

Type of variable		**Suitable methods of display**
Categorical	Binary	Frequency distribution only *
	Ordinal	Bar-chart
	Nominal	Bar-chart / Pie diagram
Numeric	Discrete	Bar-chart / Histogram / Dot-diagram
	Continuous	Bar-chart / Histogram / Dot-diagram

*There is no point in using a graphical display of data if it does not impart information in a more easily digestible form than the frequency distributions. It is therefore not recommended that binary data is displayed as a bar-chart or pie-chart and this may also apply to ordinal and nominal data with few categories.

Ref: Platt and Pharoah, Child health statistical review; 1996; Archives of Disease in Childhood, 1996, 75, 527-533.



The text states that "In the 927 patients studied, 649 were male (70.1%) and 278 female (29.9%)". The bar chart does not give any more information than the sentence in the text which references it. It is very wasteful of space, there must be more important aspects to this study than the excess of males.

Tables and Graphs for the Relationship Between Two Variables

•Construct and interpret cross tabulation for two categorical variables

•Construct and interpret clustered bar graphs for two categorical variables

•Construct and interpret scatter plots for two quantitative variables

Cross Tabulation

Cross tabulation (two-way table or contingency table) is a tabular method for simultaneously summarizing data for two categorical variables.

Table 4.1 Prestigious career survey data set

Student	Career seen as prestigious	Student gender	Student	Career seen as prestigious	Student gender
1	Doctor	F	11	Doctor	F
2	Scientist	M	12	Military Officer	M
3	Military Officer	M	13	Scientist	F
4	Military Officer	F	14	Lawyer	F
5	Doctor	M	15	Lawyer	F
6	Scientist	F	16	Military Officer	M
7	Military Officer	M	17	Doctor	F
8	Athlete	F	18	Scientist	M
9	Doctor	F	19	Doctor	F
10	Scientist	M	20	Lawyer	M

- **Cross tabulation**

Construct a cross tabulation career by gender.

Table 4.2 Crosstabulation of prestigious career survey data set

	Gender		
Career	**Female**	**Male**	**Total**
Doctor	5	1	6
Scientist	2	3	5
Military Officer	1	4	5
Lawyer	2	1	3
Athlete	1	0	1
Total	11	9	20

Cross tabulation

Cross tabulation (two-way table or contingency table) is a tabular method for simultaneously summarizing data for two categorical variables.

1.Put the categories of one variable at the top of each column and the categories of the other variable at the beginning of each row.

2.For each row and column combination, enter the number of observations that fall in the two categories.

3.The bottom of the table gives the column totals, and the right-hand column gives the row totals.

Construct a cross tabulation for the variables major and gender

a. **Do you notice any patterns? How would you characterize the relationship?**
b. **Construct the frequency distributions for the individual variables.**

a. Example of Cross tabulation A sample of 25 college students was selected And their major class, and gender were recorded

TABLE 4.7 Major, class, and gender of 25 college students

Major	Class	Gender
English	Sophomore	Male
Business	Sophomore	Female
Political Science	Senior	Male
English	Freshman	Female
Psychology	Sophomore	Female
History	Junior	Female
Philosophy	Senior	Male
Communication	Sophomore	Male
History	Junior	Female
English	Freshman	Female
Psychology	Sophomore	Female
English	Freshman	Female
History	Junior	Male
Psychology	Sophomore	Female
Economics	Senior	Male
Art	Sophomore	Female
Psychology	Junior	Male
English	Freshman	Female
Economics	Senior	Male
Business	Sophomore	Male
Business	Junior	Female
Communication	Sophomore	Male
Art	Junior	Female
Communication	Freshman	Female
Psychology	Junior	Female

Major	MALE	FEMALE	TOTAL
Art	0	2	2
Business	1	2	3
Communication	2	1	3
Economics	2	0	2
English	1	4	5
History	1	2	3
Philosophy	1	0	1
Political Science	1	0	1
Psychology	1	4	5

TOTAL	10	15	25

b. Frequency Distribution

The following table shows the relative frequencies for the three leading causes of death for Americans aged 16–24 years and 25–34 years.

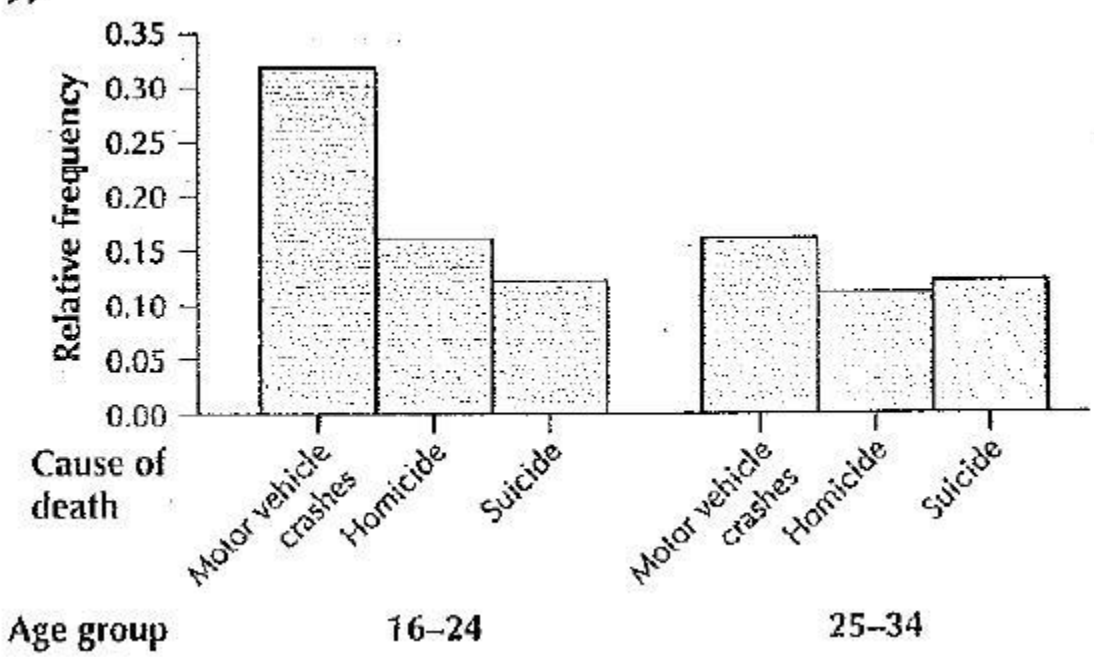
Age group	Cause of death	Percentage of all deaths for age group
16–24	Motor vehicle crashes	0.32
16–24	Homicide	0.16
16–24	Suicide	0.12
25–34	Motor vehicle crashes	0.16
25–34	Suicide	0.12
25–34	Homicide	0.11

- **Clustered Bar Graph**

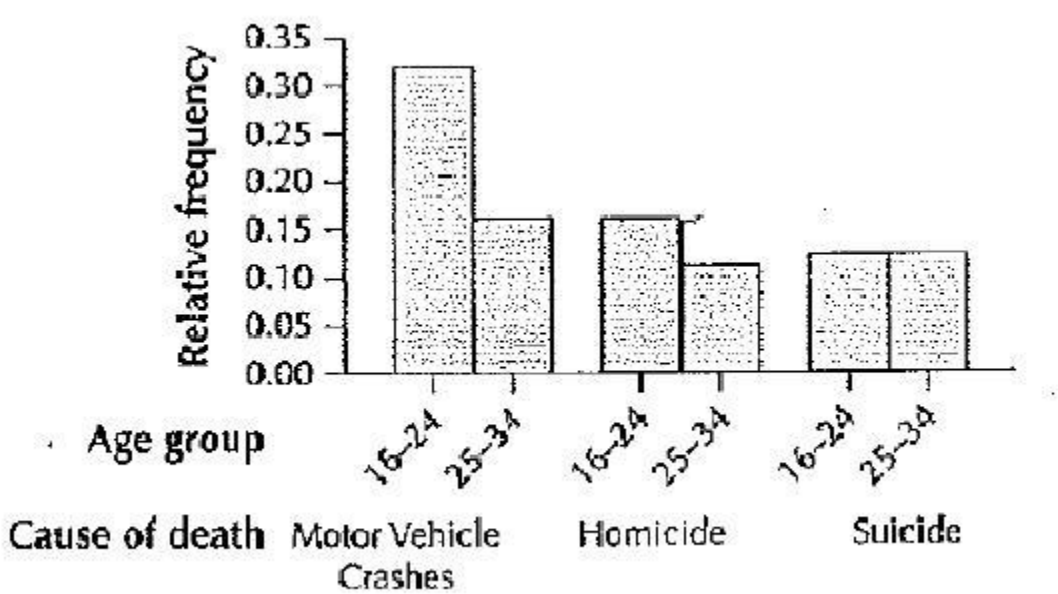
Example of Clustered Bar Graph

Construct a clustered bar graph of the cause of death, clustered by age group.

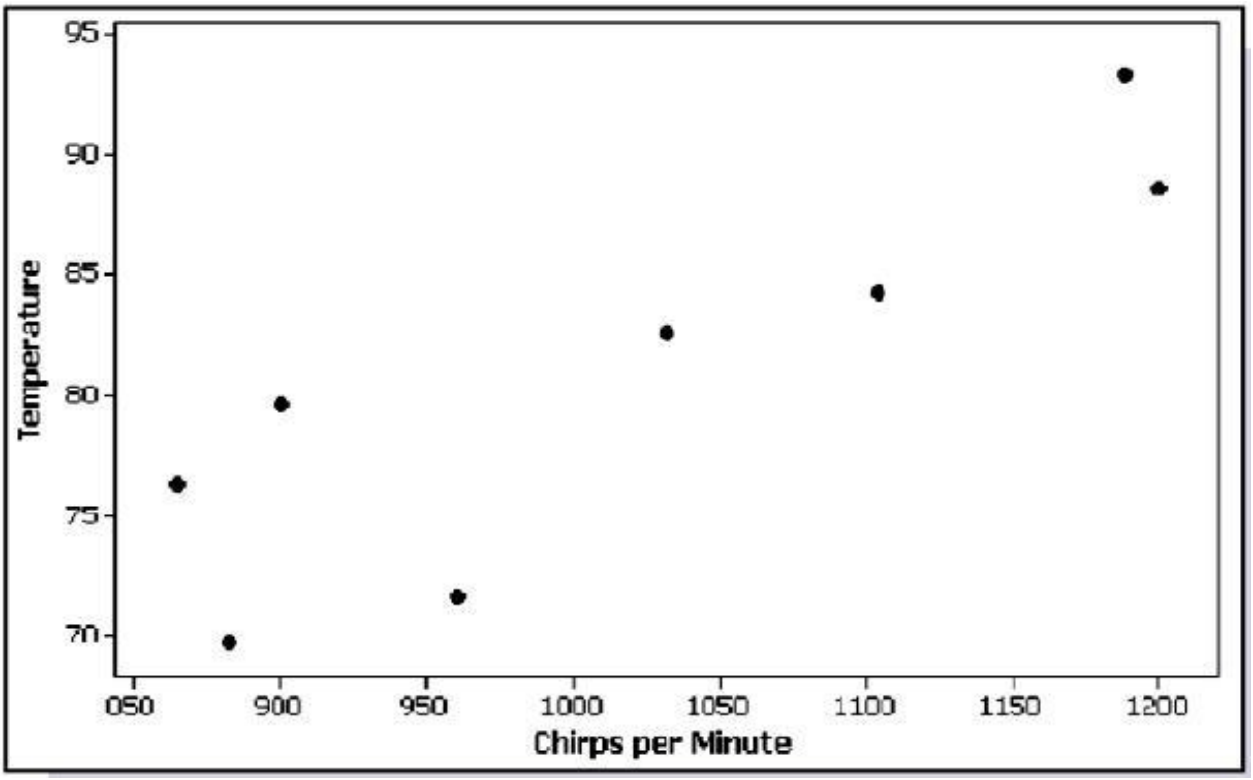
9.



10.



Example of Clustered Bar Graph



- **Scatter Plot**

A plot of paired (x,y) data with a horizontal x-axisand a vertical y-axis. Used to determine whether there is a relationship between the two variables

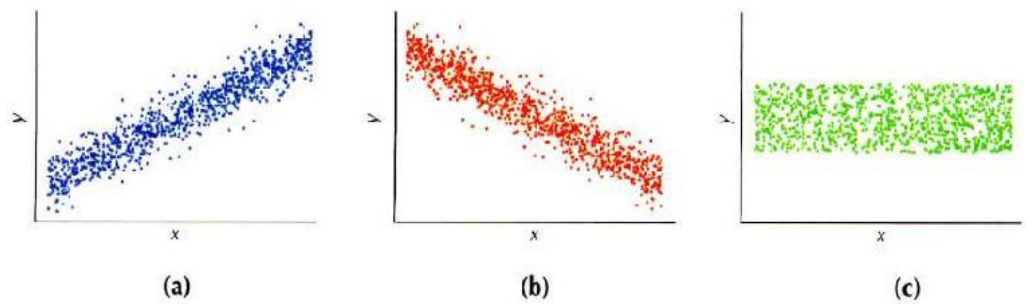


FIGURE 4.3 (a) Scatterplot of a positive relationship; (b) scatterplot of a negative relationship; (c) scatterplot of no apparent relationship.

Scatter plots of Paired Data

Example: Construct a Scatter Plot

Construct a scatterplot

x	y
10	7.46
8	6.77
13	12.74
9	7.11

11	7.81
14	8.84
6	6.08
4	5.39
12	8.15
7	6.42
5	5.73

Scatter plot with Graphing Calculator

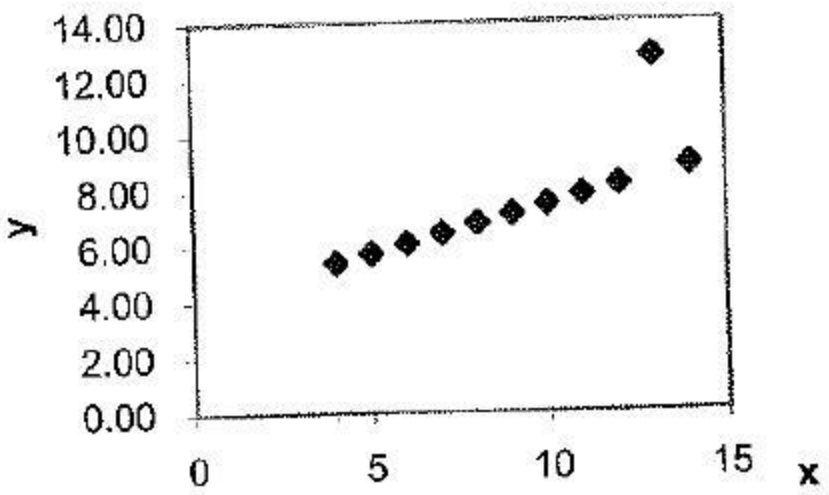
•Enter the X data values in two lists.

•Press 2nd STATPLOT and choose #1 PLOT 1. Be sure the plot is ON, the scatter plot icon is highlighted (top row, first icon)

•Enter the list of the X data values next to Xlist, and the list of the Y data values next to Ylist.

•Choose any of the three marks. Press

ZOOM and #9 ZoomStat.



Scatter plot with Graphing Calculator

Note the outlier at x=13

RELATION BETWEEN FREQUENCY DISTRIBUTIONS AND OTHER GRAPHS

FREQUENCY DISTRIBUTIONS AND THEIR GRAPHS

The main characteristics we will use to describe a data set are its center, its variability, and its shape. One way to see patterns in data is to make a graph. In this section, we will look at 3 ways to graphically summarize data: frequency distributions, frequency histograms, and a cumulative frequency graph.

Frequency distribution

A frequency distribution is a table that shows \classes" or \intervals" of data entries with a count of the number of entries in each class. The frequency f of a class is the number of data entries in the class. Each class will have a \lower class limit" and an \upper class limit" which are the lowest and highest numbers in each class. The \class width" is the distance between the lower limits of consecutive classes. The range is the di erence between the maximum and minimum data entries.

Steps for constructing a frequency distribution from a data set

1.If the number of classes is not given, decide on a number of classes to use. This number should be between 5 and 20.

2.**Find the class width:** Determine the range of the data and divide this by the number of classes. Round up to the next convenient number (if it's a whole number, also round up to the next whole number).

3.**Find the class limits:** You can use the minimum data entry as the lower limit of the rst class. To get the lower limit of the next class, add the class width. Continue until you reach the last class. Then nd the upper limits of each class (since the classes cannot overlap, and occasionally your data will include decimal numbers, remember that it's ne for the upper limits to be decimals).

4.Count the number of data entries for each class, and record the number in the row of the table for that class. (The book recommends using \tally" marks to count)

A frequency distribution is the organization of raw data in table from, using classes and frequency.

The number of miles that the employees of a large department store traveled to work each day

1	2	6	7	12	13	2	6	9	5
18	7	3	15	15	4	17	1	14	5
4	16	4	5	8	6	5	18	5	2
9	11	12	1	9	2	10	11	4	10
9	18	8	8	4	14	7	3	2	6

Class Limits (in miles)	**Frequency**
1-3	10
4-6	14
7-9	10
10-12	6
13-15	5
16-18	5

14

Total 50

How to construct a grouped frequency Distribution?

Number of classes
It should be between 5 and 20.

Some Statisticians use " 2^k "rule

k	1	2	3	4	5	6	7	8	9	10
2^k	2	4	8	16	32	64	128	256	512	1,024

2 to k rule

Essentially we would look to construct **k** classes for our frequency distribution, when the value of 2^k first exceeds the number of observations in our sample. So, if we had a sample with 39 observations, we would first consider constructing *6* classes, because 2^6= 64, the first power of 2 with a value larger than the sample size of 39.

A guide, not a dictator.

Strictly speaking the 2^k rule is a guide, not a rule. If the 2^k rule suggests you need 6 classes, also consider using 5 or 7 classes

... but certainly not 3 or 9.

- Class interval or class width

$$w \quad \frac{H \quad L}{k} \qquad >$$

H : the highest value, *L*: the smallest value

- Class interval can also be estimated based on # of observations

$$w = \frac{H - L}{1 \quad 3.322 \ln n}$$

Select the lower limit of the first class and set the limits of each class

It could be L or any value smaller than L.

It should be an even multiple of the class interval

- There should be between 5 and 20 classes.
- The classes must be continuous.
- The classes must be exhaustive.
- The classes must be mutually exclusive.
- The classes must be equal in width.

Relative frequency

- Relative frequency of a class is the frequency of that class divided by to total number of frequency.

$$RF = \frac{f}{n}$$

Example

These data represent the record high temperatures for each of the 50 states. Construct a grouped frequency distribution for the data using 7 classes.

112	100	127	120	134	118	105	110	109	112
110	118	117	116	118	122	114	114	105	109
107	112	114	115	118	117	118	122	106	110
116	108	110	121	113	120	119	111	104	111
120	113	120	117	105	110	118	112	114	114

Class limits	Class boundaries	Frequency	Relative frequency	Cumulative frequency
100-104	99.5-104.5	2	0.04	2
105-109	104.5-109.5	8	0.16	10
110-114	109.5-114.5	18	0.36	28
115-119	114.5-119.5	13	0.26	41
120-124	119.5-124.5	7	0.14	48
125-129	124.5-129.5	1	0.02	49
130-134	129.5-134.5	1	0.02	50

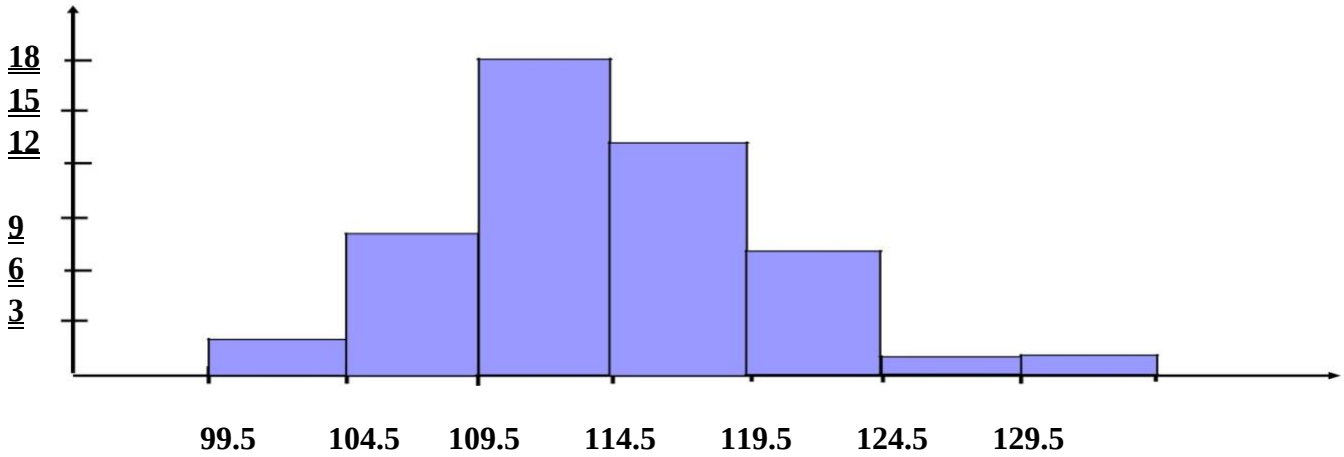
Histogram

A **histogram** is a graph that displays the data by using contiguous vertical bars (unless the frequency of a class is 0) of various heights to represent the frequencies of the classes

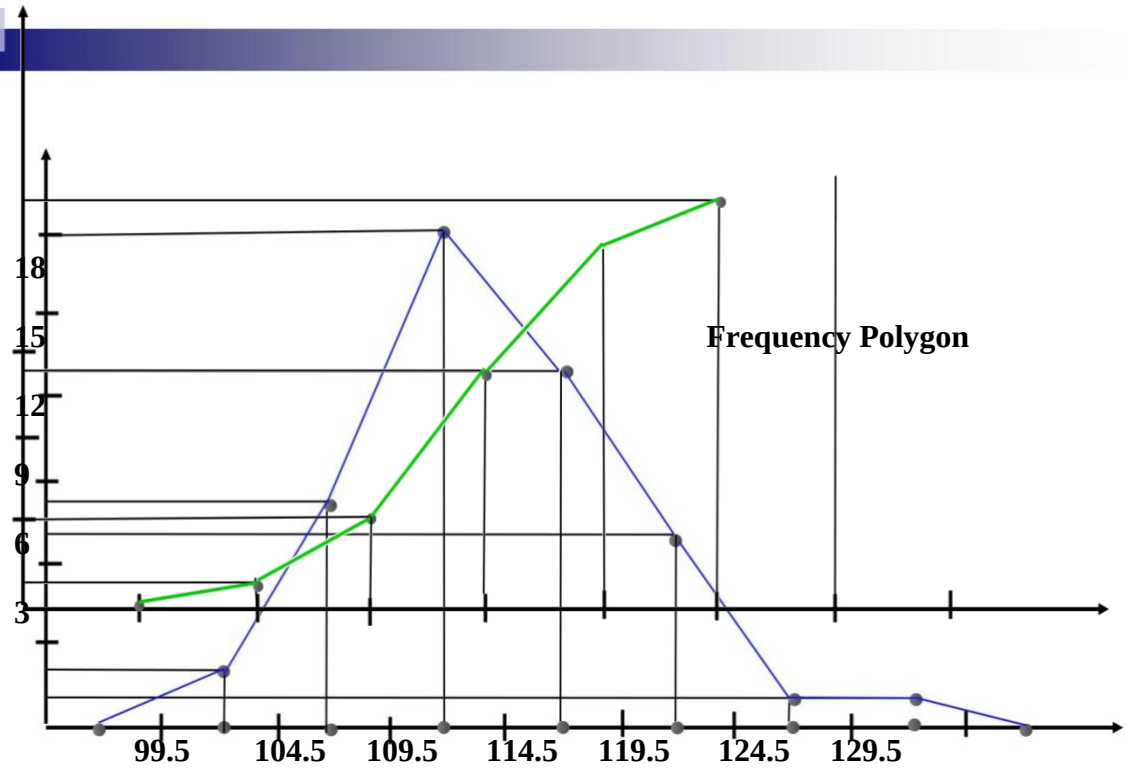
Example

Construct a histogram to represent the data shown below for the record high temperature:

Class boundaries	Frequency
99.5-104.5	2
104.5-109.5	8
109.5-114.5	18
114.5-119.5	13
119.5-124.5	7
124.5-129.5	1
129.5-134.5	1



- The largest concentration is in the class 109.5 – 114.5



The **Ogive** is a graph that represents the cumulative frequencies for the classes in a frequency distribution.

Class boundaries	Frequency	Cumulative Frequency
99.5-104.5	2	2
104.5-109.5	8	10
109.5-114.5	18	28
114.5-119.5	13	41
119.5-124.5	7	48
124.5-129.5	1	49
129.5-134.5	1	50

Cumulative Frequency Polygon

40

30

20

10

99.5 104.5 109.5 114.5 119.5 124.5 129.5

<u>Other types of Graphs</u>

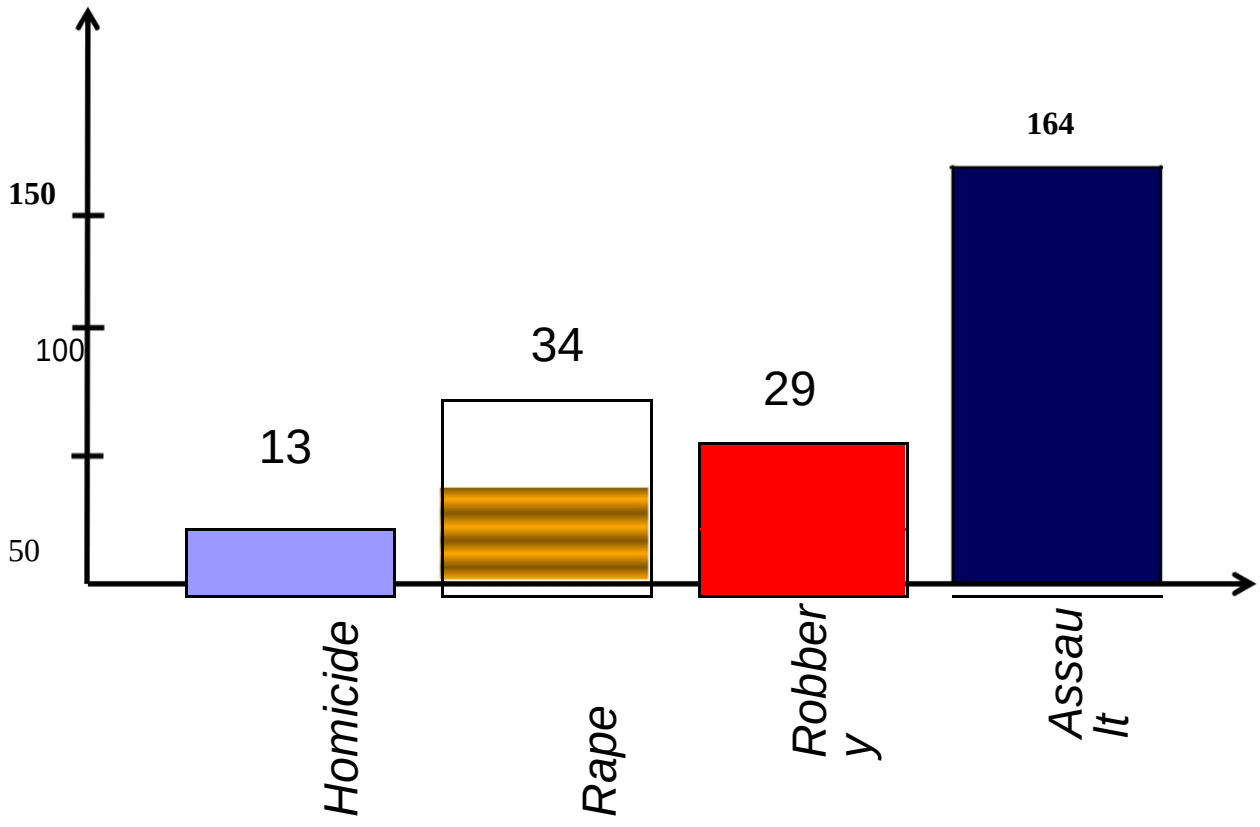
Bar Chart

Bar Chart is use to represent a frequencydistribution for a categorical variable, and the frequencies are displayed by the heights of vertical bars.

Example

The table shown here displays the number of crimes investigated by law enforcement officers in U.S. national parks during 1995. Construct a Bar chart for the data.

Type	**Number**
Homicide	13
Rape	34
Robbery	29
Assault	164



Total number of crime: 234

Pie Graph

A **pie graph** is a circle that is divided into sections or wedges according to the percentage of frequencies in each category of the distribution.

Example

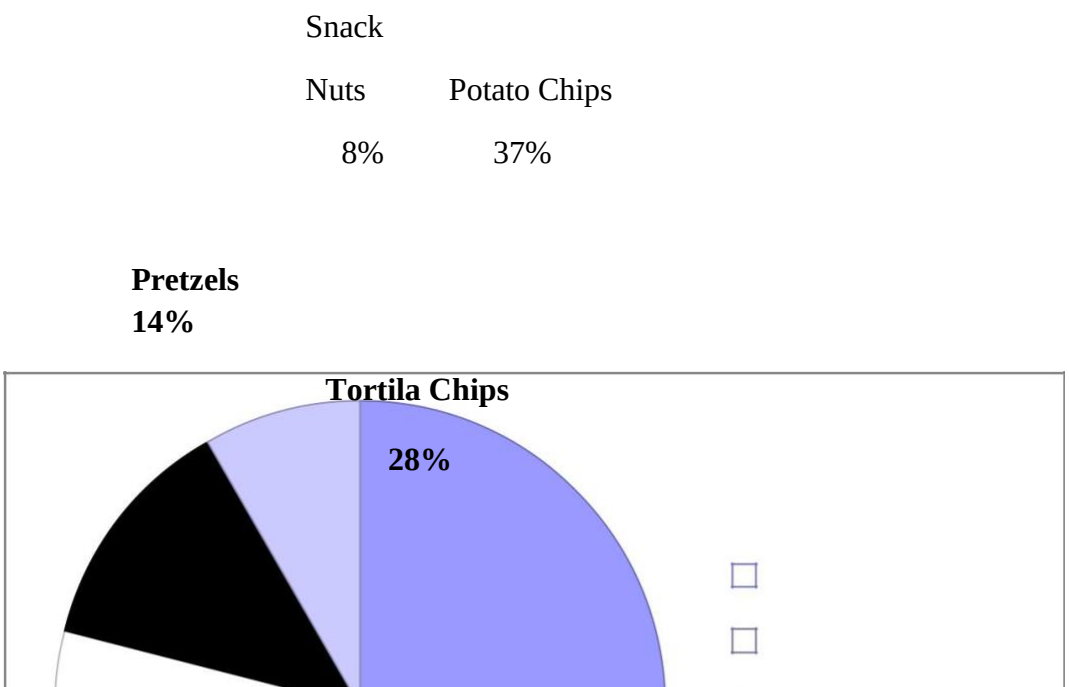
This frequency distribution shows the number of pounds of each snack food eaten during the 1998 Super Bowl. Construct a pie graph for the data.

We need to find percentages for each category and then compute the corresponding sectors so

Snack	Million pounds
Potato Chips	11.2
Tortilla Chips	8.2
Pretzels	4.3
Popcorn	3.8
Snack nuts	2.5

Snack	Million pounds	percentage	Degree
Potato Chips	11.2	37.33%	≈134º
Tortilla Chips	8.2	27.33%	≈98º
Pretzels	4.3	14.33%	≈41º
Popcorn	3.8	12.67%	≈46º
Snack nuts	2.5	8.33%	≈30º

that we divide the circle proportionally.



Stem and Leaf Plots

A **stem and leaf plot** is a data plot that uses part of the data value as the stem and part of the data value as the leaf to form groups or classes.

Example

At an outpatient testing center, the number of cardiograms performed each day for 20 days is shown. Construct a tem and leaf plot for the data.

25	31	20	32	13
14	43	02	57	23
36	32	33	32	44
32	52	44	51	45

It is helpful to arrange the data in order but it is not required.

02, 13, 14, 20, 23, 25, 31, 32, 32, 32, 32, 33, 36, 43, 44, 44, 45, 51, 52, 57

Leading digit(Stem)	Trailing digit (Leaf)
0	**2**
1	**3 4**
2	**0 3 5**
3	**1 2 2 2 2 3 6**
4	**3 4 4 5**
5	**1 2 7**

PREPARING DATA FOR ANALYSIS

Data Preparation and Analysis

Preparing Data

After data collection, the researcher must prepare the data to be analyzed. Organizing the data correctly can save a lot of time and prevent mistakes. Most researchers choose to use a database or statistical analysis program (e.g. Microsoft Excel, SPSS) that they can format to fit their needs and organize their data effectively. Once the data has been entered, it is crucial that the researcher check the data for accuracy. This can be accomplished by spot-checking a random assortment of participant data groups, but this method is not as effective as re-entering the data a second time and searching for discrepancies. This method is particularly easy to do when using numerical data because the researcher can simply use the database program to sum the columns of the spreadsheet and then look for differences in the totals. One of the best methods of checking for accuracy is to use a specialized computer program that cross-checks double-entered data for discrepancies.

Descriptive statistics describe but do not draw conclusions about the data. Each descriptive statistic summarizes multiple discrete data points using a single number. They can tell the researcher the central tendency of the variable, meaning the average score of a participant on a given study measure. The researcher can also determine the distribution of scores on a given study measure, or the range in which scores appear. Additionally, descriptive statistics can be used to tell the researcher the frequency with which certain responses or scores arise on a given study measure. For example, in the Module 1 example about the effectiveness of corrective lenses on economic productivity, the researcher might observe that the average dollars-per-week of a person with corrected vision is $500, whereas the average DPW for a person without corrected vision is $450. This amount of information is not enough information to conclude that vision correction affects economic productivity. Inferential statistics are necessary to draw conclusions of this kind. Descriptive statistics might also tell the researcher that the distribution of DPW is $351-$640 for the whole sample and that the average DPW is $445 for the sample.

Correlation

Correlation is one of the most often used (and most often misused) kinds of descriptive statistics. It is perhaps best described as “a single number that describes the degree of relationship between two variables.” If two variables tend to be “correlated,” that means that a participant’s score on one variable tends to vary with a score on the other. For example, people’s height and shoe size tend to be positively correlated. This means that for the most part, if a person is tall, they are likely to have a large shoe size, and conversely, if they are short, they are likely to have a smaller shoe

size. Correlation can also be negative. For example, warmer temperatures outside may be negatively correlated with the number of hot chocolates sold at a local coffee shop. This is to say that as the temperature goes up, hot chocolate sales tend to go down. Although causality may seem to be implied in this situation, it is important to note that on a statistical level, correlation does not imply causation. A good researcher knows that there is no way to assess from correlation alone that a causal relationship exists between two variables. In order to assert that "X caused Y," a study should be experimental, with control groups and random sampling procedures. Determining causation is a difficult thing to do, and it is a common mistake to assert a cause-and-effect relationship when the study methodology does not support this assertion.

Inferential Statistics

Inferential statistics allow the researcher to begin making inferences about the hypothesis based on the data collected. This means that, while applying inferential statistics to data, the researcher is coming to conclusions about the population at large. Inferential statistics seek to generalize beyond the data in the study to find patterns that ostensibly exist in the target population. This course will not address the specific types of inferential statistics available to the researcher, but a succinct and very useful summary of them, complete with step-by-step examples and helpful descriptions, is available here.

Statistical Significance

Researchers cannot simply conclude that there is a difference between two groups in a well-constructed study. This difference must be due to the manipulation of the independent variable. No matter how well a researcher designs the study, there always exists a degree of error in the results. This error may be due to individual differences within and between experimental groups, or the error may be due to systematic differences within the researcher's sample. Irrespective of its source, this error acts as "noise" in the data and affects participants' scores on study measures even though it is not the variable of interest. Statistical significance is aimed at determining the probability that the observed result of a study was due to the influence of something other than chance. A result is "statistically significant" at a certain level. For example, a result might be significant at $p<.05$. "P" represents the probability that the result was due to chance, and .05 represents a 5% probability that the result was due to chance. Therefore, in a well-run study, $p<.05$ means that inferential statistical analysis has indicated that the observed results have over a 95% probability of being due to the influence of the independent variable. The 5% cutoff is generally thought of as the standard for most scientific research. Note that it is theoretically impossible to ever be entirely certain that one's results are not due to chance, as the nature of science is one of observing trends and testing hypotheses, not immutable proof.

7 Steps to Prepare Data for Analysis

We researchers spend a lot of time interviewing our clients to determine their needs. Then we go about carefully creating a plan to collect the data that will be most useful. Having done that, the appropriate instrument is carefully crafted that will generate data that can ultimately be transformed into knowledge. All this up-front work necessitates and lot of time and effort. And well it should! But sooner or later we will have collected data and need to start the grunt work of data preparation.

So what is involved in data preparation? There are several simple, but sometimes overlooked steps, required to properly prepare data. They are:

Questionnaire checking: Questionnaire checking involves eliminating unacceptable questionnaires. These questionnaires may be incomplete, instructions not followed, little variance, missing pages, past cutoff date or respondent not qualified.

Editing: Editing looks to correct illegible, incomplete, inconsistent and ambiguous answers.

Coding: Coding typically assigns alpha or numeric codes to answers that do not already have them so that statistical techniques can be applied.

Transcribing: Transcribing data involves transferring data so as to make it accessible to people or applications for further processing.

Cleaning: Cleaning reviews data for consistencies. Inconsistencies may arise from faulty logic, out of range or extreme values.

Statistical adjustments: Statistical adjustments applies to data that requires weighting and scale transformations.

Analysis strategy selection: Finally, selection of a data analysis strategy is based on earlier work in designing the research project but is finalized after consideration of the characteristics of the data that has been gathered.

Not all of these steps occur in every market research study. But as situations dictate, none of these steps should be overlooked in the name of expediency or economy. Later articles will drill down in to the details of these important steps in data preparation.

UNIT IV

SOFT COMPUTING:

Computer and its role in Research:

Computer have always assisted to solve the problems faced by the mankind since the time of invention, the size of the computers have drastically reduced from that of a room to that can be accommodated in a human palm. The word computer means something which computes or a machine for performing calculations automatically, but, today computer means not merely a calculator. It does vast variety of jobs with tremendous speed and efficiency. Today people use computers in almost every walk of life. Electronic computers have now become an indispensable part of every profession: so do research. Computers have a very important role to play in all research activities.

WHY COMPUTERS?

The importance of computers in scientific research is exceptionally high and the use of a computer can help scientific research immensely, and is an almost invaluable tool [5]. There are many reasons why computers are so important in scientific research and here are some of the main reasons:

SPEED: computer can process numbers and information in a very short time. So researcher can process and analyze data quickly. By saving time researcher can conduct further research. A calculation that may take a person several hours to process will take computer mere minutes, if not seconds.

ACCURACY: Computer is incredibly accurate. Accuracy is very much important in scientific research. Wrong calculation could result an entire research or project being filled with incorrect information.

ORGANIZATION: We can store millions of pages of information by using simple folders, word processors & computer programs. Computer is more productive & safer than using a paper filing system in which anything can be easily misplaced.

CONSISTENCY: computer cannot make mistakes through “tiredness” or lack of concentration like human being. This characteristic makes it exceptionally important in scientific research.

COMPUTER IN THE RESEARCH PROCESS

Research process consists of series of actions or steps necessary to effectively carry out research and the desired sequencing of these steps. The following order concerning various steps provides a useful procedural guideline regarding the research process: (1) formulating the research problem; (2) extensive literature survey; (3) developing the hypothesis; (4) preparing the research design; (5) determining sample design; (6) collecting the data; (7) execution of the project; (8) analysis of data; (9) hypothesis testing; (10) generalisations and interpretation, and (11) preparation of the report or presentation of the results, i.e., formal write-up of conclusions reached.

There are five major phases of the research process . They are:

1. Conceptual phase
2. Design and planning phase
3. Data collection phase
4. Data Analysis phase and
5. Research Publication phase

1.Role of Computer in Conceptual Phase

The conceptual phase consists of formulation of research problem, extensive literature survey, theoretical frame work and developing the hypothesis.

Use of computers in extensive literature review: computers help for searching the literatures (for review of literature) and bibliographic reference stored in the electronic database of the world wide webs. It can thus be used for storing relevant published articles to the retrieved whenever needed. This has the advantage over searching the literatures in the form of books, journals and other newsletters at the libraries which consume considerable amount of time and effort.

2.Role of Computers in Design and Planning Phase

This phase consists of research design preparation and determining sample design. Design and planning phase also consists of population, research variables, sampling plan, reviewing research plan and pilot study.

Role of Computers for Sample Size Calculation: Several software's are available to calculate the sample size required for a proposed study. The standard deviation of the data from the pilot study is required for the sample size calculation.

3. Role of Computers in Data collection phase

This Empirical phase consists of collecting and preparing the data for analysis: In research studies, the preparation and inputting data is the most labor-intensive and time consuming aspect of the work. Typically the data will be initially recorded on a questionnaire or record for suitable for its acceptance by the computer. To do this the researcher in conjunction with the statistician and the

programmer, will convert the data into Microsoft word file or excel spreadsheet or any statistical software data file. These data can be directly opened with statistical software's for analysis.

Data collection and Storage: The data obtained from the subjects are stored in computes are word files or excel spread sheets or any statistical software data file. This has the advantage of making necessary corrections or editing the whole layout of the tables if needed, which is impossible or time consuming incase of writing in papers. Thus, computers help in data entry, data editing, data management including follow up actions etc. computers also allow for greater flexibility in recording the data while they are collected as well as greater ease during the analysis of these data. Examples of editors are WordPad, SPSS data editor, word processors, others like ultraedit etc.

Data exposition: Most researchers are anxious about seeing the data: what they look like; how they are distributed etc. you can also examine different dimension of variables or plot them in various charts using a statistical application.

4. Role of Computers in Data Analysis

This phase consist of the analysis of data, hypothesis testing and generalisations and interpretation. Data analysis phase mainly consist of statistical analysis of the data and interpretation of results.

Data analysis: many software's are now available to perform the mathematical part of the research process i.e. the calculations using various statistical methods.

Softwares like SPSS and spreadsheets are the widely used. They can be like calculating the sample size for a proposed study, hypothesis testing and calculating the power of the study. Familiarity with any one package will suffice to carry out the most intricate statistical analysis.

Computers are useful not only for statistical analysis, but also to monitor the accuracy and completeness of the data as they are collected. These software's also display the results in graphical char or graph form.

5. Role of Computer in Research Publication

This phase consists of preparation of the report or presentation of the results, i.e., formal write-up of conclusions reached. This is the research publication phase.

The research article, research paper, research thesis or research dissertation is typed in word processing software and converted to portable data format (PDF) and stored and/or published in the world wide web. Online sites are available through we can convert our word file into any format like html, pdf etc. Various online applications are also available for this purpose. Even we can prepare our document using online word processing software and can store/edit/access it from anywhere using internet.

ROLE OF COMPUTERS IN SCIENTIFIC RESEARCH

There are various computer applications used in scientific research. Some of the most important applications used in scientific research are data storage, data analysis, scientific simulations, instrumentation control and knowledge sharing.

Data Storage

Experimentation is the basis of scientific research. Every experiment in any of the natural sciences generates a lot of data that needs to be stored and analyzed to derive important conclusions, to validate or disprove hypotheses. Computers attached with experimental apparatuses, directly record data as it's generated and subject it to analysis through specially designed software. Data storage is possible in SPSS data file, lotus spreadsheet, excel spreadsheet, ASCII/DOS text file etc.

Data Analysis

Analyzing tons of statistical data is made possible using specially designed algorithms that are implemented by computers. This makes the extremely time-consuming job of data analysis to be a matter of a few minutes. In genetic engineering, computers have made the sequencing of the entire human genome possible. Data from different sources can be stored and accessed via computer networks set up in research labs, which makes collaboration simpler.

Scientific Simulations

One of the prime uses of computers in pure science and engineering projects is the running of simulations. A simulation is a mathematical modeling of a problem and a virtual study of its possible solutions. Problems which do not yield themselves to experimentation can be studied through simulations carried out on computers. For example, astrophysicists carry out structure formation simulations, which are aimed at studying how large-scale structures like galaxies are formed. Space missions to the Moon, satellite launches and interplanetary missions are first simulated on computers to determine the best path that can be taken by the launch vehicle and spacecraft to reach its destination safely.

Instrumentation Control

Most advanced scientific instruments come with their own on-board computer, which can be programmed to execute various functions. For example, the Hubble Space Craft has its own onboard computer system which is remotely programmed to probe the deep space. Instrumentation control is one of the most important applications of computers.

Knowledge Sharing Through Internet

Lastly, in the form of Internet, computers have provided an entirely new way to share knowledge. Today, anyone can access the latest research papers that are made available for free on websites. Sharing of knowledge and collaboration through the Internet, has made international cooperation on scientific projects possible. Through various kinds of analytical software programs, computers are contributing to scientific research in every discipline, ranging from biology to astrophysics, discovering new patterns and providing novel insights. When the work in neural network based artificial intelligence advances and computers are granted with the ability to learn and think for themselves, future advances in technology and research will be even more rapid.

USE OF SATTISTICAL SOFTWARE SPSS, GRETL etc IN RESEACH:

Statistical software are specialized computer programs for statistical analysis.

SPSS

SPSS, which stands for statistical package for the social sciences, is an application that can aid in quantitative data handling. Before SPSS, researchers had to run statistical tests on data sets by hand. However, SPSS automates this process. Not only does SPSS allow you to run statistical tests, you can use SPSS for other purposes as well.

- Provides Data view & variable view
- Measures of central tendency & dispersion
- Statistical inference
- Correlation & Regression analysis
- Analysis of variance
- Non parametric test
- Hypothesis tests: T-test, chi-square, z-test, ANOVA, Bipartite variable….
- Multivariate data analysis
- Frequency distribution
- Data exposition by using various graphs like line, scatter, bar, ogive, histogram, pie chart….

Data Collection and Organization

SPSS is often used as a data collection tool by researchers. The data entry screen in
SPSS looks much like any other spreadsheet software. You can enter variables and quantitative data and save the file as a data file. Furthermore, you can organize your data in SPSS by assigning properties to different variables. For example, you can designate a variable as a nominal variable, and that information is stored in SPSS. The next time you access the data file, which could be weeks, months or even years, you'll be able to see exactly how your data is organized.

Data Output

Once data is collected and entered into the data sheet in SPSS, you can create an output file from the data. For example, you can create frequency distributions of your data to determine whether your data set is normally distributed. The frequency distribution is displayed in an output file. You can export items from the output file and place them into a research article you're writing. Therefore, instead of recreating a table or graph, you can take the table or graph directly from the data output file from SPSS.

Statistical Tests

The most obvious use for SPSS is to use the software to run statistical tests. SPSS has all of the most widely used statistical tests built-in to the software. Therefore, you won't have to do any mathematical equations by hand. Once you run a statistical test, all associated outputs are displayed in the data output file. You can also transform your data by performing advanced statistical transformations. This is especially useful for data that is not normally distributed.

GRETL:

Gretl, short form for GNU Regression, Econometrics and Time-series Library, is an open-source statistical software package with the main focus being on statistical methods for econometric analyses.

It has a graphical user interface (GUI) and can be used together with TRAMO/SEATS, R, Stata, Python, Octave, Ox and Julia. It is written in C, uses GTK+ as widget toolkit for creating its GUI, and uses gnuplot for generating graphs. As a complement to the GUI it also has a command-line interface.

EVOLUTIONARY ALGORITHMS:

Evolutionary Algorithms (EA) consist of several heuristics, which are able to solve optimization tasks by imitating some aspects of natural evolution. They may use different levels of abstraction, but they are always working on whole populations of possible solutions for a given task. EAs are an approved set of heuristics, which are flexible to use and postulate only neglectible requirements on the optimization task.
As a practical application, technical trading rules found by the use of EA will be presented.

Genetic Algorithm:
For difficult non-linear functions GA has been shown to perform exceedingly well in obtaining global solutions. Basically, an objective function, such as minimization of the sum of squared errors or sum of absolute errors, is chosen for optimizing the neural network. Using the chosen objective function, each candidate point out of the initial population of randomly chosen starting points is used to evaluate the objective function. These values are then used in assigning probabilities for each of the points in the population. For minimization, as in the case of sum of squared errors, the highest probability is assigned to the point with the lowest objective function value. Once all points have been assigned a probability, a new population of points is drawn from the original population with replacement. The points are chosen randomly with the probability of selection equal to its assigned probability value. Thus, those points generating the lowest sum of squared errors are the most likely to be represented in the new population. The points comprising this new population are then randomly paired for the crossover operation. Each point is a vector (string) of n parameters (weights). A position along the vectors is randomly selected for each pair of points and the preceding parameters are switched between the two points. This crossover operation results in each new point having parameters from both parent points. Finally, each weight has a small probability of being replaced with a value randomly chosen from the parameter space. This operation is referred to as mutation. Mutation enhances the GA by intermittently injecting a random point in order to better search the entire parameter space. This allows the GA to possibly escape from local optima if the new point generated is a better solution than has previously been found, thus providing a more robust solution. This resulting set of points now becomes the new population, and the process repeats until convergence.

Since this method simultaneously searches in many directions, the probability of finding a global

optimum greatly increases. The algorithm's similarity to natural selection inspires its name. As the GA progresses through generations, the parameters most favorable for optimizing the objective function will reproduce and thrive in future generations, while poorly performing parameters die out, as in "survival of the fittest".

SIMULATED ANNEALING:

Annealing is referred to as tempering certain alloys of metal, glass, or crystal by heating above its melting point, holding its temperature, and then cooling it very slowly until it solidifies into a perfect crystalline structure. This physical/chemical process produces high-quality materials. The simulation of this process is known as simulated annealing . The defect-free crystal state corresponds to the global minimum energy configuration. There is an analogy of SA with an optimization procedure. The physical material states correspond to problem solutions, the energy of a state to cost of a solution, and the temperature to a control parameter.The Metropolis algorithm is a simple method for simulating the evolution to the thermal equilibrium of a solid for a given temperature is a variant of the Metropolis algorithm, where the temperature is changing from high to low. SA is basically composed of two stochastic processes: one process for the generation of solutions and the other for the acceptance of solutions. The generation temperature is responsible for the correlation between generated probing solutions and the original solution. SA is a descent algorithm modified by random ascent moves in order to escape
local minima which are not global minima. The annealing algorithm simulates a nonstationary finite state Markov chain whose state space is the domain of the cost function to be minimized. Importance sampling is the main principle that underlies SA. It has been used in statistical physics to choose sample states of a particle system model to efficiently estimate some physical quantities. Importance sampling probabilistically favors states with lower energies. SA is a general- purpose, serial algorithm for finding a global minimum for a continuous function. It is also a popular Monte Carlo algorithm for any optimization problem including COPs. The solutions by this technique are close to the global minimum within a polynomial upper bound for the computational time and are independent of the initial conditions. Some parallel algorithms for SA have been proposed aiming to improve the accuracy of the solutions by applying parallelism .

Basic Stimulated Annealing:

According to statistical thermodynamics, P α. the probability of a physical system being in state a with energy E α at temperature T satisfies the Boltzmann distribution1 where KB is the Boltzmann's constant, T is the absolute temperature, and Z is the partition function, defined by the summation being taken over all states with energy $E\beta$ at temperature T. At high T, the Boltzmann distribution exhibits uniform preference for all the states, regardless of the energy. When T approaches zero, only the states with minimum energy haven on zero probability of occurrence.
In SA, the constant KB is omitted. At high T, the system ignores small changes in the energy and approaches thermal equilibrium rapidly, that is, it performs a coarse search of the space of global states and finds a good minimum. As T is lowered,
the system responds to small changes in the energy, and performs a fine search in the neighborhood of the already determined minimum and finds a better minimum. At T = 0, any change in the system states does not lead to an increase in the energy, and thus, the system must reach equilibrium if T = 0.

When performing SA, theoretically a global minimum is guaranteed to be reached with high probability. The artificial thermal noise is gradually decreased in time. T is a control parameter called computational temperature, which controls the magnitude of the erturbations of the energy function E(x). The probability of a state change is determined by the Boltzmann distribution of the energy difference of the two states:

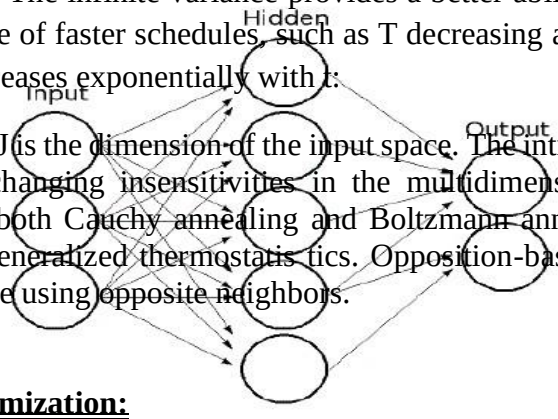
SA can be performed as Algorithm

I. Initialize the system configuration.
Randomize x(O).
2. Initialize T with a large value.
3. Repeat:
a. Repeat:
i. Apply random perturbations to the state x = x + Ax.
ii. Evaluate AE(x) = E(x + Ax) — E(x):
if AE(x) <0, keep the new state;
otherwise, accept the new state with probability P =
until the number of accepted transitions is below a threshold level.
b. SetT=T-AT.
until T is small enough.

The basic SA procedure is known as Boltzmann annealing. The cooling schedule for T is critical to the efficiency of SA. If T is reduced too rapidly, a premature convergence to a local minimum may occur. In contrast, if it is too slow, the algorithm is very slow to converge. Based on a Markov-chain analysis on the SA process, Geman and Geman have proved that a simple necessary and sufficient condition on the cooling schedule for the algorithm state to converge in probability to the set of globally minimum cost states is that T must be decreased according to ensure convergence to the global minimum with probability one, where T0 is a sufficiently large initial temperature. Given a sufficiently large number of iterations at each temperature, SA is proved to converge almost surely to the global optimum ,T0 is proved to be greater than or equal to the depth of the deepest local minimum which is not a global minimum state. In order to guarantee Boltzmann annealing to converge to the global minimum with probability one, T(t)needs to decrease logarithmically with time. This is practically too slow. In practice, one usually applies, in Step 3b, a fast schedule
T(t) = α T (t−1) with 0.85 ≤ α ≤0.96, to achieve a suboptimal solution.
However, due to its Monte Carlo nature, SA would require for some problems even more iterations than complete enumeration in order to guarantee convergence to an exact solution. For example, for an n-city TSP, SA using the logarithmic cooling $2n-1$ schedule needs a computational complexity of O(nn), which is far more than O((n−1)!)for complete enumeration and O(n22n)for dynamic programming .Thus, one has to apply heuristic fast cooling schedules to improve the convergence speed.

Variants of Simulated Annealing:

Standard SA is a stochastic search method, and the convergence to the global optimum is too slow for a reliable cooling schedule. Many methods, such as Cauchy annealing , simulated reannealing , generalized SA and SA with known global value have been proposed to accelerate SA search. There are also global optimization methods that make use of the idea of annealing .Cauchy annealing replaces the Boltzmann distribution with the Cauchy distribution, also known as the Cauchy–Lorentz distribution. The infinite variance provides a better ability to escape from local minima and allows for the use of faster schedules, such as T decreasing according to T(t)=T0t.In simulated reannealing ,T decreases exponentially with t:



where the constant c1>0, and J is the dimension of the input space. The introduction of reannealing also permits adaptation to changing insensitivities in the multidimensional parameter space. Generalized SA generalizes both Cauchy annealing and Boltzmann annealing within a unified framework inspired by the generalized thermostatis tics. Opposition-based SA improves SA in accuracy and convergence rate using opposite neighbors.

<u>Neural Network based Optimization:</u>

In machine learning,artificial neural networks (ANNs) are a family of models inspired by biological neural networks and are used to approximate functions that can depend on a large number of inputs and are generally unknown. ANNs are presented as systems of interconnected "neurons" which exchange messages between each other. The connections have weights that can be tuned, making neural nets adaptive to inputs and capable of learning. The ability of the ANNsto accurately approximate unknown functions, underlies this growing popularity. Neural networks have been used to solve a wide variety of tasks that are hard to solve using Ordinary rule-based programming, including computer vision and speech recognition, Character Recognition, Image Compression, Stock Market Prediction, Traveling Saleman's Problem, and Medicine, Electronic Nose, Security, and Loan Applications. The word network in the term 'artificial neural network' refers to the inter– connections between the neurons in the different layers of each system. An example system has three layers .The first layer has input neurons which send data via synapses to the second layer of neurons (hidden layer), and then via more synapses to the third layer of output neurons. More complex systems will have more layers of neurons, some having increased layers of input neurons and output neurons. The synapses store parameters called "weights" that manipulate the data in the calculations.

An ANN is typically defined by three types of parameters:

1. Architecture(The interconnection pattern between the different layers of neurons)
2. The learning process for updating the weights of the interconnections
3. The activation function that converts a neuron's weighted input to its output activation.

Mathematically, a neuron's network function f(x) is defined as a composition of other functions gi(x) , which can further be defined as a composition of other functions. This can be conveniently represented as a network structure, with arrows depicting the dependencies between variables. A widely used type of composition is the nonlinear weighted sum,

ANN with one hidden layer

Where f(x) = K($\Sigma i\ \omega i\ gi\ (x)$), where K (commonly referred to as the activation function, is some predefined function, such as the hyperbolic tangent. It will be convenient for the following to refer to a collection of functions as simply a vector g = (g1,g2,……,gn).

What has attracted the most interest in neural networks is the possibility of learning. Given a specific task to solve, and a class of functions F, learning means using a set of observations to find f* ϵ F which solves the task in some optimal sense.This entails defining a cost function (objective function) C : F → R such that, for the optimal solution f*, $c\ (f\ *\ ) \leq c(f) \forall f \epsilon F$ i.e., no solution has a cost less than the cost of the optimal.

Artificial Neural Network Optimization:

Generally when we talk about ANN optimization, the objective function is mean square error function (loss/cost function). We have to find optimize values weights of neural network to minimize the objective function. Although, gradient based search techniques such as back-propagation are currently the most widely used optimization techniques for training neural networks, it has been shown that these gradient techniques are severely limited in their ability to find global solutions. Global search techniques have been identified as a potential solution to this problem. Other techniques can be used. Two well-known global search techniques, Simulated Annealing and the Genetic Algorithm can be used as well.

Because of its ease of use, an overwhelming majority of these applications have used some variation of the gradient technique, back propagation (BP) [5,6] for optimizing the networks. Although, back propagation has unquestionably been a major factor for the success of past neural network applications, it is plagued with inconsistent and inpredictable performances.

Recently, demonstrated that for a variety of complex functions the genetic algorithm was able to achieve superior solutions for neural network optimization than back propagation and found that

another global search heuristic, Tabu Search (TS), is able to systematically achieve superior solutions for optimizing the neural network than those achieved by back propagation. In addition to GA and TS another well-known global search heuristic is simulated annealing. Simulated annealing has been shown to perform well for optimizing a wide variety of complex problems. Authors in compared BP, SA, and GA for ANN. They concluded that although the most frequently used algorithm for optimizing neural networks is back propagation, it is likely to obtain local solutions. They showed that simulated annealing, a global search algorithm, performs better than back propagation, but is also uses a point to point search. Both BP and SA have multiple user determined parameters which may significantly impact the solution. Since there are no established rules for selecting these parameters, solution outcome is based on chance. The genetic algorithm appears to be able to systematically obtain superior solutions to simulated annealing for optimizing neural networks. These solutions provide the researcher with superior estimates of interpolation data. The genetic algorithm's process of moving from one population of points to another enables it to discard potential local solutions and also to achieve the superior solutions in a computationally more efficient manner.

OPTIMIZATION OF FUZZY SYSTEMS:

Introduction:

Large, small, pure, polluted, satisfactory, unsatisfactory, sufficient, insufficient, excellent, good, fair, poor and soon are words often used to describe various attributes or performance measures of water resources systems. These descriptors do not have 'crisp', well-defined boundaries that separate them from others. A particular mix of economic and environmental impacts may be more acceptable to some and less acceptable to others. Plan A is better than Plan B. The water quality and temperature is Good for swimming. These qualitative, or 'fuzzy', statements convey information despite the imprecision of the italicized adjectives. This chapter illustrates how fuzzy descriptors can be incorporated into optimization models of water resources systems. Before this can be done some definitions are needed.

Fuzzy Membership Functions:

Consider a set A of real or integer numbers ranging from say 18 to 25. Thus A [18, 25]. In classical (crisp) set theory, any number x is either in or not in the set A . The statement 'x belongs to A' is either true or false depending on the value of x. The set A is referred to as a crisp set.If one is not able to say for certain whether or not any number x is in the set, then the set A could be referred to as fuzzy. The degree of truth attached to that statement is defined by a membership function. This function ranges from 0 (completely false) to 1 (completely true).

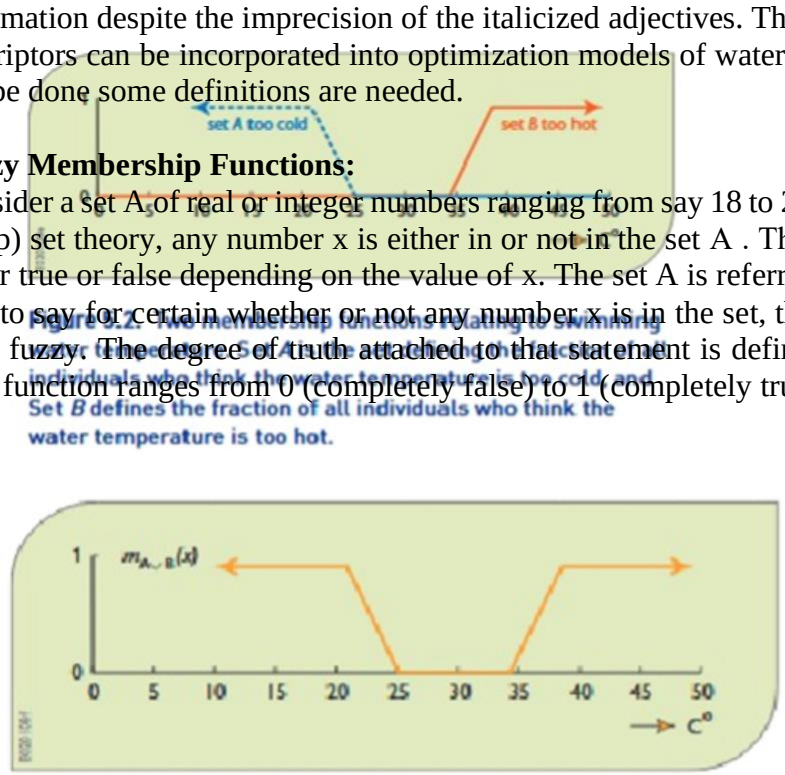


Figure 5.3. Membership function for water temperatures that are considered too cold or too hot.

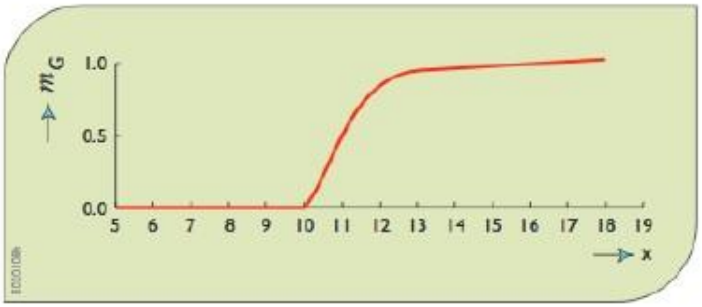


Figure 5.5. Membership function defining the fraction of individuals who think a particular value of *x* is 'substantially' greater than 10.

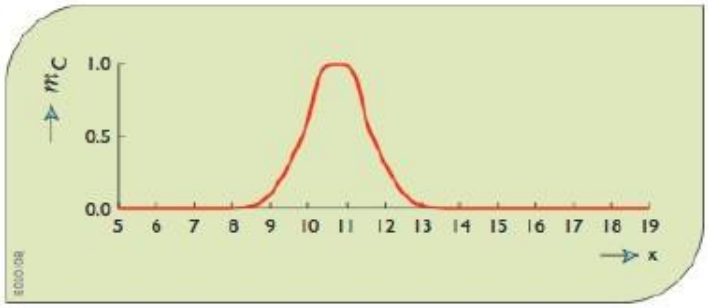


Figure 5.6. Membership function representing the vicinity of 11.

Consider the statement, 'The water temperature should be suitable for swimming'. Just what temperatures are suitable will depend on the person asked. It would be difficult for anyone to define precisely those temperatures that are suitable if it is understood that temperatures outside that range are absolutely not suitable.

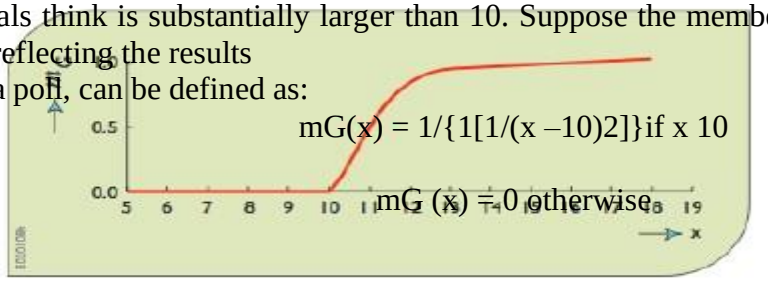
Optimization in Fuzzy Environments:

Consider the problem of finding the maximum value of x given that x cannot exceed 11. This is written as: Maximize U = x

subject to: $x \leq 11$(5)
The obvious optimal solution, x = 11, is shown in Figure 5.4.

Now suppose the objective is to obtain a value of x substantially larger than 10 while making sure that the maximum value of x should be in the vicinity of 11. This is no longer a crisp optimization problem; rather, it is a fuzzy one. What is perceived to be substantially larger than 10 could be defined by a membership function, again representing the results of an opinion poll of what

individuals think is substantially larger than 10. Suppose the membership function for this goal, mG(x), reflecting the results
of such a poll, can be defined as:



$$mG(x) = 1/\{1[1/(x - 10)2]\} \text{ if } x\ 10$$

$$mG(x) = 0 \text{ otherwise}$$

This function is shown in Figure 5.5.

Figure 5.5. Membership function defining the fraction of individuals who think a particular value of x is 'substantially' greater than 10.

The constraint on x is that it 'should be in the vicinity of 11'. Suppose the results of a poll asking individuals to state what they consider to be in the vicinity of 11 results in the following constraint membership function, mC(x):

$$mC(x) = 1/[1\ (x - 11)4](7)$$

This membership function is shown in Figure 5.6.

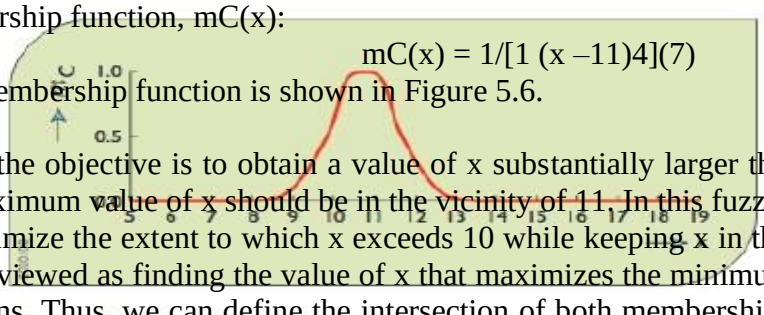


Figure 5.6. Membership function representing the vicinity of 11.

Recall the objective is to obtain a value of x substantially larger than 10 while making sure that the maximum value of x should be in the vicinity of 11. In this fuzzy environment the objective is to maximize the extent to which x exceeds 10 while keeping x in the vicinity of 11. The solution can be viewed as finding the value of x that maximizes the minimum values of both membership functions. Thus, we can define the intersection of both membership functions and find the value of x that maximizes that intersection membership function.

The intersection membership function is:

$$\begin{aligned} mD(x) &= \text{minimum}\{mG(x)\ ,\ mC(x)\} \\ &= \text{minimum}\{1/(1 + [1/(x-10)2]), \\ &\quad 1/(1 +(x-11)4)\} \text{ if } x>10 \\ &= 0 \text{ otherwise}(8) \end{aligned}$$

This intersection set, and the value of x that maximizes its value, is shown in Figure 5.7.

This fuzzy decision is the value of x that maximizes the intersection membership function mD(x), or equivalently

Maximize mD(x) = max min{mG(x), mC(x)}

Using LINGO®, the optimal solution is x = 11.75 and mD(x) = 0.755.

Unit V

STRUCTURE OF RESEARCH REPORT

"Research report is a research document that contains basic aspects of the research project".

Introduction:

Mostly, research work is presented in a written form. The practical utility of research study depends heavily on the way it is presented to those who are expected to act on the basis of research findings. Research report is a written document containing key aspects of research project.

Research report is a medium to communicate research work with relevant people. It is also a good source of preservation of research work for the future reference. Many times, research findings are not followed because of improper presentation. Preparation of research report is not an easy task. It is an art. It requires a good deal of knowledge, imagination, experience, and expertise. It demands a considerable time and money.

Research report involves relevant information on the research work carried out. It may be in form of hand-written, typed, or computerized.

Report Format:

There is no one best format for all reports. Format depends on several relevant variables. One must employ a suitable format to create desirable impression with clarity. Report must be attractive. It should be written systematically and bound carefully. A report must use the format (often called structure) that best fit the needs and wants of its readers. Normally, following format is suggested as a basic outline, which has sufficient flexibly to meet the most situations.

Research report is divided into three parts as:

I. First Part (Formality Part):

(i) Cover page

(ii) Title page

(iii) Certificate or statement

(iv) Index (brief contents)

(v) Table of contents (detailed index)

(vi) Acknowledgement

(vii) List of tables and figures used

(viii) Preface/forwarding/introduction

(ix) Summary report

II. Main Report (Central Part of Report):

(i) Statement of objectives

(ii) Methodology and research design

(iii) Types of data and its sources

(iv) Sampling decisions

(v) Data collection methods

(vi) Data collection tools

(vii) Fieldwork

(viii) Analysis and interpretation (including tables, charts, figures, etc.)

(ix) Findings

(x) Limitations

(xi) Conclusions and recommendations

(xii) Any other relevant detail

III. Appendix (Additional Details):

(i) Copies of forms used

(ii) Tables not included in findings

(iii) A copy of questionnaire

(iv) Detail of sampling and rate of response

(v) Statement of expenses

(vi) Bibliography – list of books, magazines, journals, and other reports

(vii) Any other relevant information

Key Considerations/Factors:

While preparing research report, following issues must be considered:

(i) Objectives

(ii) Type of problem/subject

(iii) Nature and type of research

(iv) Audience or users of research work

(v) Size of report

(vi) Form of writing – handwritten, typed, or computerized.

(vii) Time and cost

(viii) Language

(ix) Contents of report

(x) Order of contents

(xi) Number of copies

(xii) Format – type and size of paper; lengths width, and depth of report; and pattern of writing including paragraph, indent, numbering, font size and type, colouring, etc.

(xiii) Binding (for soft, and, particularly, for hard copy) – type, quality of material, colour, etc., related issues.

The Importance of Research

The purpose of research is to inform action. Thus, your study should seek to contextualize its findings within the larger body of research. Research must always be of high quality in order to produce knowledge that is applicable outside of theresearch setting.

Study Implications

The purpose of research is to inform action. Thus, your study should seek to contextualize its findings within the larger body of research. Research must always be of high quality in order to produce knowledge that is applicable outside of the research setting. Furthermore, the results of your study may have implications for policy and future project implementation.

One problem that often plagues progress in global health is the slow translation of research into practice. Often, a disconnect exists between those who conduct research and those who are positioned to implement the research findings. The underlying problem is that "the production of evidence is organized institutionally with highly centralized mechanisms, whereas the application of that science is highly decentralized. This social distance prevails because scientists are more oriented to the international audiences of other scientists for which they publish than to the needs of practitioners, policy makers, or the local public."

Thus, as researchers, it is imperative to take steps to overcome this barrier. Publishing your study may be one initial step to make your research known to the global community. Other proactive measures can be taken to encourage the uptake of evidence-based interventions. For example, you

can present your research findings at various venues, such as Unite for Sight's Global Health and Innovation Conference. Furthermore, you can send the results of your study to local officials, policy-makers, and community leaders.

Goals of Research

There are relatively few published studies about eye care in developing countries, and Unite For Sight encourages all volunteers to consider developing a research study to contribute important knowledge to the eye care community on a global scale. Pursuing a research project will be a challenging and rewarding experience, and this opportunity enables you to pursue an in-depth original study about a topic of interest.

Well-conducted research is vital to the success of global heath endeavors. Not only does research form the foundation of program development and policies all over the world, but it can also be translated into effective global health programs. Research draws its power from the fact that it is empirical: rather than merely theorizing about what might be effective or what could work, researchers go out into the field and design studies that give policymakers hard data on which they can base their decisions. Furthermore, good research utilizes methodologies that can be replicated, produces results that are examinable by peers, and creates knowledge that can be applied to real-world situations. Researchers work as a team to enhance our knowledge of how to best address the world's problems.

The "Iterative" Process of Research

Ultimately, the key to a successful research project lies in iteration: the process of returning again and again to the research questions, methods, and data, which leads to new ideas, revisions and improvements. It is easy to think of research as a step-by-step "1, 2, 3" process, but it is important to be flexible and open to change. Oftentimes, by discussing the research project with advisers and peers, one will find that new research questions need to be added, variables need to be omitted, and other changes need to be made. As a proposed study is examined and reexamined from different perspectives, it may begin to transform and take a different shape. This is to be expected and is a component of a good research study. In addition, it is important to examine study methods and data from different viewpoints to ensure a comprehensive approach to the research question. In conclusion, there is no one formula for developing a successful study, but it is important to realize that the research process is cyclical and iterative.

COMPONENTS OF RESEARCH REPORT

The six components of a research report are as follows: An abstract, introduction, methodology, results, discussion, and references.

The Abstract

The abstract is an overview of the research study and is typically two to four paragraphs in length. Think of it as an executive summary that distills the key elements of the remaining sections into a few sentences.

Introduction

The introduction provides the key question that the researcher is attempting to answer and a review of any literature that is relevant. In addition, the researcher will provide a rationale for why the research is important and will present a hypothesis that attempts to answer the key question. Lastly, the introduction should summarize the state of the key question following the completion of the research. For example, are there any important issues or questions still open?

Methodology

The methodology section of the research report is arguably the most important for two reasons. First it allows readers to evaluate the quality of the research and second, it provides the details by which another researcher may replicate and validate the findings.

Typically the information in the methodology section is arranged in chronological order with the most important information at the top of each section.

Ideally the description of the methodology doesn't force you to refer to other documents; however if the author is relying on existing methods, they will be referenced.

Results

In longer research papers, the results section contains the data and perhaps a short introduction. Typically the interpretation of the data and the analysis is reserved for the discussion section.

Discussion

The discussion section is where the results of the study are interpreted and evaluated against the existing body or research literature. In addition, should there be any anomalies found in the results, this is where the authors will point them out. Lastly the discussion section will attempt to connect the results to the bigger picture and show how the results might be applied.

References

This section provides a list of each author and paper cited in the research report. Any fact, idea, or direct quotation used in the report should be cited and referenced.

<u>Types of Research Studies</u>

Research can be classified into two categories: Basic research, which is done in a lab or a clinical setting and applied research, which is done with real subjects in real-world situations.And from these categories of research, we have the following general types of studies:

Animal Study: An animal or in vivo study is a study in which animals are used as subjects. A common use of an animal study is with a clinical trial (see below) and as a precursor to evaluating a medical intervention on humans. However, it is critical to recognize that results from animal studies should not be extrapolated to draw conclusions on what WILL happen in humans.

Case Study: A case study provides significant and detailed information about a single participant or a small group of participants. "Case studies are often referred to interchangeably with ethnography, field study, and participant observation." Unlike other studies which rely heavily on statistical analysis, the case study is often undertaken to identify areas for additional research and exploration.

Clinical Trial Study: A clinical trial study is often used in the areas of health and medical treatments that will presumably yield a positive effect. Typically a small group of people or animals are selected based upon the presence of a specific medical condition. This group is used to evaluate the effectiveness of a new medication or treatment, differing dosages, new applications of existing treatments. Due to the risk involved with many new medical treatments, the initial subjects in a clinical trial may be animals and not humans. After positive outcomes are obtained, research then can proceed to a human study where the treatment is compared against results from the existing standard of care.

Correlational Study: Correlational studies evaluate the relationship between variables and determine if there is a positive correlation, a negative correlation, or no correlation. Please note, a positive correlation does not mean one thing causes another. Correlational studies are typically used in naturalistic observations, surveys, and with archival research.

Cross-sectional Survey: Also know as the synchronic study, a cross-sectional survey collects data at a single point in time but the questions asked of a participant may be about current and past experiences. They are often done to evaluate some aspect of public health policy.

Epidemiological Study: Epidemiological studies evaluate the factors and associations linked to diseases. Types of epidemiological studies include case series studies, case control studies, cohort studies, longitudinal studies, and outbreak investigations.

Epidemiological studies are often beneficial in identifying areas for a more control research evaluation; however all to often, readers of epidemiological research miscategorize links and associations as causes. In addition, a common problem with epidemiological studies is that they rely on memory recall which can be quite unreliable.

Experimental Study: In an experimental study, specific treatments are applied to a sample or group and the results are observed.

Literature Review: A literature review is an exhaustive search of all of the relevant literature related to a specific research topic.

Longitudinal Study: A specific type of epidemiological study, the longitudinal study follows subjects over a long period of time, asking a specific research question with repeated samples of data gathered across the duration of the study. These studies are often used as the basis for specific experimental studies. For example, the Framingham Heart Study has evaluated people from the town of Framingham, Massachusetts since 1948 looking for patterns in heart disease.

Meta-analysis: A meta-analysis is a statistical process in which the results of multiple studies evaluating a similar research objective are collected and pooled together. They are often used to determine the effectiveness of healthcare interventions and experiments.

TYPES OF REPORTS

INTRODUCTION

REPORT:

A report or account is an informational work, such as writing, speech, television or film, made with the intention of relaying information or recounting events in a presentable form.

A report is made with the specific intention of relaying information or recounting certain events in a way that is concise, factual and relevant to the audience at hand. Reports may be conveyed through a written medium, speech, television, or film. In professional spheres, reports are a common and vital communication tool. Additionally, reports may be official or unofficial, and can be listed publicly or only available privately depending on the specific scenario. The audience for a report can vary dramatically, from an elementary school classroom to a boardroom on Wall Street.

Reports fill a vast array of informational needs for a spectrum of audiences. They may be used to keep track of information, evaluate a strategy, or make decisions. Written reports are documents which present focused and salient content, generally to a specific audience. An example of an official report would be a police report, which could have legally binding consequences. Other types of reports, such as Consumer Reports, inform the public about the quality of products available on the market. Reports are used in government, business, education, science, and other fields, often to display the results of an experiment, investigation or inquiry.

Attributes

One of the most common formats for presenting reports is IMRAD—introduction, methods, results, and discussion. This structure, standard for the genre, mirrors traditional publication of

scientific research and summons the ethos and credibility of that discipline. Reports are not required to follow this pattern and may use alternative methods such as the problem-solution format, wherein the author first lists an issue and then details what must be done to fix the problem. Transparency and a focus on quality are keys to writing a useful report. Accuracy is also important. Faulty numbers in a financial report could lead to disastrous consequences.

Standard Elements

Reports use features such as tables, graphics, images, voice, or specialized vocabulary in order to persuade a specific audience to undertake an action or inform the reader of the subject at hand. Some common elements of written reports include headings to indicate topics and help the reader locate relevant information quickly, and visual elements such as charts, tables and figures, which are useful for breaking up large sections of text and making complex issues more accessible. Lengthy written reports will almost always contain a table of contents, appendices, footnotes, and references. A bibliography or list of references will appear at the end of any credible report and citations are often included within the text itself. Complex terms are explained within the body of the report or listed as footnotes in order to make the report easier to follow. A short summary of the report's contents, called an abstract, may appear in the beginning so that the audience knows what the report will cover. Online reports often contain hyperlinks to internal or external sources as well.

Verbal reports differ from written reports in the minutiae of their format, but they still educate or advocate for a course of action. Quality reports will be well researched and the speaker will list their sources if at all possible.

Types

Some examples of reports are:

Annual Reports

Auditor's reports

Book reports

Bound reports

Retail reports

Census reports

Credit reports

Demographic reports

Expense reports

Experience report

Inspection reports

Military reports

Police reports

Policy reports

Informal reports

Progress reports

Investigate reports

Scientific reports

Trip reports

White papers

Appraisal reports

Workplace reports

REPORT WRITING:TYPES, FORMAT, STRUCTURE AND RELEVANCE

1. **ENGLISH PRESENTATION REPORT WRITING**: TYPES, FORMATS, STRUCTURE and RELEVANCE

2. **REPORT** :It is any informational work made with an intention to relay information or recounting certain events in a presentable manner. Reports are often conveyed in writing, speech, television, or film. Report is an administrative necessity. Most official form of information or work are completed via report. Report is always written in a sequential manner in order of occurrence

3. **TYPES OF REPORT**: FORMAL INFORMAL

• The informal report functions to inform, analyze, and recommend.

• It usually takes the form of a memo, letter or a very short international document like a monthly financial report, monthly activities report, research and development report, etc.

• This report differs from the formal report in length and formality.

• It is written according to organization style and rules, but usually does not include the preliminary (front) and supplemental (back) material.

• The informal report is usually more controversial in tone and typically deals with everyday problems and issues addressed to a narrow readership inside the organization.

4.TYPES OF INFORMAL REPORT

There are many embodiments of the informal report:

i. Progress report

ii. Sales activity report

iii. Personnel evaluation

iv. Financial report

v. Feasibility report

vi. Literature review

vii. Credit report

5. The formal report is the collection and interpretation of data and information.

• The formal report is complex and used at an official level.

• It is often a written account of a major project.

• Examples of subject matter include new technologies, the advisability of launching a new project line, results of a study or experiment, an annual report, or a year old review of developments in the field.

6. TYPES OF FORMAL REPORT

They can be categorized as:

Informational reports

Analytical reports

Recommendation reports

7.INFORMATIONAL REPORTS

• Informational reports present results so readers can understand a particular problem or situation.

• Example: Manager of a city's website might prepare an informational report for the city council; the report would provide statistics on the number of people who pay their city water and sewage bills online etc.

• Informational reports might:

A. Present information on the status of current research or of a project.

B. Present an update of the operation in your division.

C. Explain how your organization or division does something.

D. Present the results of a questionnaire or research.

8.ANALYTICAL REPORTS

• This type goes a step beyond presenting results. Analytical reports present results, analyze those results, and draw conclusions based on those results.

• These reports attempt to describe why or how something happened and then to explain what it means.

• Like informational reports, analytical reports can be formal or informal.

• Explain what cause a problem or situation – Present the results of a traffic study showing accidents at an intersection – the report explains what it means.

• Explain the potential results of a particular course of action.

• Suggest which option, action, or procedure is best.

9. **RECOMMENDATION REPORTS**

• This type advocate a particular course of action. This usually present the results and conclusions that support the recommendations.

• This type is identical to analytical report.

• For example, your analytical report suggests using treatment X is more efficient than treatments Y and Z. However, that does not mean that you will use treatment X as cost and other considerations might recommend treatment Y.

• What should we do about a problem?

• Should we or can we do something?

• Should we change the method or technology we use to do something?

10.THE PLAN FOR PREPARING A FORMAL REPORT

• Identify the readers

• Determine your purpose

• Formulate specific questions

• Conduct research to answer the questions

• Draw valid conclusions (for analytical or recommendation reports)

• Decide on recommendations (for recommendations)

• Write the report

11.Most widely used Information written in a logical manner or sequence All the case studies or formal narration are written in this format , it follows a sequential manner of description All the steps are to be followed accordingly. **NARRATIVE REPORT**

12. Summary of reports about specialized law enforcement and police problems May be either narrative, chronological, or both It may not follow a single pattern due to the different case types and description. **SPECIALIZED REPORT**

13.The report defines complete description of the accident It describes the location and the loss occurred. It is important for documentation It is required to start any investigation. **ACCIDENT REPORT**

14. The purpose is to describe the way things are, this type is mainly used in letter, non fiction book, information leaflet, catalouge etc. The opening contains a general classification, a description of chosen subject then the paragraphs about different aspects of the subject and lastly conclusion **NON CHRONOLOGICAL REPORT**

15. **FORMAT OF A REPORT** □ Although, there is no set report writing format, however, there are general sections that should be included. □ Unlike essays, reports are written in sections with headings and sub-headings, which are usually numbered. There are numerous possible formats available for writing a report, and it mostly depends on the context of topic

16.. Below given are the **components of a report** in which they would occur:

1. **Title page:** It should include the title, your name and the name of the tutor to whom it is being submitted, date of submission, your course/department. The logo of the organisation should also be printed.

2. **Acknowledgements:** A list of people and organisations who have helped you in the compilation of report and other related work.

3. Contents page: A clear, well-formatted list of all the sections and sub-sections of the report. Page numbers should be marked correctly.

4. Abstract: A summary of the major points, conclusions, and recommendations should be written to give a general overview of report.

5. Introduction: The first page of the report needs to have an introduction. You will explain the problem and show the reader why the report is being made.

6. Body: This is the main section of the report. There needs to be several sections, with each having a subtitle. The various sections include Review of Literature, Materials and Methods and Results. A discussion section can also be included at the end of the body to go over by findings and their significance.

7. **Conclusion:** A conclusion should draw out the implications of your findings, with deductions based on the facts described in your main body. The significance and relevance of study is discussed in this section.

8. References: This is a list giving the full details of all the sources to which you have made reference within your text.

17. RELEVANCE OF REPORT WRITING

1. REPORT AS A MEANS OF INTERNAL COMMUNICATION A report acts as an effective means of communication within the organization. It provides feedback to employees. It is prepared for the information and guidance of others connected with the matter / problem.

2. REPORT FACILITATES DECISION MAKING AND PLANNING Report provide reliable data which can be used in the planning and decision making process. It acts as a treasure house of reliable information for long term planning and decision making.

3. REPORT DISCLOSES UNKNOWN INFORMATION Reports provide information, which may not be known previously. The committee members collect data, draw conclusions and provide information which will be new to all concerned parties.Even new business opportunities are visible through unknown information available in the reports.

4. REPORT GIVES INFORMATION TO EMPLOYEES .Reports are available to managers and departments for internal use. They are widely used by the departments for guidance. Report provide a feedback to employees and are useful for their self-improvement.

<u>LAYOUT OF RESEARCH REPORT</u>

INTRODUCTION

LAYOUT OF THE RESEARCH REPORT.

Anybody, who is reading the research report, must necessarily be conveyed enough about the study so that he can place it in its general scientific context, judge the adequacy of its methods and thus form an opinion of how seriously the findings are to be taken. For this purpose there is the need of proper layout of the report. The layout of the report means as to what the research report should contain.

A comprehensive layout of the research report should comprise preliminary pages, the main text and the end matter. Let us deal with them separately.

Preliminary Pages

In its preliminary pages the report should carry a title and date, followed by acknowledgements in the form of 'Preface' or 'Foreword'. Then there should be a table of contents followed by list of tables and illustrations so that the decision-maker or anybody interested in reading the report can easily locate the required information in the report.

Main Text

The main text provides the complete outline of the research report along with all details. Title of the research study is repeated at the top of the first page of the main text and then follows the other details on pages numbered consecutively, beginning with the second page. Each main section of the report should begin on a new page. The main text of the report should have the following sections:

- **Introduction**
- **Statement of findings and recommendations**
- **The results**
- **The implications drawn from the results; and**
- **The summary.**

Introduction:

The purpose of introduction is to introduce the research project to the readers. It should contain a clear statement of the objectives of research i.e., enough background should be given to make clear to the reader why the problem was considered worth investigating. A brief summary of other relevant research may also be stated so that the present study can be seen in that context. The hypotheses of study, if any, and the definitions of the major concepts employed in the study should be explicitly stated in the introduction of the report.
The methodology adopted in conducting the study must be fully explained. The scientific reader

would like to know in detail about such thing: How was the study carried out? What was its basic design? If the study was an experimental one, then what were the experimental manipulations? If the data were collected by means of questionnaires or interviews, then exactly what questions were asked (The questionnaire or interview schedule is usually given in an appendix)? If measurements were based on observation, then what instructions were given to the observers? Regarding the sample used in the study the reader should be told: Who were the subjects? How many were there? How were they selected? All these questions are crucial for estimating the probable limits of generalizability of the findings. The statistical analysis adopted must also be clearly stated. In addition to all this, the scope of the study should be stated and the boundary lines be demarcated. The various limitations, under which the research project was completed, must also be narrated.

Statement of findings and recommendations:

After introduction, the research report must contain a statement of findings and recommendations in non-technical language so that it can be easily understood by all concerned. If the findings happen to be extensive, at this point they should be put in the summarised form.

Results:

A detailed presentation of the findings of the study, with supporting data in the form of tables and charts together with a validation of results, is the next step in writing the main text of the report. This generally comprises the main body of the report, extending over several chapters. The result section of the report should contain statistical summaries and reductions of the data rather than the raw data. All the results should be presented in logical sequence and splitted into readily identifiable sections. All relevant results must find a place in the report. But how one is to decide about what is relevant is the basic question. Quite often guidance comes primarily from the research problem and from the hypotheses, if any, with which the study was concerned. But ultimately the researcher must rely on his own judgement in deciding the outline of his report. "Nevertheless, it is still necessary that he states clearly the problem with which he was concerned, the procedure by which he worked on the problem, the conclusions at which he arrived, and the bases for his conclusions.

Implications of the results:

Toward the end of the main text, the researcher should again put down the results of his research clearly and precisely. He should, state the implications that flow from the results of the study, for the general reader is interested in the implications for understanding the human behaviour. Such implications may have three aspects as stated below:

A statement of the inferences drawn from the present study which may be expected to apply in similar circumstances.

The conditions of the present study which may limit the extent of legitimate generalizations of the inferences drawn from the study.

The relevant questions that still remain unanswered or new questions raised by the study along with suggestions for the kind of research that would provide answers for them. It is considered a good practice to finish the report with a short conclusion which summarises and recapitulates the main points of the study. The conclusion drawn from the study should be clearly related to the hypotheses that were stated in the introductory section. At the same time, a forecast of the probable future of the subject and an indication of the kind of research which needs to be done in that particular field is useful and desirable.

Summary: It has become customary to conclude the research report with a very brief summary, resting in brief the research problem, the methodology, the major findings and the major conclusions drawn from the research results.

End Matter

At the end of the report, appendices should be enlisted in respect of all technical data such as questionnaires, sample information, mathematical derivations and the like ones. Bibliography of sources consulted should also be given. Index (an alphabetical listing of names, places and topics along with the numbers of the pages in a book or report on which they are mentioned or discussed) should invariably be given at the end of the report. The value of index lies in the fact that it works as a guide to the reader for the contents in the report.

MECHANISM OF WRITING A RESEARCH REPORT

INTRODUCTION

There are very definite and set rules which should be followed in the actual preparation of the research report or paper. Once the techniques are finally decided, they should be scrupulously adhered to, and no deviation permitted. The criteria of format should be decided as soon as the materials for the research paper have been assembled. The following points deserve mention so far as the mechanics of writing a report are concerned:

Size and physical design:

The manuscript should be written on unruled paper 81/2× 11in size. If it is to be written by hand, then black or blue-black ink should be used. A margin of at least one and one-half inches should be allowed at the left hand and of at least half an inch at the right hand of the paper. There should

also be one-inch margins, top and bottom. The paper should be neat and legible. If the manuscript is to be typed, then all typing should be double-spaced on one side of the page only except for the insertion of the long quotations.

Procedure:

Various steps in writing the report should be strictly adhered (All such steps have already been explained earlier in this chapter).

Layout:

Keeping in view the objective and nature of the problem, the layout of the report should be thought of and decided and accordingly adopted (The layout of the research report and various types of reports have been described in this chapter earlier which should be taken as a guide for report-writing in case of a particular problem).

Treatment of quotations:

Quotations should be placed in quotation marks and double spaced, forming an immediate part of the text. But if a quotation is of a considerable length (more than four or five type written lines) then it should be single-spaced and indented at least half an inch to the right of the normal text margin.

The footnotes:

Regarding footnotes one should keep in view the followings:

The footnotes serve two purposes viz., the identification of materials used in quotations in the report and the notice of materials not immediately necessary to the body of the research text but still of supplemental value. In other words, footnotes are meant for cross references, citation of authorities and sources, acknowledgement and elucidation or explanation of a point of view. It should always be kept in view that footnote is not an end nor a means of the display of scholarship. The modern tendency is to make the minimum use of footnotes for scholarship does not need to be displayed.

Footnotes are placed at the bottom of the page on which the reference or quotation which they identify or supplement ends. Footnotes are customarily separated from the textual material by a space of half an inch and a line about one and a half inches long.

Footnotes should be numbered consecutively, usually beginning with 1 in each chapter separately. The number should be put slightly above the line, say at the end of a quotation. At the foot of the page, again, the footnote number should be indented and typed a little above the line. Thus, consecutive numbers must be used to correlate the reference in the text with its corresponding note at the bottom of the page, except in case of statistical tables and other numerical material, where symbols such as the asterisk (*) or the like one may be used to prevent confusion.

Footnotes are always typed in single space though they are divided from one another by double space.

Documentation style: Regarding documentation, the first footnote reference to any given work should be complete in its documentation, giving all the essential facts about the edition used. Such documentary footnotes follow a general sequence. The common order may be described as under:

Regarding the single-volume reference

Author's name in normal order (and not beginning with the last name as in a bibliography) followed by a comma;

Title of work, underlined to indicate italics;

Place and date of publication;

Pagination references (The page number).

Example
John Gassner, Masters of the Drama, New York: Dover Publications, Inc. 1954, p. 315.

Regarding multivolumed reference

Author's name in the normal order;

Title of work, underlined to indicate italics;

Place and date of publication;

Number of volume;

Pagination references (The page number).

Regarding works arranged alphabetically

For works arranged alphabetically such as encyclopedias and dictionaries, no pagination reference is usually needed. In such cases the order is illustrated as under:
Example 1
"Salamanca," Encyclopaedia Britannica, 14th Edition.
Example 2
"Mary Wollstonecraft Godwin," Dictionary of national biography.
But if there should be a detailed reference to a long encyclopedia article, volume and pagination reference may be found necessary.

Regarding periodicals reference

Name of the author in normal order;

Title of article, in quotation marks;

Name of periodical, underlined to indicate italics;

Volume number;

Date of issuance;

Pagination.

Regarding anthologies and collections reference Quotations from anthologies or collections of literary works must be acknowledged not only by author, but also by the name of the collector.

Regarding second-hand quotations reference
In such cases the documentation should be handled as follows:

Original author and title;

"quoted or cited in,";

Second author and work.

Example
J.F. Jones, Life in Ploynesia, p. 16, quoted in History of the Pacific Ocean area, by R.B. Abel, p. 191.

Case of multiple authorship
If there are more than two authors or editors, then in the documentation the name of only the first given and the multiple authorship is indicated by "et al." or "and others". Subsequent references to the same work need not be so detailed as stated above. If the work is cited again without any other work intervening, it may be indicated as ibid, followed by a comma and the page number. A single page should be referred to as p., but more than one page be referred to as pp. If there are several pages referred to at a stretch, the practice is to use often the page number, for example, pp. 190ff, which means page number 190 and the following pages; but only for page 190 and the following page '190f'. Roman numerical is generally used to indicate the number of the volume of a book. Op. cit. (opera citato, in the work cited) or Loc. cit. (loco citato, in the place cited) are two of the very convenient abbreviations used in the footnotes. Op. cit. or Loc. cit. after the writer's name would suggest that the reference is to work by the writer which has been cited in detail in an earlier footnote but intervened by some other references.

Punctuation and abbreviations in footnotes: The first item after the number in the footnote is the author's name, given in the normal signature order. This is followed by a comma. After the comma, the title of the book is given: the article (such as "A", "An", "The" etc.) is omitted and only the first word and proper nouns and adjectives are capitalized. The title is followed by a comma. Information concerning the edition is given next. This entry is followed by a comma. The

place of publication is then stated; it may be mentioned in an abbreviated form, if the place happens to be a famous one such as Lond. for London, N.Y. for New York, N.D. for New Delhi and so on. This entry is followed by a comma. Then the name of the publisher is mentioned and this entry is closed by a comma. It is followed by the date of publication if the date is given on the title page. If the date appears in the copyright notice on the reverse side of the title page or elsewhere in the volume, the comma should be omitted and the date enclosed in square brackets [c 1978], [1978]. The entry is followed by a comma. Then follow the volume and page references and are separated by a comma if both are given. A period closes the complete documentary reference. But one should remember that the documentation regarding acknowledgements from magazine articles and periodical literature follow a different form as stated earlier while explaining the entries in the bibliography.

Certain English and Latin abbreviations are quite often used in bibliographies and footnotes to eliminate tedious repetition. The following is a partial list of the most common abbreviations frequently used in report-writing (the researcher should learn to recognise them as well as he should learn to use them):

anon., anonymous
ante., before
art., article
aug., augmented
bk., book
bull., bulletin
cf., compare
ch., chapter
col., column
diss., dissertation
ed., editor, edition, edited.
ed. cit., edition cited
e.g., exempli gratia: for example
eng., enlarged
et.al., and others
et seq., et sequens: and the following
ex., example
f., ff., and the following
fig(s)., figure(s)
fn., footnote
ibid., ibidem: in the same place (when two or more successive footnotes refer to the same work, it is not necessary to repeat complete reference for the second footnote. Ibid. may be used. If different pages are referred to, pagination must be shown).
id., idem: the same

ill., illus., or
illust(s). illustrated, illustration(s)
Intro., intro., introduction
l, or ll, line(s)
loc. cit., in the place cited; used as op.cit., (when new reference
loco citato: is made to the same pagination as cited in the previous note)
MS., MSS., Manuscript or Manuscripts
N.B., nota bene: note well
n.d., no date
n.p., no place
no pub., no publisher
no(s)., number(s)
o.p., out of print
op. cit: in the work cited (If reference has been made to a work
opera citato and new reference is to be made, ibid., may be used, if intervening
reference has been made to different works, op.cit. must be used. The
name of the author must precede.
p. or pp., page(s)
passim: here and there
post: after
rev., revised
tr., trans., translator, translated, translation
vid or vide: see, refer to
viz., namely
vol. or vol(s)., volume(s)
vs., versus: against

Use of statistics, charts and graphs:

A judicious use of statistics in research reports is often considered a virtue for it contributes a great deal towards the clarification and simplification of the material and research results. One may well remember that a good picture is often worth more than thousand words. Statistics are usually presented in the form of tables, charts, bars and line-graphs and pictograms. Such presentation should be self explanatory and complete in itself. It should be suitable and appropriate looking to the problem at hand. Finally, statistical presentation should be neat and attractive.

The final draft:

Revising and rewriting the rough draft of the report should be done with great care before writing the final draft. For the purpose, the researcher should put to himself questions like: Are the sentences written in the report clear? Are they grammatically correct? Do they say what is meant'? Do the various points incorporated in the report fit together logically? "Having at least one

colleague read the report just before the final revision is extremely helpful. Sentences that seem crystal-clear to the writer may prove quite confusing to other people; a connection that had seemed self evident may strike others as a non-sequitur. A friendly critic, by pointing out passages that seem unclear or illogical, and perhaps suggesting ways of remedying the difficulties, can be an invaluable aid in achieving the goal of adequate communication.

Bibliography:

Bibliography should be prepared and appended to the research report as discussed earlier.

Preparation of the index:

At the end of the report, an index should invariably be given, the value of which lies in the fact that it acts as a good guide, to the reader. Index may be prepared both as subject index and as author index. The former gives the names of the subject-topics or concepts along with the number of pages on which they have appeared or discussed in the report, whereas the latter gives the similar information regarding the names of authors. The index should always be arranged alphabetically. Some people prefer to prepare only one index common for names of authors, subject-topics, concepts and the like ones.

PRECAUTIONS FOR WRITING RESEARCH REPORTS

Research report is a channel of communicating the research findings to the readers of the report. A good research report is one which does this task efficiently and effectively. As such it must be prepared keeping the following precautions in view:

While determining the length of the report (since research reports vary greatly in length), one should keep in view the fact that it should be long enough to cover the subject but short enough to maintain interest. In fact, report-writing should not be a means to learning more and more about less and less.

A research report should not, if this can be avoided, be dull; it should be such as to sustain reader's interest.

Abstract terminology and technical jargon should be avoided in a research report. The report should be able to convey the matter as simply as possible. This, in other words, means that report should be written in an objective style in simple language, avoiding expressions such as "it seems," "there may be" and the like.

Readers are often interested in acquiring a quick knowledge of the main findings and as such the report must provide a ready availability of the findings. For this purpose, charts, graphs and the statistical tables may be used for the various results in the main report in addition to the summary of important findings.

The layout of the report should be well thought out and must be appropriate and in accordance with the objective of the research problem.

The reports should be free from grammatical mistakes and must be prepared strictly in accordance with the techniques of composition of report-writing such as the use of quotations, footnotes, documentation, proper punctuation and use of abbreviations in footnotes and the like.

The report must present the logical analysis of the subject matter. It must reflect a structure wherein the different pieces of analysis relating to the research problem fit well.

A research report should show originality and should necessarily be an attempt to solve some intellectual problem. It must contribute to the solution of a problem and must add to the store of knowledge.

Towards the end, the report must also state the policy implications relating to the problem under consideration. It is usually considered desirable if the report makes a forecast of the probable future of the subject concerned and indicates the kinds of research still needs to be done in that particular field.

Appendices should be enlisted in respect of all the technical data in the report.

Bibliography of sources consulted is a must for a good report and must necessarily be given.

Index is also considered an essential part of a good report and as such must be prepared and appended at the end.

Report must be attractive in appearance, neat and clean, whether typed or printed.

Calculated confidence limits must be mentioned and the various constraints experienced in conducting the research study may also be stated in the report.

Objective of the study, the nature of the problem, the methods employed and the analysis techniques adopted must all be clearly stated in the beginning of the report in the form of introduction.

REFERENCING IN ACADEMIC WRITING

The role of referencing

Academic writers need to support their arguments with evidence, and readers of academic writing need to evaluate the validity of that evidence. A reference is the bracketed or footnoted piece of information that tells the reader where the supporting evidence used in academic writing comes from. The role of the reference is twofold: firstly, it informs the reader of the source of your ideas so that he or she can distinguish between your words and ideas and those of others;

secondly, accurate referencing and lists of references are necessary to allow the reader to evaluate the information and read further into the area.

When to reference

You should use a reference when you have included an idea in your essay or report which is not your own original idea and which is not common knowledge. You must reference the idea whether you have presented the idea in the author's original words, as a quote, or have paraphrased or summarised the author's idea into your own words. You don't need to include a reference when the idea or concept is common knowledge in your discipline.

A GUIDE TO REFERENCING IN ACADEMIC WRITING

Why do we cite our references?

To credit the sources of our information and ideas.

To enable the reader to locate further information if required

To demonstrate the breadth of our reading and knowledge.

When to reference

When you are directly quoting from somebody else’s work.

When you are referring to or paraphrasing somebody else’s theory or idea.

Different styles of referencing

It is essential that you find out from your lecturer which style of referencing your department uses. Most departments at UCC use the Harvard style, but the APA style (American Psychological Association) is often used in social science subjects, so please check! This guide will cover the Harvard Style of referencing in detail, but if you do need the APA style you can find guidelines here: http://www.bibme.org/citation-guide/apa/

The Harvard Style of Referencing

The Harvard referencing style requires:

In-text references.

In the main body of the text after quoting from or referring to another person’s work. The author’s name followed by the year of publication.

End-of-Text references.

A detailed list at the end of your essay of all the references cited throughout the text.

Bibliography

A list of any other books, chapters or articles you may have read which are relevant to the subject matter of your essay, which you have not referred to specifically.

Examples on In-Text References in the Harvard Style

Example 1

Two authors of the same book, their research is being referred to. Note that the full stop comes after the brackets.

In conjunction with their perceived low social status, the key factors that influence the use of contraception among African women are the dominance of the husband in the marriage and his opposition to family planning (Beekle and McCabe 2006).

Example 2

Two separate pieces of research with the same findings are being referenced, one conducted by Kitzinger in 1992, one by Stanton et al in 2002. The earlier piece of research is always cited first. If two pieces of research have the same date, put them in alphabetical order according to the author's surname. 'Et al' means Stanton is one of two or more authors of that piece of research.

Recent research shows that women's experiences of and response to pregnancy and childbirth vary significantly and can have a long term positive or negative effect (Kitzinger 1992, Stanton et al. 2002).

Examples of different ways of using the author's name as part of a sentence:

Example 3

Findley (2003) suggests that loneliness is rarely considered as appropriate for intervention research; however, the results of such studies are promising.

Example 4

Findley (2003) and Wikstrom (2002) agree that . . .

Example 5

Beekle and McCabe (2006) state that in order for women to be able to make choices about family planning they need comprehensive information about all the methods of modern contraception, their benefits and side effects.

Example 6

According to Moreau et al. (2006) the main reasons for women discontinuing oral contraception included a lack of knowledge and high levels of anxiety.

Using Quotations (i.e. the actual words an author used)

Quotations must be used exactly as they appear in the original text and be placed within quotation marks. When quoting a piece of original text it is important to include the page number as well as the author's name and the date of publication.

Examples of short quotations as part of a sentence

Example 7

According to Papa et al. (2008, p.338) "the large, complex organization provides a rich medium for the development of coalitions and the occurrence of intergroup conflict".

Example 8

Huynh et al. (2008, p.195) are of the view that "the concept of emotional labour should be introduced into pre-registration programmes."

Quotations of two or more sentences (block quotations) should be presented as follows:

Use a smaller font i.e. if using font size 12 for the main body of your text, use font size 11 for the quote. Include the page number(s) of the original source text at the end of the quote. Do not use bold or italic

in this section, unless it appears in the original text. Leave an extra line of space between the main body of text and the quote. There is no need to use quotation marks/inverted commas with blockquotations.

Example 9

Sandall et al. (2009) highlight issues related to mid-wife led versus other models of care for childbearing women as follows:

Little is known about whether women feel that they are part of the decision making process; sense of control; maternal self-confidence; post-traumatic stress disorder; coping after birth. There is wide variation in the instruments used to measure women's views of and experiences of care. There is a need to develop meaningful, robust, valid and reliable methods to assess psychosocial outcomes and wellbeing in pregnant and childbearing women. (p.12)

Using primary and secondary sources

Wherever possible, it is much better to cite original work by an author (primary source), but if necessary, you can include a quote cited in a text by another author (a secondary source).

Example 10

A citation of a secondary source within a text.Weber (1969), as cited by Papa et al. (2008, p.83) provides what he refers to as the ideal modern organization.

Example 11 How to list a secondary source in your reference list.

Papa M.J., Daniels T.D. & Spiker B.K. (2008) Organizational Communication: Perspectives and Trends. Sage, Los Angeles.

How to Write a Reference List

You must include a list of all the references used in your text at the end of your assignment.

This should appear on a separate page under the title References

The list should be in alphabetical order, using the surname of the author(s).

Don't use numbers or bullet points.

Referencing a book

Author(s) surname plus initials in full. The names of all authors must be listed.

If it is an edited book, editor's name(s) are followed by a comma, then ed or eds followed by a full stop, in brackets.

Year of publication in brackets. If a book has been reprinted, give the original publication date.

Title in italics.

Volume number if the book has more than one volume number.

Edition number if the book has more than one edition, followed by "edn."

Publisher's name

Place of publication: if there is more than one place name listed, use the first one.

Morse J., (2000) Exploring pragmatic utility: concept analysis by critically appraising the literature.

In Concept Development in Nursing: Foundations, Techniques and Applications, 2nd edn. (Rogers B L., & Knafl K.A., eds), W.B. Saunders, Philadephia, pp.333-352

Referencing a Journal Article

Author(s) surname plus initial(s) in full. The names of the authors must be listed.

Year of publication in brackets.

Title of article.

Title of journal in full in italics.

Volume number in bold.

Issue number in brackets, if given.